

BOOK 7

WAVES

7 Waves

An understanding of colour from Cambridge IGCSE/O Level Physics or equivalent is assumed.

7.1 Progressive waves

Candidates should be able to:

- 1 describe what is meant by wave motion as illustrated by vibration in ropes, springs and ripple tanks
- 2 understand and use the terms displacement, amplitude, phase difference, period, frequency, wavelength and speed
- 3 understand the use of the time-base and y-gain of a cathode-ray oscilloscope (CRO) to determine frequency and amplitude
- 4 derive, using the definitions of speed, frequency and wavelength, the wave equation $v = f\lambda$
- 5 recall and use $v = f\lambda$
- 6 understand that energy is transferred by a progressive wave
- 7 recall and use $intensity = power/area$ and $intensity \propto (amplitude)^2$ for a progressive wave

7.2 Transverse and longitudinal waves

Candidates should be able to:

- 1 compare transverse and longitudinal waves
- 2 analyse and interpret graphical representations of transverse and longitudinal waves

7.3 Doppler effect for sound waves

Candidates should be able to:

- 1 understand that when a source of sound waves moves relative to a stationary observer, the observed frequency is different from the source frequency (understanding of the Doppler effect for a stationary source and a moving observer is not required)
- 2 use the expression $f_o = f_s v / (v \pm v_s)$ for the observed frequency when a source of sound waves moves relative to a stationary observer

7.4 Electromagnetic spectrum

Candidates should be able to:

- 1 state that all electromagnetic waves are transverse waves that travel with the same speed c in free space
- 2 recall the approximate range of wavelengths in free space of the principal regions of the electromagnetic spectrum from radio waves to γ -rays
- 3 recall that wavelengths in the range 400–700 nm in free space are visible to the human eye

7.5 Polarisation

Candidates should be able to:

- 1 understand that polarisation is a phenomenon associated with transverse waves
- 2 recall and use Malus's law ($I = I_0 \cos^2 \theta$) to calculate the intensity of a plane polarised electromagnetic wave after transmission through a polarising filter or a series of polarising filters

1 9702/1/M/J/02/Q25

Which of the following summarises the change in wave characteristics on going from infra-red to ultraviolet in the electromagnetic spectrum?

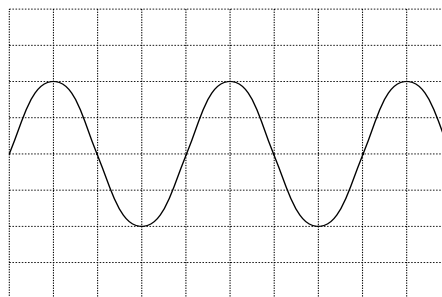
	frequency	speed (in a vacuum)
A	decreases	decreases
B	decreases	remains constant
C	increases	remains constant
D	increases	increases

2 9702/1/M/J/02/Q26

The diagram shows a cathode-ray oscilloscope trace of a sound wave. The time-base is calibrated at 2.0 ms cm^{-1} .

What is the frequency of the sound wave?

- A** 62.5 Hz **B** 125 Hz
C 250 Hz **D** 500 Hz



3 9702/1/M/J/02/Q27

Which statement correctly relates the intensity of a sound wave to the vibrations of the molecules?

- A** intensity $\propto$ amplitude **C** intensity $\propto$ displacement
B intensity $\propto$ (amplitude)² **D** intensity $\propto$ (displacement)²

4 9702/1/O/N/02/Q26

Which value is a possible wavelength for radiation in the microwave region of the electromagnetic spectrum?

- A** $3 \times 10^{-2} \text{ m}$ **B** $3 \times 10^{-5} \text{ m}$ **C** $3 \times 10^{-8} \text{ m}$ **D** $3 \times 10^{-10} \text{ m}$

5 9702/11/M/J/15/Q26

A wave of amplitude a has an intensity of 3.0 W m^{-2} .

What is the intensity of a wave of the same frequency that has an amplitude $2a$?

- A** 4.2 W m^{-2} **B** 6.0 W m^{-2} **C** 9.0 W m^{-2} **D** 12 W m^{-2}

6 9702/01/M/J/03/Q23

Which of the following is true for all transverse waves?

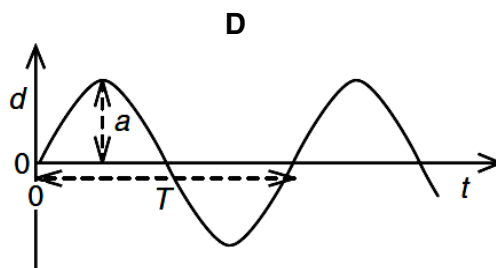
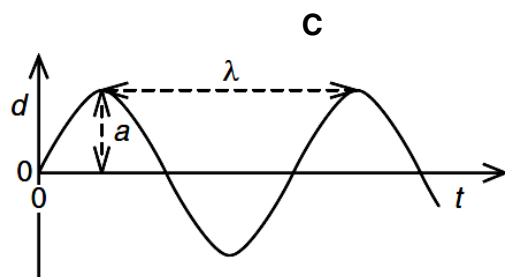
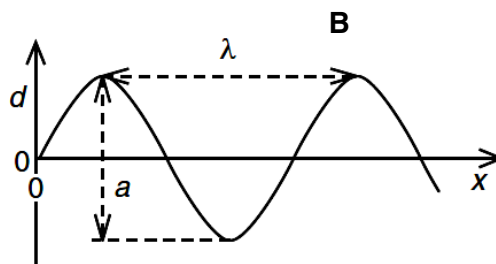
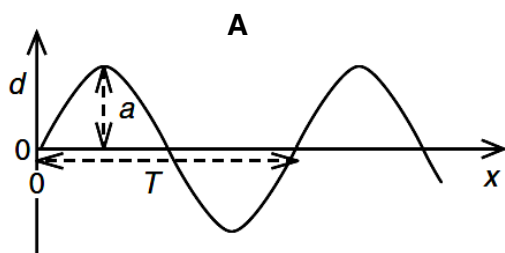
- A** They are all electromagnetic.
B They can all be polarised.
C They can all travel through a vacuum.
D They all involve the oscillation of atoms.

7 9702/1/O/N/02/Q26**

The four graphs represent a progressive wave on a stretched string. Graphs **A** and **B** show how the displacement d varies with distance x along the string at one instant. Graphs **C** and **D** show how the displacement d varies with time t at a particular value of x .

The labels on the graphs are intended to show the wavelength λ , the period T , and the amplitude a of the wave, but only one graph is correctly labelled.

Which graph is correctly labelled?

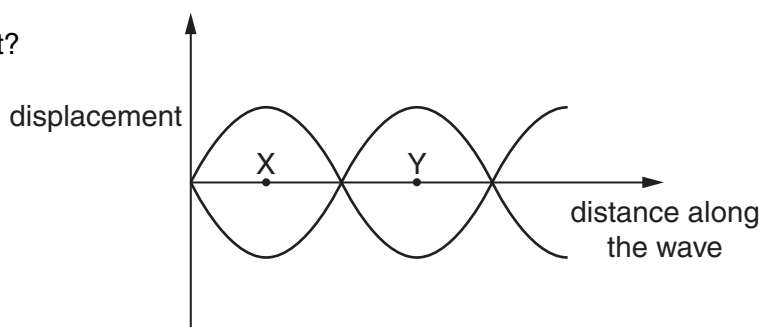


8 9702/01/M/J/03/Q24

The graph represents a stationary wave at two different times.

What does the distance XY represent?

- A** half the amplitude
- B** half the frequency
- C** half the period
- D** half the wavelength



9 9702/01/M/J/03/Q25*

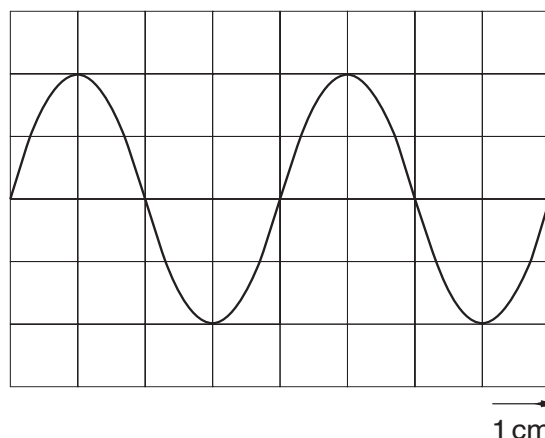
Electromagnetic waves of wavelength λ and frequency f travel at speed c in a vacuum.

Which of the following describes the wavelength and speed of electromagnetic waves of frequency $f/2$?

	wavelength	speed in a vacuum
A	$\lambda/2$	$c/2$
B	$\lambda/2$	c
C	2λ	c
D	2λ	$2c$

10 9702/01/M/J/03/Q26

A sound wave is displayed on the screen of a cathode-ray oscilloscope. The time base of the c.r.o. is set at 2.5 ms/cm.

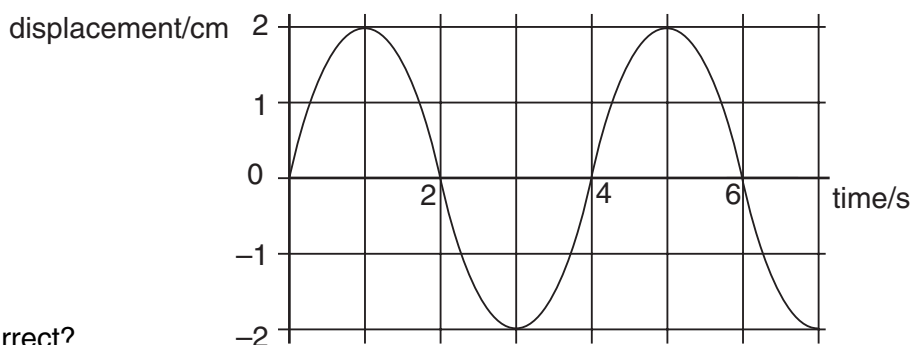


What is the frequency of the sound wave?

- A** 50 Hz **B** 100 Hz
C 200 Hz **D** 400 Hz

11 9702/01/O/N/03/Q23**

The graph shows how the displacement of a particle in a wave varies with time.



Which of the following is correct?

- A** The wave has an amplitude of 2 cm and could be either transverse or longitudinal.
B The wave has an amplitude of 2 cm and must be transverse.
C The wave has an amplitude of 4 cm and could be either transverse or longitudinal.
D The wave has an amplitude of 4 cm and must be transverse.

12 9702/01/O/N/03/Q24

A stationary sound wave has a series of nodes. The distance between the first and the sixth node is 30.0 cm.

What is the wavelength of the sound wave?

- A** 5.0 cm **B** 6.0 cm **C** 10.0 cm **D** 12.0 cm

13 9702/01/O/N/03/Q25

Which of the following applies to a progressive transverse wave?

	transfers energy	can be polarised
A	no	no
B	no	yes
C	yes	no
D	yes	yes

14 9702/01/O/N/03/Q26

Which of the following may be used to produce stationary waves?

- A blowing air over the top of an empty bottle
- B making a loud sound near a mountain
- C passing monochromatic light through a double slit
- D passing water waves through a narrow slit

15 9702/01/M/J/04/Q24

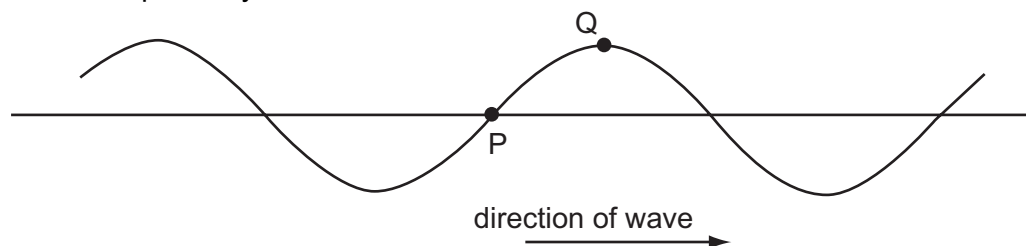
Which observation indicates that sound waves are longitudinal?

- A Sound can be reflected from a solid surface.
- B Sound cannot be polarised.
- C Sound is diffracted around corners.
- D Sound is refracted as it passes from hot air to cold air.

16 9702/01/M/J/04/Q25

The diagram shows a transverse wave on a rope. The wave is travelling from left to right.

At the instant shown, the points P and Q on the rope have zero displacement and maximum displacement respectively.



Which of the following describes the direction of motion, if any, of the points P and Q at this instant?

	point P	point Q
A	downwards	stationary
B	stationary	downwards
C	stationary	upwards
D	upwards	stationary

17 9702/01/M/J/04/Q26*

A plane wave of amplitude A is incident on a surface of area S placed so that it is perpendicular to the direction of travel of the wave. The energy per unit time reaching the surface is E .

The amplitude of the wave is increased to $2A$ and the area of the surface is reduced to $\frac{1}{2}S$.

How much energy per unit time reaches this smaller surface?

- A $4E$
- B $2E$
- C E
- D $\frac{1}{2}E$

18 9702/01/O/N/04/Q24

Which of the following is a longitudinal wave?

- A a light wave travelling through air
- B a radio wave from a broadcasting station
- C a ripple on the surface of water
- D a sound wave travelling through air

19 9702/01/O/N/04/Q25

A stationary sound wave is set up along the line joining two loudspeakers.

Which measurement is sufficient on its own to enable you to deduce the wavelength of the wave?

- A the amplitude of the sound wave
- B the distance between the two loudspeakers
- C the distance between two adjacent antinodes
- D the frequency of the sound wave

20 9702/01/O/N/04/Q26

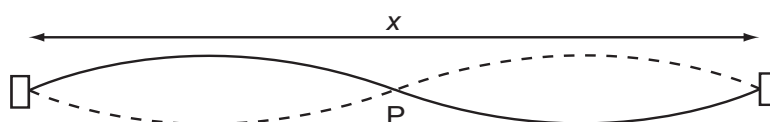
A wave of amplitude 20 mm has intensity I_X . Another wave of the same frequency but of amplitude 5 mm has intensity I_Y .

What is $\frac{I_X}{I_Y}$?

- A 2
- B 4
- C 16
- D 256

21 9702/01/O/N/04/Q27

The diagram represents a stationary wave on a stretched string.



What is represented by point P and by the length x ?

	point P	length x
A	antinode	one wavelength
B	antinode	two wavelengths
C	node	one wavelength
D	node	two wavelengths

22 9702/01/M/J/05/Q23

What do **not** travel at the speed of light in a vacuum?

- A electrons
- B microwaves
- C radio waves
- D X-rays

23 9702/01/M/J/05/Q24

The number of wavelengths of visible light in one metre is of the order of

- A 10^4 .
- B 10^6 .
- C 10^8 .
- D 10^{10} .

24 9702/01/M/J/05/Q25

A health inspector is measuring the intensity of a sound. Near a loudspeaker his meter records an intensity I . This corresponds to an amplitude A of the sound wave. At another position the meter gives an intensity reading of $2I$.

What is the corresponding sound wave amplitude?

- A $\frac{A}{\sqrt{2}}$
- B $\sqrt{2} A$
- C $2A$
- D $4A$

25 9702/01/O/N/05/Q22

Polarisation is a phenomenon associated with a certain type of wave.

Which condition **must** be fulfilled if a wave is to be polarised?

- A It must be a light wave.
- B It must be a longitudinal wave.
- C It must be a radio wave.
- D It must be a transverse wave.

26 9702/12/O/N/11/Q27*

A sound wave is set up in a long tube, closed at one end. The length of the tube is adjusted until the sound from the tube is loudest.

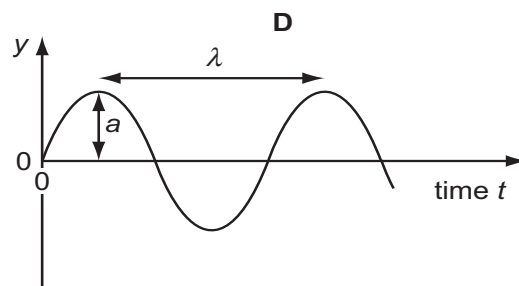
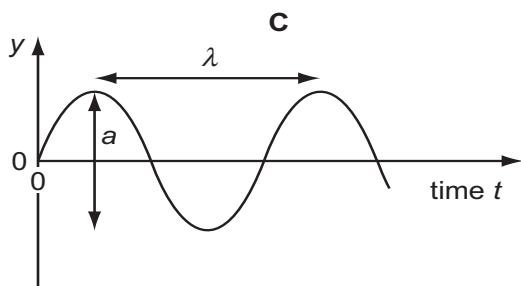
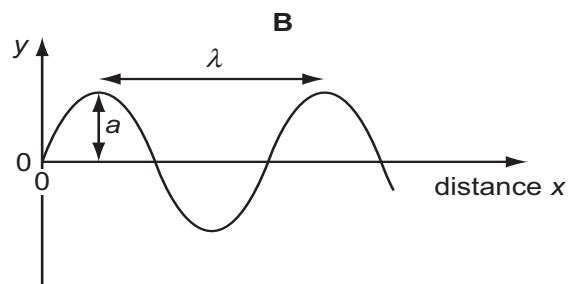
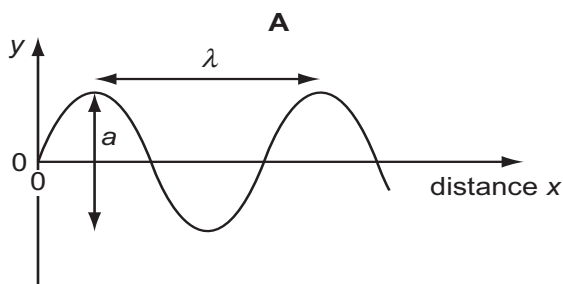
What is the nature of the sound wave in the tube?

- A** longitudinal and progressive **C** transverse and progressive
B longitudinal and stationary **D** transverse and stationary

27 9702/01/O/N/05/Q23

A sound wave has displacement y at distance x from its source at time t .

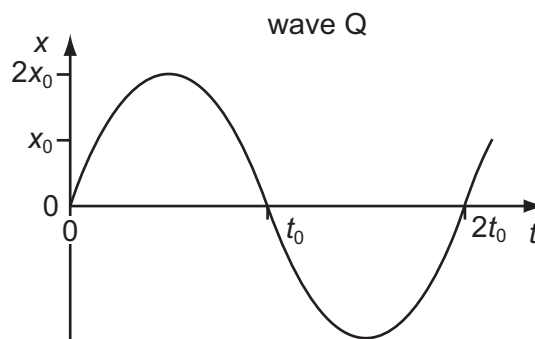
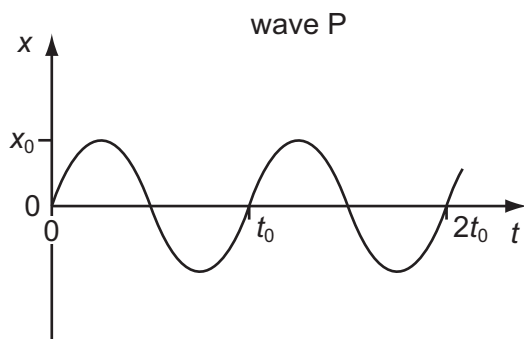
Which graph correctly shows the amplitude a and the wavelength λ of the wave?



28 9702/01/O/N/05/Q24

The intensity of a progressive wave is proportional to the square of the amplitude of the wave. It is also proportional to the square of the frequency.

The variation with time t of displacement x of particles in a medium, when two progressive waves P and Q pass separately through the medium, are shown on the graphs.



The intensity of wave P is I_0 .

What is the intensity of wave Q?

- A** $\frac{1}{2}I_0$ **B** I_0 **C** $8I_0$ **D** $16I_0$

29 9702/01/O/N/05/Q25

A sound wave of frequency 150 Hz travels in water at a speed of 1500 m s^{-1} . It then travels through the surface of the water and into air, where its speed is 300 m s^{-1} .

Which line in the table gives the correct values for the wavelengths of the sound in water and in air?

	wavelength in water / m	wavelength in air / m
A	0.10	0.10
B	0.10	0.50
C	10	2.0
D	10	50

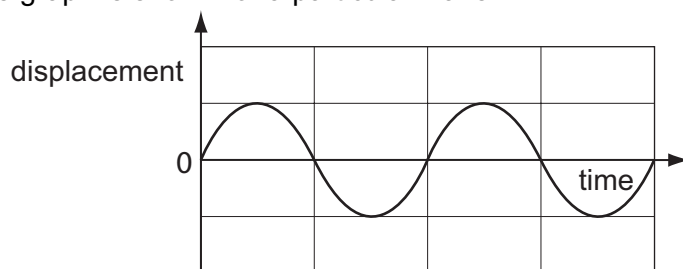
30 9702/01/M/J/06/Q23

Which phenomenon is associated with transverse waves but **not** longitudinal waves?

A polarisation B reflection C refraction D superposition

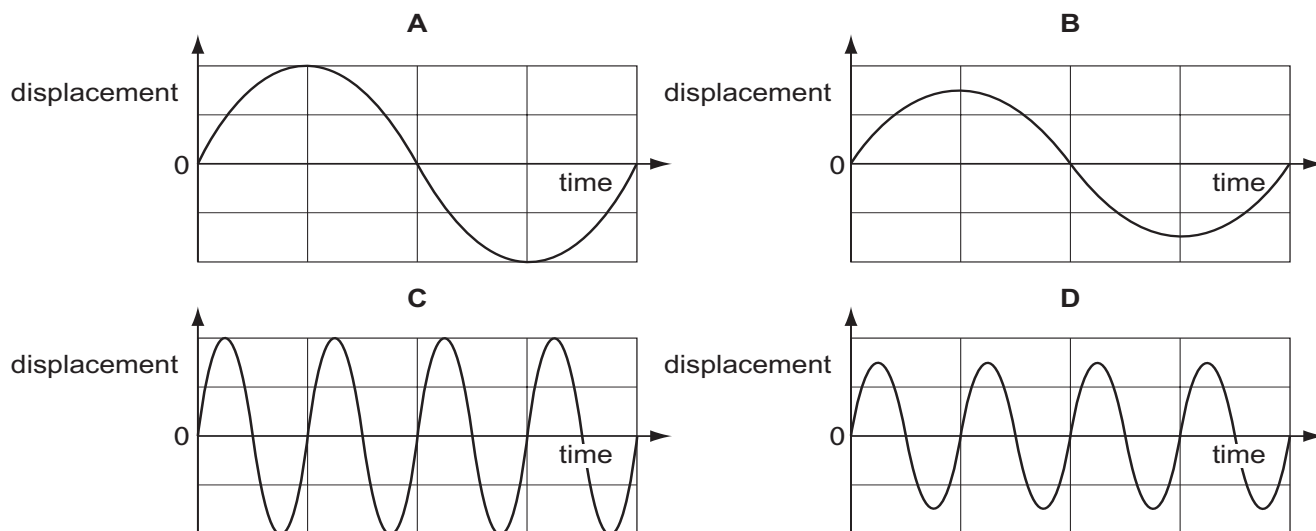
31 9702/01/M/J/06/Q24

A displacement-time graph is shown for a particular wave.



A second wave of similar type has twice the intensity and half the frequency.

When drawn on the same axes, what would the second wave look like?



32 9702/01/M/J/06/Q25

The frequency of a certain wave is 500 Hz and its speed is 340 m s^{-1} .

What is the phase difference between the motions of two points on the wave 0.17 m apart?

A $\frac{\pi}{4}$ rad B $\frac{\pi}{2}$ rad C $\frac{3\pi}{4}$ rad D π rad

33 9702/01/M/J/06/Q26

Where, in a standing wave, do the vibrations of the medium occur?

- A** only at the nodes **C** at all points between the nodes
B only at the antinodes **D** at all points between the antinodes

34 9702/01/O/N/06/Q18

An electric railway locomotive has a maximum mechanical output power of 4.0 MW. Electrical power is delivered at 25 kV from overhead wires. The overall efficiency of the locomotive in converting electrical power to mechanical power is 80 %.

What is the current from the overhead wires when the locomotive is operating at its maximum power?

- A** 130 A **B** 160 A **C** 200 A **D** 250 A

35 9702/01/O/N/06/Q24

A wave motion is described by the oscillation of particles.

What is the name given to the number of complete oscillations of a particle in one second?

- A** amplitude **B** frequency **C** wavelength **D** wave speed

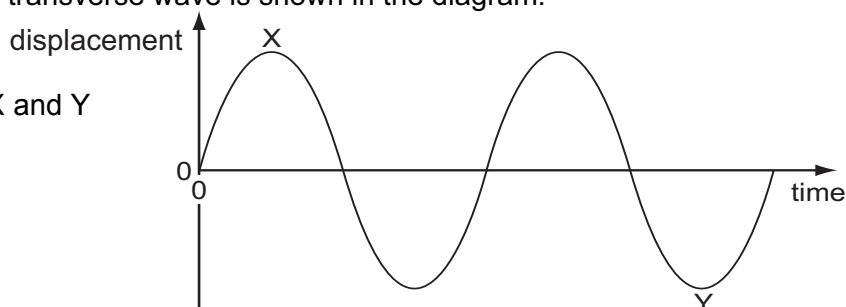
36 9702/01/O/N/06/Q25

A displacement-time graph for a transverse wave is shown in the diagram.

The phase difference between X and Y can be expressed as $n\pi$.

What is the value of n ?

- A** 1.5 **B** 2.5
C 3.0 **D** 6.0



37 9702/01/O/N/06/Q26

Continuous water waves are diffracted through a gap in a barrier in a ripple tank.

Which change will cause the diffraction of the waves to increase?

- A** increasing the frequency of the waves **C** reducing the wavelength of the waves
B increasing the width of the gap **D** reducing the width of the gap

38 9702/01/O/N/06/Q28

What may be used to produce stationary waves?

- A** blowing air over the top of an empty bottle
B making a loud sound near a mountain
C passing monochromatic light through a double slit
D passing water waves through a narrow slit

39 9702/01/M/J/07/Q21

Which of the following types of wave can be polarised?

- A** a longitudinal progressive wave **C** a transverse stationary wave
B a longitudinal stationary wave **D** a transverse sound wave

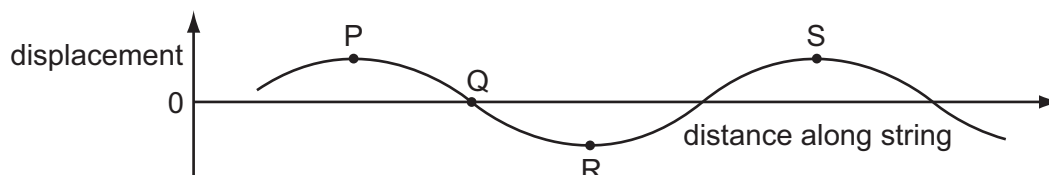
40 9702/01/M/J/07/Q22

Sound wave X has intensity 10^{12} times greater than that of sound wave Y.
By how much is the amplitude of X greater than the amplitude of Y?

- A 10^6 times C 5×10^{11} times
B 3.16×10^6 times D 10^{12} times

41 9702/01/M/J/07/Q23

The graph shows the shape at a particular instant of part of a transverse wave travelling along a string.

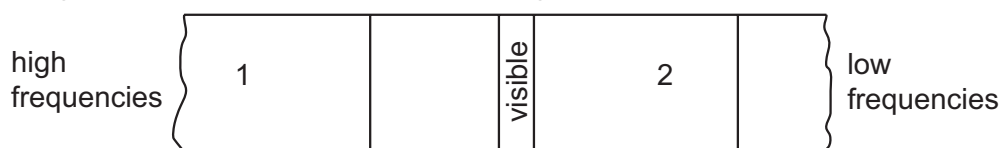


Which statement about the motion of points in the string is correct?

- A The speed at point P is a maximum. C The energy at point R is entirely kinetic.
B The displacement at point Q is always zero. D The acceleration at point S is a maximum.

42 9702/01/M/J/07/Q24

The diagram illustrates part of the electromagnetic spectrum.

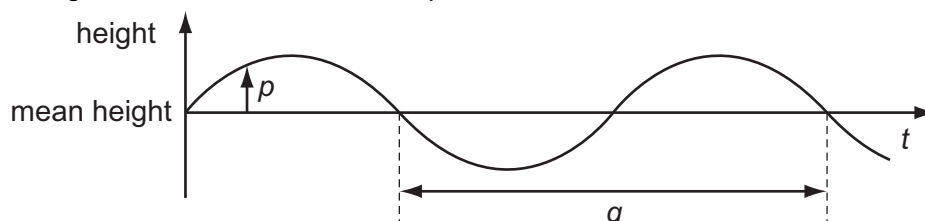


Which labels are correct for the regions marked 1 and 2?

	1	2
A	infrared	X-rays
B	microwaves	X-rays
C	ultraviolet	microwaves
D	X-rays	infrared

43 9702/01/M/J/08/Q25

The graph shows how the height of a water surface at a point in a harbour varies with time t as waves pass the point.



What are p and q ?

	p	q
A	displacement	wavelength
B	displacement	period
C	amplitude	wavelength
D	amplitude	period

44 9702/01/O/N/07/Q21

What is the relationship between the intensity I and the amplitude a of a wave?

- A $\frac{I}{a} = \text{constant}$ B $\frac{I}{a^2} = \text{constant}$ C $Ia = \text{constant}$ D $Ia^2 = \text{constant}$

45 9702/01/O/N/07/Q22

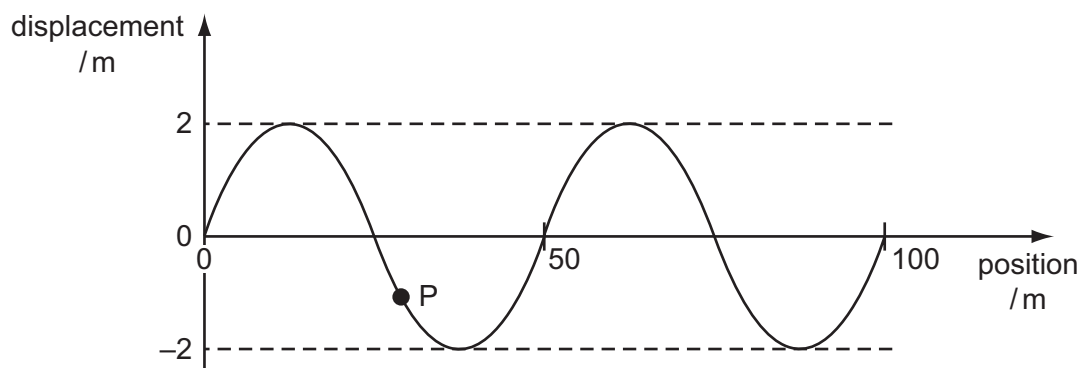
An electromagnetic wave has a frequency of 10^8 Hz.

In which region of the electromagnetic spectrum does the wave occur?

- A infra-red B radio C ultraviolet D visible

46 9702/01/O/N/07/Q23

The graph represents a sinusoidal wave in the sea, travelling at a speed of 8.0 m s^{-1} , at one instant of time. The maximum speed of the oscillating particles in the wave is $2\pi af$, where a is the amplitude and f is the frequency.



An object P of mass $2.0 \times 10^{-3} \text{ kg}$ floats on the surface.

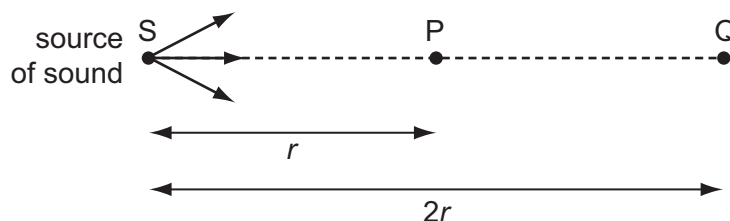
What is the maximum kinetic energy of P due to the wave? Assume that its motion is vertical.

- A 0.026 mJ B 4.0 mJ C 39 mJ D 64 mJ

47 9702/01/M/J/08/Q26

The intensity I of a sound at a point P is inversely proportional to the square of the distance x of P from the source of the sound. That is

$$I \propto \frac{1}{x^2}.$$



Air molecules at P, a distance r from S, oscillate with amplitude $8.0 \mu\text{m}$.

Point Q is situated a distance $2r$ from S.

What is the amplitude of oscillation of air molecules at Q?

- A $1.4 \mu\text{m}$ B $2.0 \mu\text{m}$ C $2.8 \mu\text{m}$ D $4.0 \mu\text{m}$

48 9702/01/M/J/08/Q27

Sound waves, emitted by a small loudspeaker, are reflected by a wall.

The frequency f of the waves is adjusted until a stationary wave is formed with the antinode nearest the wall at a distance x from the wall.

Which expression gives f in terms of x and the speed of sound c ?

- A $f = \frac{4c}{x}$ B $f = \frac{2c}{x}$ C $f = \frac{c}{2x}$ D $f = \frac{c}{4x}$

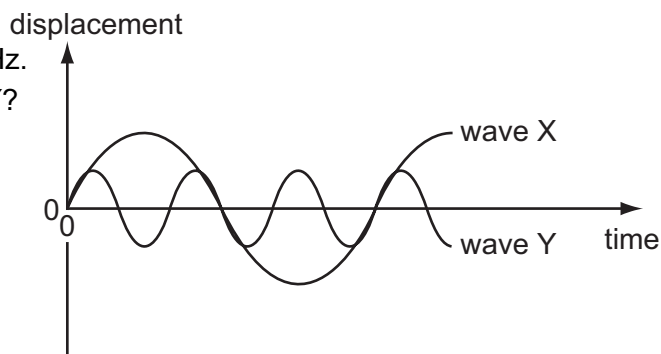
49 9702/01/O/N/08/Q24

The diagram shows two waves X and Y.

Wave X has amplitude 8 cm and frequency 100 Hz.

What are the amplitude and frequency of wave Y?

	amplitude / cm	frequency / Hz
A	2	33
B	2	300
C	4	33
D	4	300



50 9702/01/O/N/08/Q25

Light can exhibit all of the properties listed.

Which property can sound **not** exhibit?

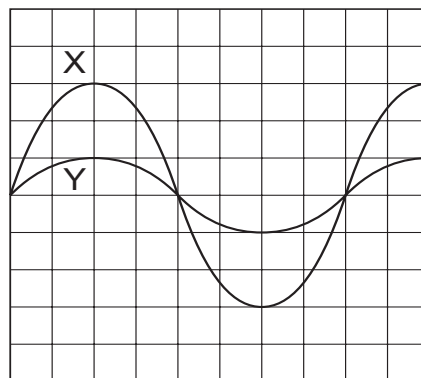
- A interference B polarisation C refraction D total internal reflection

51 9702/01/O/N/08/Q26

The diagram represents the screen of a cathode-ray oscilloscope displaying two sound waves labelled X and Y.

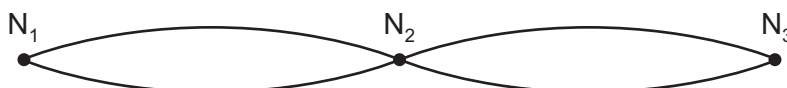
What is the ratio $\frac{\text{intensity of sound wave X}}{\text{intensity of sound wave Y}}$?

- A $\frac{9}{1}$ B $\frac{3}{1}$
C $\frac{\sqrt{3}}{1}$ D $\frac{1}{1}$



52 9702/12/O/N/09/Q24

The diagram shows a standing wave on a string. The standing wave has three nodes N_1 , N_2 and N_3 .



Which statement is correct?

- A All points on the string vibrate in phase.
B All points on the string vibrate with the same amplitude.
C Points equidistant from N_2 vibrate with the same frequency and in phase.
D Points equidistant from N_2 vibrate with the same frequency and the same amplitude.

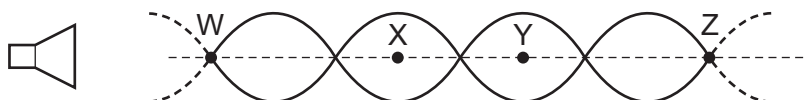
53 9702/01/M/J/09/Q23

Which wave properties change when light passes from air into glass?

- A colour and speed C speed and wavelength
B frequency and wavelength D wavelength and colour

54 9702/01/M/J/09/Q24

The diagram represents the pattern of stationary waves formed by the superposition of sound waves from a loudspeaker and their reflection from a metal sheet (not shown).



W, X, Y and Z are four points on the line through the centre of these waves.

Which statement about these stationary waves is correct?

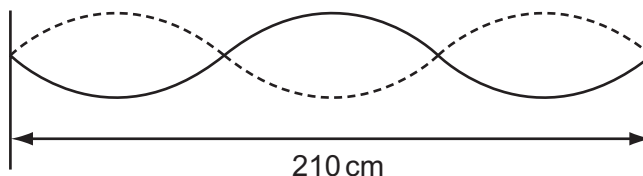
- A An antinode is formed at the surface of the metal sheet.
B A node is a quarter of a wavelength from an adjacent antinode.
C The oscillations at X are in phase with those at Y.
D The stationary waves oscillate at right angles to the line WZ.

55 9702/01/M/J/09/Q26

A stationary wave of frequency 80.0 Hz is set up on a stretched string of length 210 cm.

What is the speed of the waves that produce this stationary wave?

- A 56.0 ms^{-1} B 112 ms^{-1}
C 5600 ms^{-1} D $11\,200 \text{ ms}^{-1}$



56 9702/12/O/N/09/Q22

The order of magnitude of the frequency of the longest-wavelength ultraviolet waves can be expressed as 10^x Hz .

What is the value of x ?

- A 13 B 15 C 17 D 19

57 9702/12/O/N/09/Q23

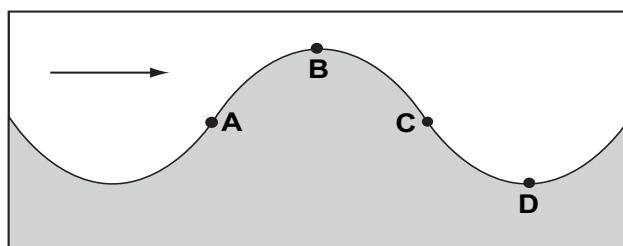
The light from two lasers passes through a vacuum. One laser emits red light and the other emits green light.

Which property of the two laser beams must be different?

- A amplitude B frequency C plane of polarisation D speed

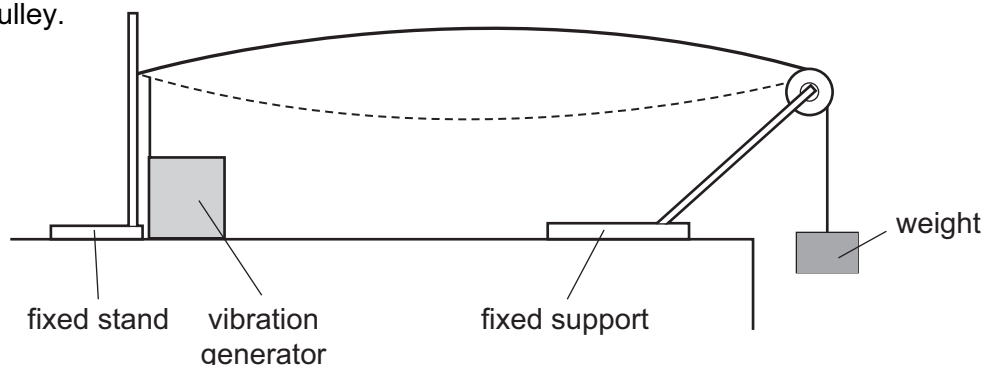
58 9702/11/O/N/10/Q24

The diagram shows a vertical cross-section through a water wave moving from left to right. At which point is the water moving upwards with maximum speed?



59 9702/12/M/J/10/Q22

The diagram shows a steel wire clamped at one end and tensioned at the other by a weight hung over a pulley.



A vibration generator is attached to the wire near the clamped end. A stationary wave with one loop is produced. The frequency of the vibration generator is f .

Which frequency should be used to produce a stationary wave with two loops?

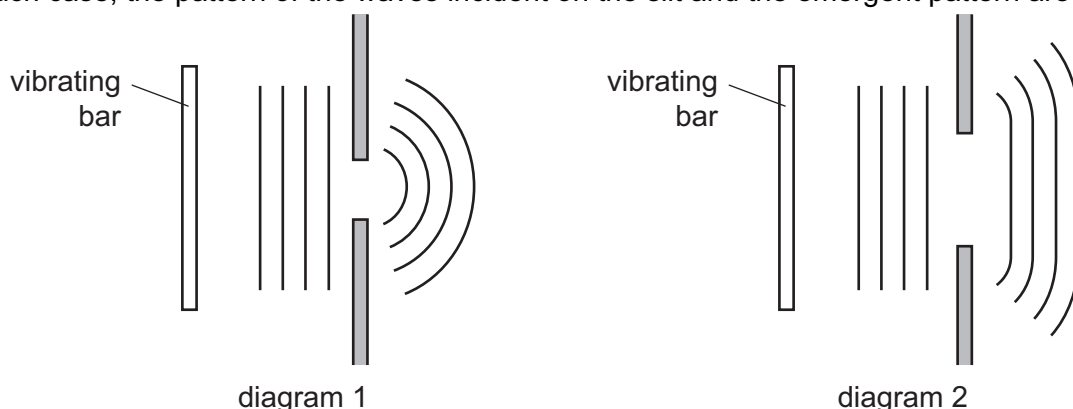
- A $\frac{f}{4}$ B $\frac{f}{2}$ C $2f$ D $4f$

60 9702/12/M/J/10/Q23

Diagram 1 shows a ripple tank experiment in which plane waves are diffracted through a narrow slit in a metal sheet.

Diagram 2 shows the same tank with a slit of greater width.

In each case, the pattern of the waves incident on the slit and the emergent pattern are shown.



Which action would cause the waves in diagram 1 to be diffracted less and so produce an emergent pattern closer to that shown in diagram 2?

- A increasing the frequency of vibration of the bar
B increasing the speed of the waves by making the water in the tank deeper
C reducing the amplitude of vibration of the bar
D reducing the length of the vibrating bar

61 9702/12/M/J/10/Q24*

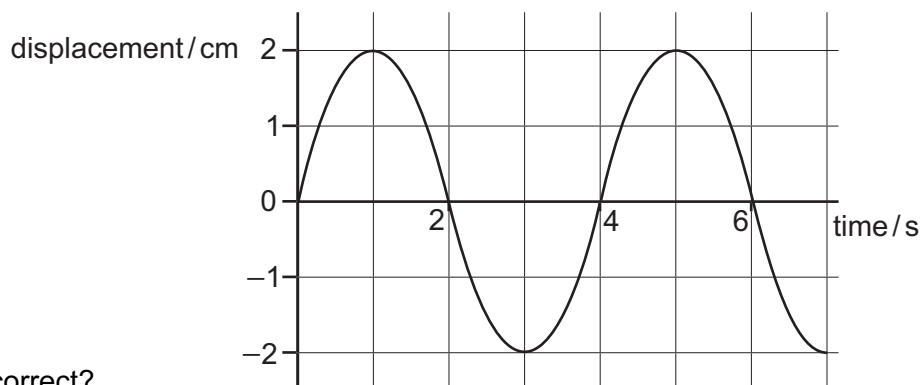
Electromagnetic waves from an unknown source in space were found to be significantly diffracted when passing through gaps of the order of 10^{-5} m.

Which type of wave are they most likely to be?

- A radio waves B microwaves C infra-red waves D ultraviolet waves

62 9702/11/O/N/10/Q23

The graph shows how the displacement of a particle in a wave varies with time.

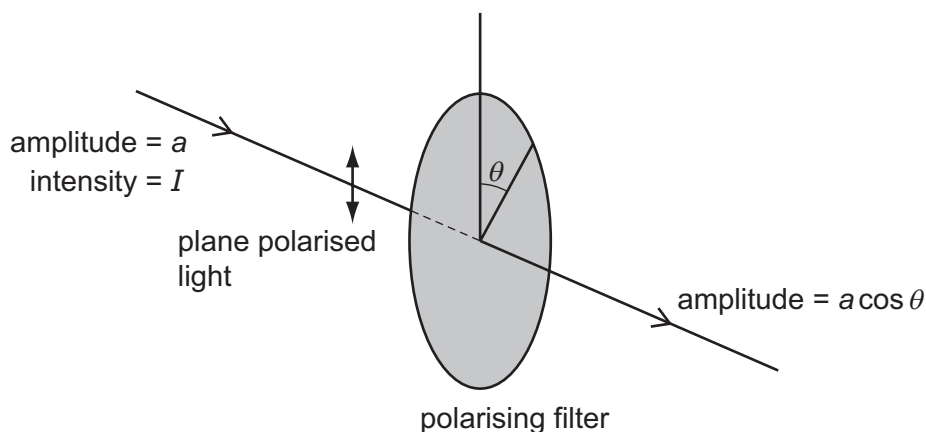


Which statement is correct?

- A The wave has an amplitude of 2 cm and could be either transverse or longitudinal.
- B The wave has an amplitude of 2 cm and must be transverse.
- C The wave has an amplitude of 4 cm and could be either transverse or longitudinal.
- D The wave has an amplitude of 4 cm and must be transverse.

63 9702/11/O/N/10/Q25

When plane-polarised light of amplitude a is passed through a polarising filter as shown, the amplitude of the light emerging is $a \cos \theta$.



The intensity of the initial beam is I .

What is the intensity of the emerging light when θ is 60.0° ?

- A $0.250I$
- B $0.500I$
- C $0.750I$
- D $0.866I$

64 9702/12/O/N/10/Q23

The amplitude of a wave is A and its intensity is I .

Which amplitude is necessary for the intensity to be doubled to $2I$?

- A A^2
- B $\sqrt{A}$
- C $\sqrt{2}A$
- D $2A$

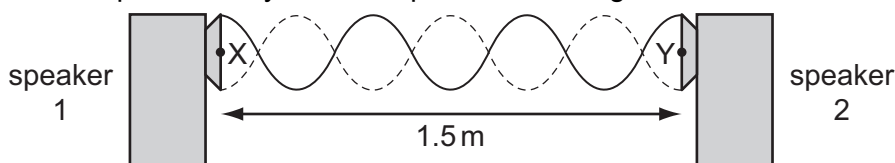
65 9702/12/O/N/10/Q24

Which value is a possible wavelength for radiation in the ultra-violet region of the electromagnetic spectrum?

- A $3 \times 10^{-2} \text{ m}$
- B $3 \times 10^{-5} \text{ m}$
- C $3 \times 10^{-8} \text{ m}$
- D $3 \times 10^{-10} \text{ m}$

66 9702/11/O/N/10/Q26

A stationary wave is produced by two loudspeakers emitting sound of the same frequency.



When a microphone is moved between X and Y, a distance of 1.5 m, six nodes and seven antinodes are detected.

What is the wavelength of the sound?

- A** 0.50 m **B** 0.43 m **C** 0.25 m **D** 0.21 m

67 9702/12/O/N/10/Q25

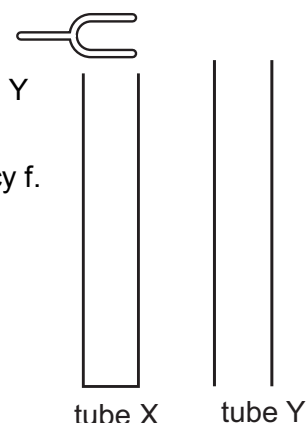
The diagram shows two tubes.

The tubes are identical except tube X is closed at its lower end while tube Y is open at its lower end. Both tubes have open upper ends.

A tuning fork placed above tube X causes resonance of the air at frequency f . No resonance is found at any lower frequency than f with tube X.

Which tuning fork will produce resonance when placed just above tube Y?

- A** a fork of frequency $\frac{f}{2}$ **C** a fork of frequency $\frac{3f}{2}$
B a fork of frequency $\frac{2f}{3}$ **D** a fork of frequency $2f$

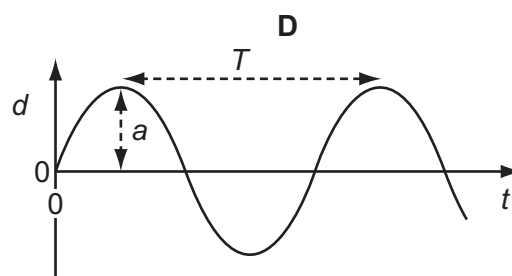
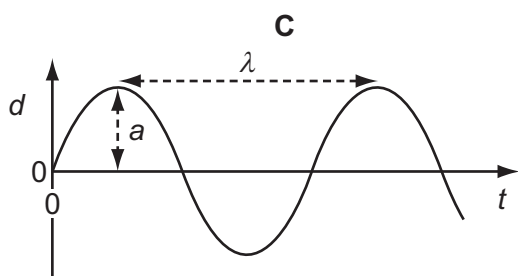
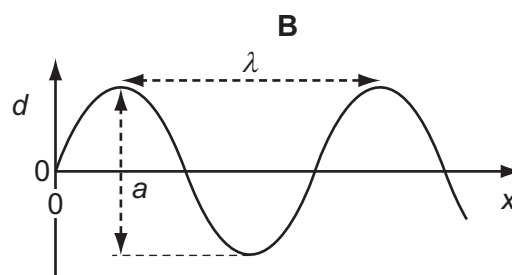
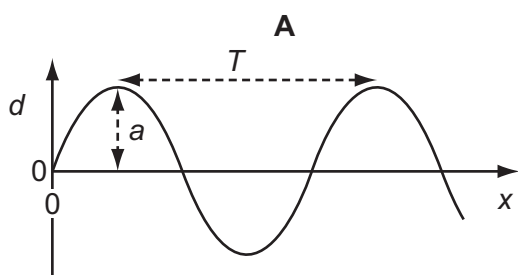


68 9702/11/M/J/11/Q22

The four graphs represent a progressive wave on a stretched string. Graphs **A** and **B** show how the displacement d varies with distance x along the string at one instant. Graphs **C** and **D** show how the displacement d varies with time t at a particular value of x .

The labels on the graphs are intended to show the wavelength λ , the period T and the amplitude a of the wave, but only one graph is correctly labelled.

Which graph is correctly labelled?



69 9702/11/M/J/11/Q23

Which statement about sound waves in air at constant temperature is correct?

- A Amplitude is inversely proportional to velocity.
- B Frequency is inversely proportional to wavelength.
- C Velocity is proportional to wavelength.
- D Wavelength is proportional to amplitude.

70 9702/11/M/J/11/Q24

A source of sound of constant power P is situated in an open space. The intensity I of sound at distance r from this source is given by $I = \frac{P}{4\pi r^2}$.

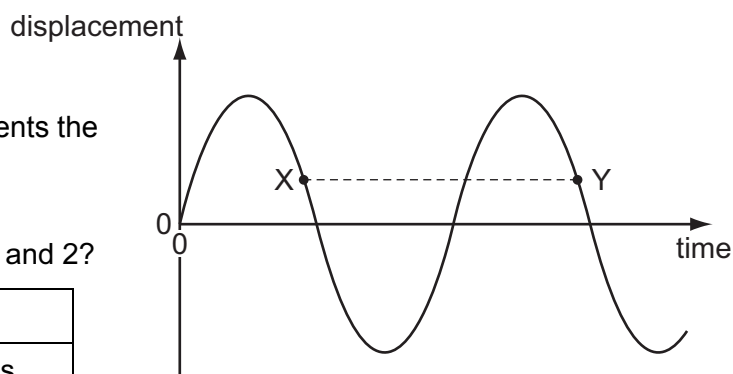
How does the amplitude a of the vibrating air molecules vary with the distance r from the source?

- A $a \propto \frac{1}{r}$
- B $a \propto \frac{1}{r^2}$
- C $a \propto r$
- D $a \propto r^2$

71 9702/12/M/J/11/Q26

A transverse progressive wave is set up on a string.

The graph shows the variation with time of displacement for a point on this string.



The separation XY on the graph represents the1..... of the wave.

X and Y have equal2..... .

Which words correctly complete gaps 1 and 2?

	1	2
A	time period	amplitudes
B	time period	displacements
C	wavelength	amplitudes
D	wavelength	displacements

72 9702/12/M/J/11/Q27

If a wave can be polarised, it **must** be

- A a longitudinal wave.
- B an electromagnetic wave.
- C a sound wave.
- D a transverse wave.

73 9702/12/M/J/11/Q28

In which situation does diffraction occur?

- A A wave bounces back from a surface.
- B A wave passes from one medium into another.
- C A wave passes through an aperture.
- D Waves from two identical sources are superposed.

74 9702/12/O/N/11/Q24

A wave that can be polarised must be

- A longitudinal.
- B progressive.
- C stationary.
- D transverse.

75 9702/12/O/N/15/Q27

P is a source emitting infra-red radiation and Q is a source emitting ultra-violet radiation. The figures in the table are suggested values for the wavelengths emitted by P and Q.

Which row is correct?

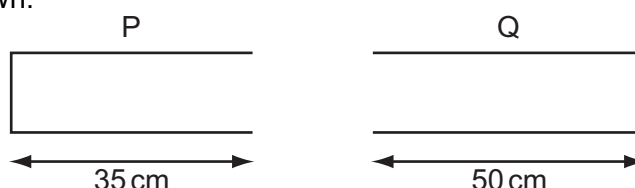
	wavelength emitted by P / m	wavelength emitted by Q / m
A	5×10^{-5}	5×10^{-8}
B	5×10^{-5}	5×10^{-10}
C	5×10^{-7}	5×10^{-8}
D	5×10^{-7}	5×10^{-10}

76 9702/11/O/N/11/Q29

Travelling waves of wavelength 20 cm are created in the air columns in a closed pipe P and an open pipe Q. The lengths of the pipes are shown.

In which pipe or pipes are stationary waves formed?

- A** P and Q **B** P only
C Q only **D** neither P nor Q



77 9702/12/O/N/11/Q25

Which statement about electromagnetic radiation is correct?

- A** Waves of wavelength 5×10^{-9} m are high-energy gamma rays.
B Waves of wavelength 3×10^{-8} m are ultra-violet waves.
C Waves of wavelength 5×10^{-7} m are infra-red waves.
D Waves of wavelength 9×10^{-7} m are light waves.

78 9702/12/O/N/11/Q27

A sound wave is set up in a long tube, closed at one end. The length of the tube is adjusted until the sound from the tube is loudest.

What is the nature of the sound wave in the tube?

- A** longitudinal and progressive **C** transverse and progressive
B longitudinal and stationary **D** transverse and stationary

79 9702/12/M/J/12/Q26

A surveyor's device emits a laser pulse.

What is the time taken for the pulse to travel from the device to a wall 150 m away, where it is reflected, and then return to the device?

- A** 0.05 ns **B** 0.10 ns **C** 0.50 μ s **D** 1.0 μ s

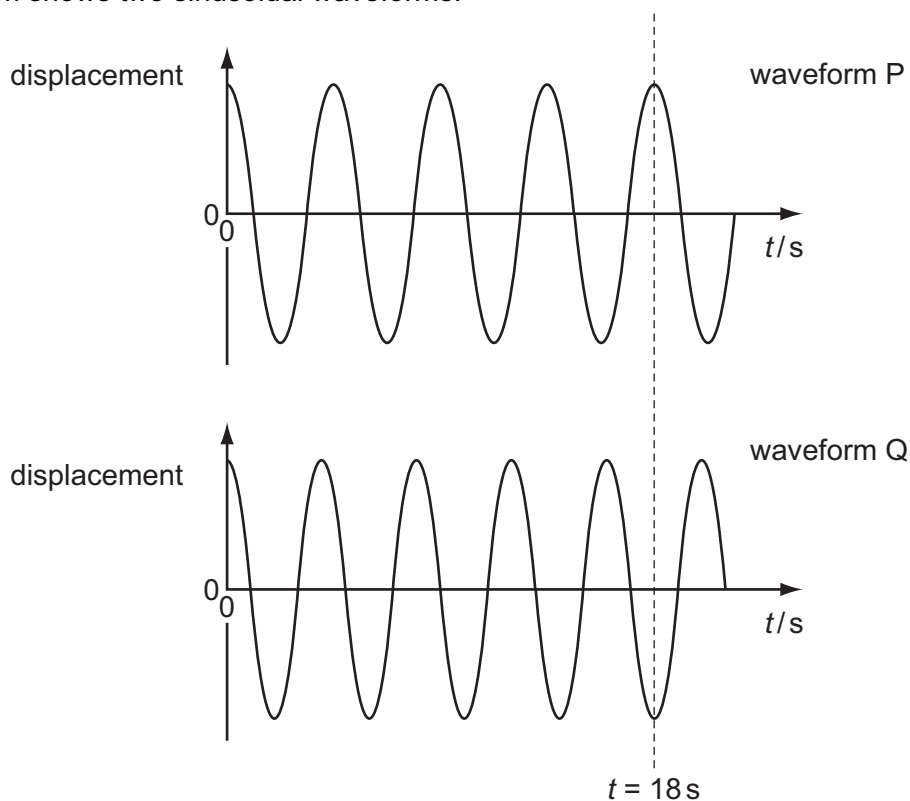
80 9702/13/M/J/12/Q29

Which observation indicates that sound waves are longitudinal?

- A** Sound can be reflected from a solid surface.
B Sound cannot be polarised.
C Sound is diffracted around corners.
D Sound is refracted as it passes from hot air to cold air.

81 9702/12/O/N/11/Q26

The diagram shows two sinusoidal waveforms.



At time $t = 0$ the waves are in phase. At the dotted line, $t = 18 s$.

At which time is the phase difference between the two oscillations $\frac{1}{8}$ of a cycle?

- A** 4.0 s **B** 4.5 s **C** 8.0 s **D** 9.0 s

82 9702/12/M/J/12/Q27

The period of an electromagnetic wave is 1.0 ns.

What are the frequency and wavelength of the wave?

	frequency / Hz	wavelength / m
A	1.0	3.0×10^8
B	1.0×10^6	300
C	1.0×10^9	0.30
D	1.0×10^{12}	3.0×10^{-4}

83 9702/13/M/J/12/Q23

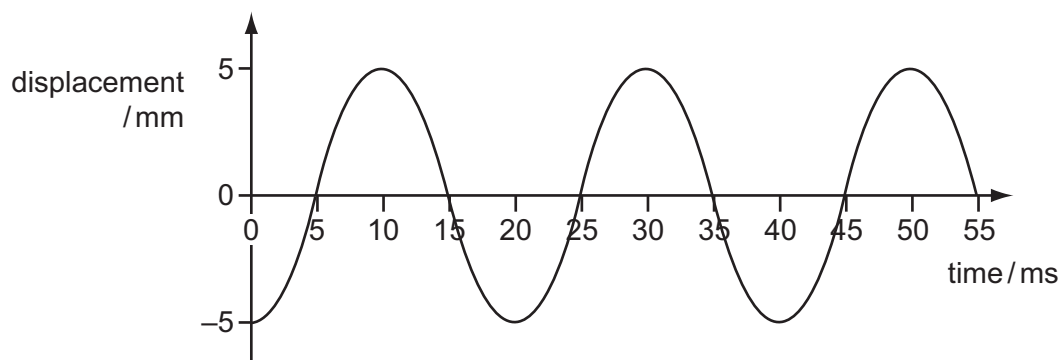
To produce a stationary wave, two waves must travel in opposite directions through the same space.

Which statement about the properties of the two waves must also be true?

- A** The waves must have equal frequency, but a different speed and wavelength.
B The waves must have equal speed, but a different wavelength and frequency.
C The waves must have equal speed, frequency and wavelength.
D The waves must have equal wavelength, but a different speed and frequency.

84 9702/13/M/J/12/Q26

The diagram shows a displacement-time graph for a progressive wave.

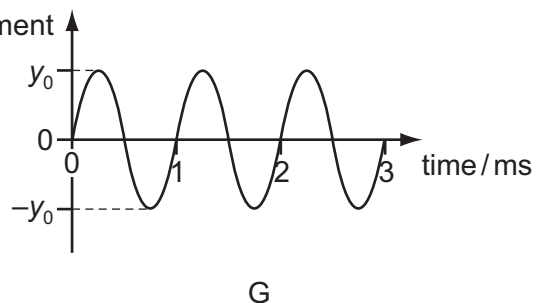
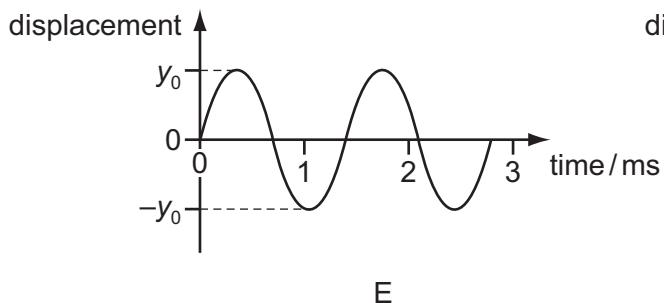


What are the amplitude and frequency of the wave?

	amplitude / mm	frequency / Hz
A	5	40
B	5	50
C	10	40
D	10	50

85 9702/13/M/J/12/Q27

Two waves E and G are shown. The waves have the same speed.



Which statement is correct?

- A** Wave E has a greater amplitude than wave G.
- B** Wave E has a greater intensity than wave G.
- C** Wave E has a smaller frequency than wave G.
- D** Wave E has a smaller wavelength than wave G.

86 9702/13/O/N/12/Q29

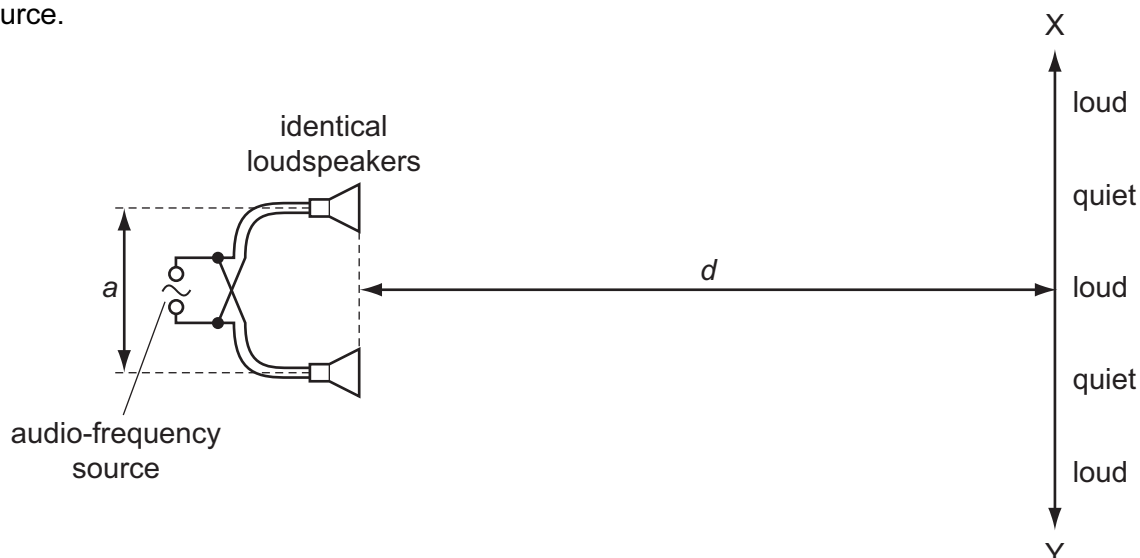
A health inspector is measuring the intensity of a sound. Near a loudspeaker, his meter records an intensity I . This corresponds to an amplitude A of the sound wave. At another position, the meter gives an intensity reading of $2I$.

What is the corresponding amplitude of the sound wave?

- A** $\frac{A}{\sqrt{2}}$
- B** $\sqrt{2}A$
- C** $2A$
- D** $4A$

87 9702/12/O/N/12/Q28*

The diagram shows two identical loudspeakers driven in phase by a common audio-frequency source.



When a student moves along line XY , she notices that there are variations in the loudness of the sound. The regions in which the sound is heard are alternately loud and quiet as indicated on the diagram.

How may the distance between loud regions be reduced?

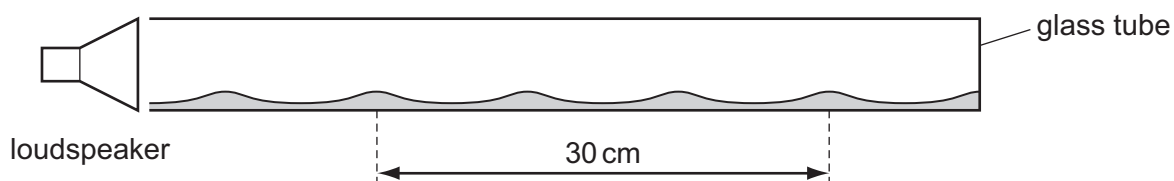
- A decreasing the distance a between the speakers
- B increasing distance d
- C increasing the frequency of the audio-frequency source
- D increasing the power output from the audio-frequency source

88 9702/12/O/N/12/Q29

A horizontal glass tube, closed at one end, has a layer of dust laid inside it on its lower side. Sound is emitted from a loudspeaker that is placed near the open end of the tube.

The frequency of the sound is varied and, at one frequency, a stationary wave is formed inside the tube so that the dust forms small heaps.

The distance between four heaps of dust is 30 cm.



The speed of sound in the tube is 330 ms^{-1} .

What is the frequency of the sound emitted by the loudspeaker?

- A 1650 Hz
- B 2200 Hz
- C 3300 Hz
- D 6600 Hz

89 9702/11/M/J/13/Q25

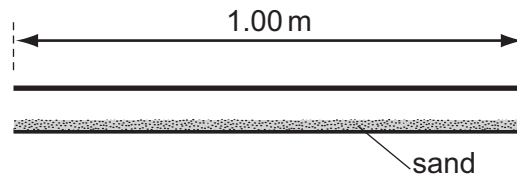
A wave has a speed of 340 ms^{-1} and a period of 0.28 ms.

What is its wavelength?

- A 0.095 m
- B 95 m
- C $1.2 \times 10^3 \text{ m}$
- D $1.2 \times 10^6 \text{ m}$

90 9702/13/O/N/12/Q26

The diagram shows an air-filled pipe open at both ends. The length of the pipe is 1.00 m and the lower surface of the inside of the pipe is covered with a layer of fine sand.



When a source of sound of a single frequency is put near one end of the pipe, the air in the pipe is found to resonate and a pattern in the sand shows that a standing wave containing three nodes is formed within the pipe.

The speed of sound in air is 330 m s^{-1} .

What is the frequency of the sound?

- A 330 Hz B 495 Hz C 990 Hz D 1320 Hz

91 9702/13/O/N/12/Q27

A stationary sound wave is formed in a measuring cylinder by blowing across the top, as shown.

Which statement is correct?

- A The fundamental frequency of the stationary wave decreases when some water is added to the cylinder.
 B The stationary wave in the cylinder is caused by the superposition of two waves moving in opposite directions.
 C The stationary wave in the cylinder is polarised.
 D The stationary wave will have an antinode at the bottom of the cylinder.



92 9702/11/M/J/13/Q24

A light wave of amplitude A is incident normally on a surface of area S . The power per unit area reaching the surface is P .

The amplitude of the light wave is increased to $2A$. The light is then focussed on to a smaller area $\frac{1}{3}S$.

What is the power per unit area on this smaller area?

- A $36P$ B $18P$ C $12P$ D $6P$

93 9702/11/M/J/13/Q26

Which line in the table summarises the change in wave characteristics on going from infra-red to ultraviolet in the electromagnetic spectrum?

	frequency	speed in a vacuum
A	decreases	decreases
B	decreases	remains constant
C	increases	remains constant
D	increases	increases

94 9702/11/M/J/13/Q27

Light of wavelength 600 nm is incident on a pair of slits. Fringes with a spacing of 4.0 mm are formed on a screen.

What will be the fringe spacing when the wavelength of the light is changed to 400 nm and the separation of the slits is doubled?

- A** 1.3 mm **B** 3.0 mm **C** 5.3 mm **D** 12 mm

95 9702/12/M/J/13/Q24

The order of magnitude of the frequency of the shortest wavelength of visible light waves can be expressed as 10^x Hz.

What is the value of x ?

- A** 12 **B** 13 **C** 14 **D** 15

96 9702/12/M/J/13/Q26

What is correct for all transverse waves?

- A** They are all electromagnetic. **C** They can all travel through a vacuum.
B They can all be polarised. **D** They all involve the oscillation of atoms.

97 9702/13/M/J/13/Q22

Which statement about different types of electromagnetic wave is correct?

- A** The frequency of infra-red waves is less than the frequency of blue light.
B The frequency of radio waves is greater than the frequency of gamma rays.
C The wavelength of red light is less than the wavelength of ultraviolet waves.
D The wavelength of X-rays is greater than the wavelength of microwaves.

98 9702/13/M/J/13/Q23

Orange light has a wavelength of 600 nm.

What is the frequency of this light?

- A** 180 GHz **B** 180 Hz **C** 500 THz **D** 500 kHz

99 9702/13/M/J/13/Q24

Electromagnetic waves of wavelength λ and frequency f travel at speed c in a vacuum.

What describes the wavelength and speed of electromagnetic waves of frequency $f/2$?

	wavelength	speed in a vacuum
A	$\lambda/2$	$c/2$
B	$\lambda/2$	c
C	2λ	c
D	2λ	$2c$

100 9702/13/M/J/13/Q25

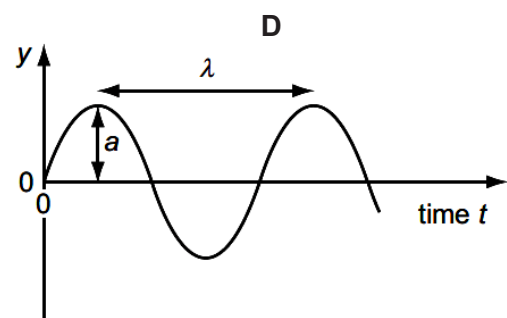
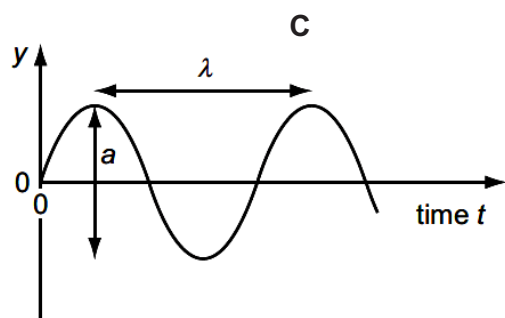
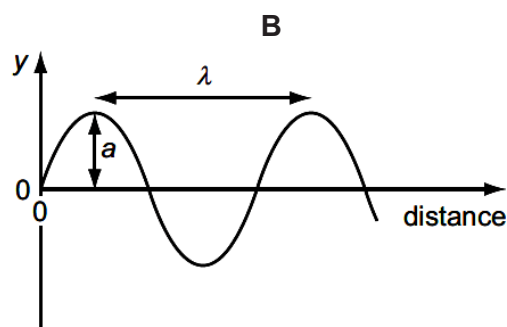
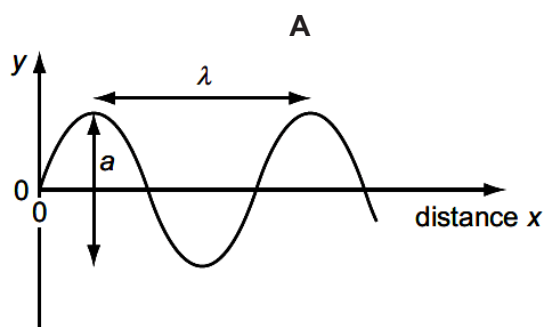
When the liquid crystal display of a calculator is observed through a polarising film, the display changes as the film is rotated.

Which property describes the radiation from the calculator display?

- A** unpolarised **C** a transverse wave
B a longitudinal wave **D** a wave with a 3 cm wavelength

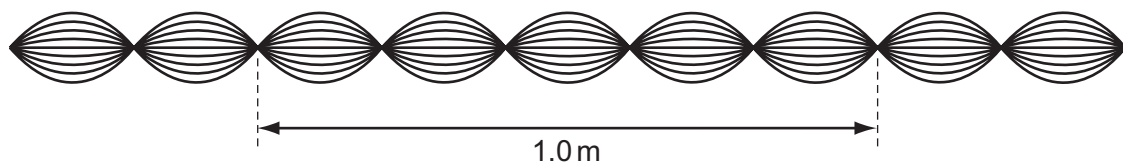
101 9702/13/M/J/13/Q26

A sound wave has displacement y at distance x from its source at time t .
Which graph correctly shows the amplitude a and the wavelength λ of the wave?



102 9702/12/O/N/13/Q25

The diagram shows a sketch of a wave pattern, over a short period of time.



Which description of this wave is correct?

- A** The wave is longitudinal, has a wavelength of 20 cm and is stationary.
- B** The wave is transverse, has a wavelength of 20 cm and is stationary.
- C** The wave is transverse, has a wavelength of 40 cm and is progressive.
- D** The wave is transverse, has a wavelength of 40 cm and is stationary.

103 9702/12/O/N/13/Q26

Which statement about a light wave and a sound wave is correct?

- A** Both can be polarised.
- B** Both can travel through free space.
- C** Both have a frequency inversely proportional to their wavelength.
- D** Both have an intensity proportional to their amplitude.

104 9702/13/O/N/13/Q24

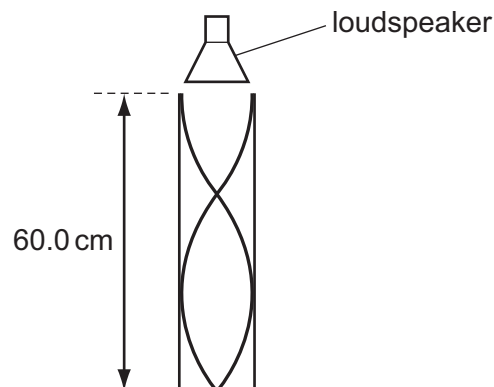
Electromagnetic waves from an unknown source in space were found to be significantly diffracted when passing through gaps of the order of 10^{-5} m.

Which type of wave are they most likely to be?

- A** radio waves **B** microwaves **C** infra-red waves **D** ultraviolet waves

105 9702/12/O/N/13/Q27

The sound from a loudspeaker placed above a tube causes resonance of the air in the tube. A stationary wave is formed with two nodes and two antinodes as shown.



The speed of sound in air is 330 m s^{-1} .

What is the frequency of the sound?

- A** 413 Hz **B** 550 Hz
C 830 Hz **D** 1650 Hz

106 9702/11/M/J/14/Q22

Which statement about longitudinal waves is correct?

- A** Longitudinal waves include radio waves travelling through air.
B Particles in a longitudinal wave vibrate at right-angles to the direction of transfer of wave energy.
C Some types of longitudinal wave can be polarised.
D Stationary waves can be produced by the superposition of longitudinal waves.

107 9702/11/M/J/14/Q23

The order of magnitude of the frequency of the longest-wavelength ultraviolet waves can be expressed as 10^x Hz .

What is the value of x ?

- A** 13 **B** 15 **C** 17 **D** 19

108 9702/11/M/J/14/Q24

The speed v of waves in deep water is given by the equation

$$v^2 = \frac{g\lambda}{2\pi}$$

where λ is the wavelength of the waves and g is the acceleration of free fall.

A student measures the wavelength λ and the frequency f of a number of these waves.

Which graph should he plot to give a straight line through the origin?

- A** f^2 against λ **B** f against λ^2 **C** f against $\frac{1}{\lambda}$ **D** f^2 against $\frac{1}{\lambda}$

109 9702/11/M/J/14/Q26

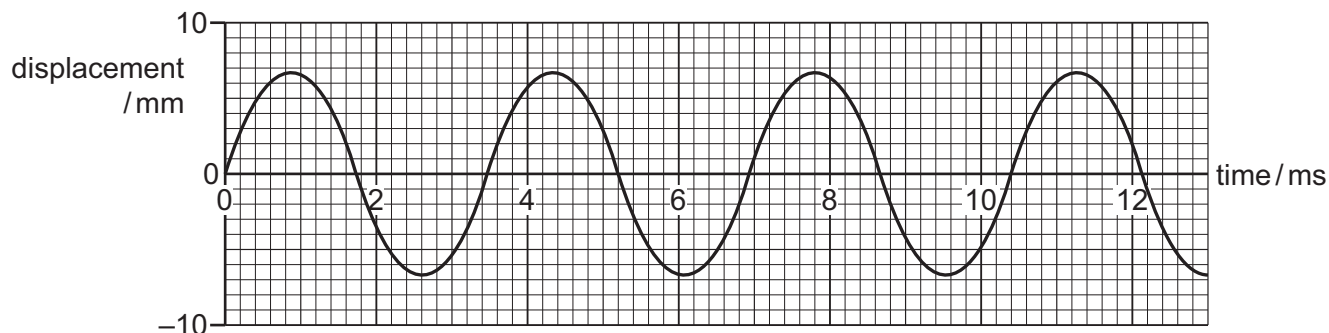
A parallel beam of white light passes through a diffraction grating. Orange light of wavelength 600 nm in the fourth order diffraction maximum coincides with blue light in the fifth order diffraction maximum.

What is the wavelength of the blue light?

- A** 450 nm **B** 480 nm **C** 500 nm **D** 750 nm

110 9702/12/M/J/14/Q22

What, to two significant figures, are the period, the frequency and the amplitude of the wave represented by the graph?

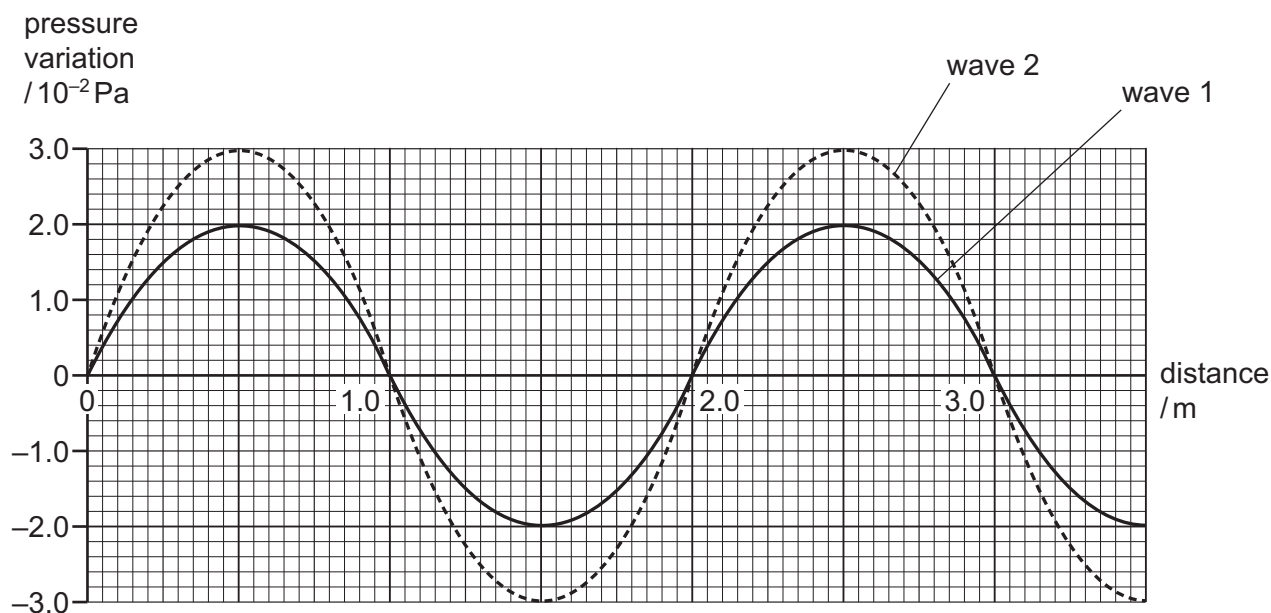


	period /s	frequency /Hz	amplitude /m
A	0.0027	370	0.0067
B	0.0031	320	0.013
C	0.0035	290	0.0067
D	0.0042	240	0.013

111 9702/12/M/J/14/Q23

A sound wave consists of a series of moving pressure variations from the normal, constant air pressure.

The graph shows these pressure variations for two waves at one instant in time.



Wave 1 has an intensity of $1.6 \times 10^{-6} \text{ W m}^{-2}$.

What is the intensity of wave 2?

- A** $2.4 \times 10^{-6} \text{ W m}^{-2}$ **B** $3.0 \times 10^{-6} \text{ W m}^{-2}$ **C** $3.6 \times 10^{-6} \text{ W m}^{-2}$ **D** $4.5 \times 10^{-6} \text{ W m}^{-2}$

112 9702/13/M/J/14/Q25

A cathode-ray oscilloscope (c.r.o.) displays a waveform corresponding to a sound wave.

In order to determine the frequency of the sound wave, which part of the displayed waveform must be measured and which c.r.o. setting must be known?

	on-screen measurement	c.r.o. setting
A	amplitude	time-base
B	amplitude	Y-gain
C	wavelength	time-base
D	wavelength	Y-gain

113 9702/13/M/J/14/Q26

What is the approximate range of frequencies of infra-red radiation?

- A** $1 \times 10^3 \text{ Hz}$ to $1 \times 10^9 \text{ Hz}$ **C** $1 \times 10^{11} \text{ Hz}$ to $1 \times 10^{14} \text{ Hz}$
B $1 \times 10^9 \text{ Hz}$ to $1 \times 10^{11} \text{ Hz}$ **D** $1 \times 10^{14} \text{ Hz}$ to $1 \times 10^{17} \text{ Hz}$

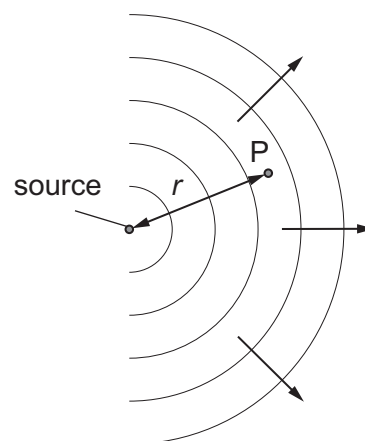
114 9702/13/M/J/14/Q27

A small source emits spherical waves.

The wave intensity I at any point P, a distance r from the source, is inversely proportional to r^2 .

What is the relationship between the wave amplitude a and the distance r ?

- A** $a^2 \propto \frac{1}{r}$ **B** $a \propto \frac{1}{r}$
C $a \propto \frac{1}{r^2}$ **D** $a \propto \frac{1}{r^4}$



115 9702/11/O/N/14/Q22

Which statement about electromagnetic radiation is correct?

- A** Waves of wavelength $5 \times 10^{-9} \text{ m}$ are high-energy gamma rays.
B Waves of wavelength $3 \times 10^{-8} \text{ m}$ are ultra-violet waves.
C Waves of wavelength $5 \times 10^{-7} \text{ m}$ are infra-red waves.
D Waves of wavelength $9 \times 10^{-7} \text{ m}$ are light waves.

116 9702/11/O/N/14/Q25

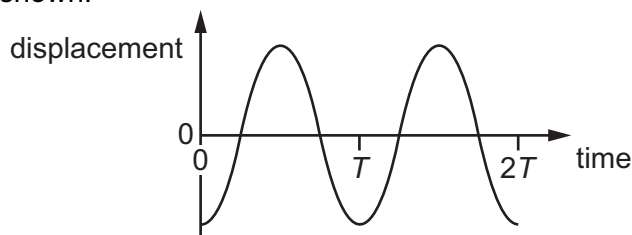
A stationary sound wave is produced in a tube.

Which statement describes the wave speed?

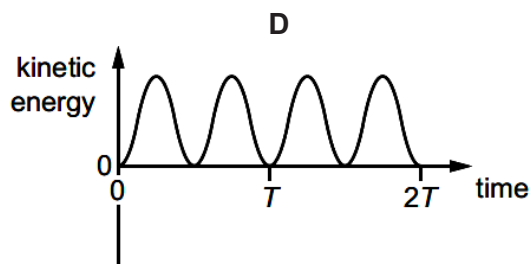
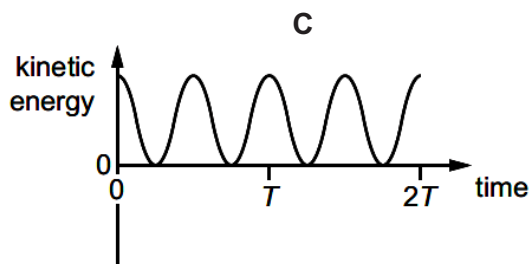
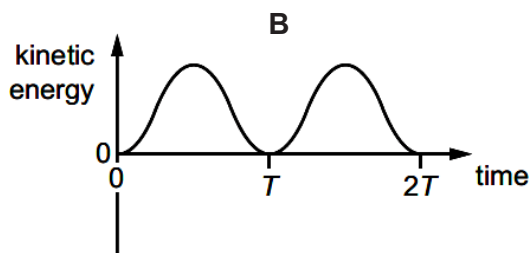
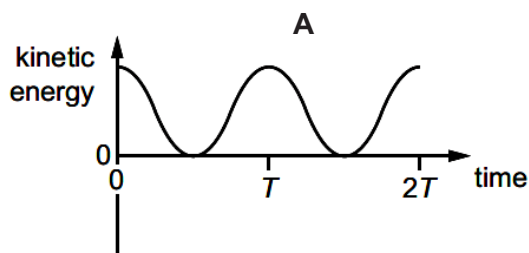
- A** It is the distance between two adjacent nodes divided by the period of the wave.
B It is the speed at which energy is transferred from one antinode to an adjacent antinode.
C It is the speed of a particle at an antinode.
D It is the speed of one of the progressive waves that are producing the stationary wave.

117 9702/11/O/N/14/Q23

When sound travels through air, the air particles vibrate. A graph of displacement against time for a single air particle is shown.



Which graph best shows how the kinetic energy of the air particle varies with time?



118 9702/13/O/N/14/Q26

Which statement about waves is correct?

- A** All electromagnetic waves travel at the same speed in a vacuum.
- B** Longitudinal waves can be polarised.
- C** The amplitude of a wave is directly proportional to the energy transferred by the wave.
- D** The frequency of infra-red light is greater than the frequency of ultra-violet light.

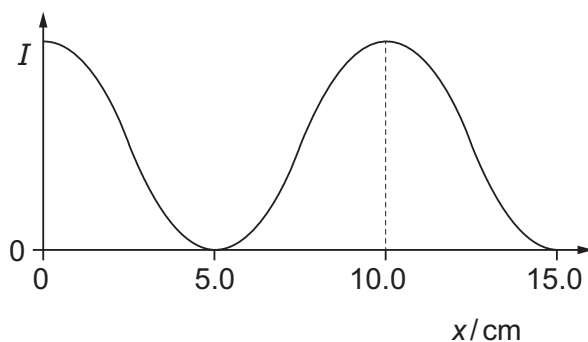
119 9702/13/O/N/14/Q27

The variation with distance x of the intensity I along a stationary sound wave in air is shown by the following graph.

The speed of sound in air is 340 m s^{-1} .

What is the frequency of the sound wave?

- A** 1700 Hz **B** 2270 Hz
- C** 3400 Hz **D** 6800 Hz



120 9702/11/O/N/14/Q24

Which statement describes a situation when polarisation could **not** occur?

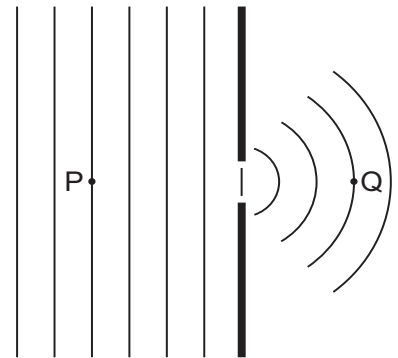
- A Light waves are reflected.
- B Light waves are scattered.
- C Microwaves pass through a metal grid.
- D Sound waves pass through a metal grid.

121 9702/13/O/N/14/Q28

Plane wavefronts in a ripple tank pass through a gap as shown.

Which property of the wave will be different at Q compared with P?

- A velocity
- B frequency
- C amplitude
- D wavelength



122 9702/13/O/N/14/Q29

An organ pipe of length l is open at both ends. Notes are produced by the pipe when stationary waves are set up.

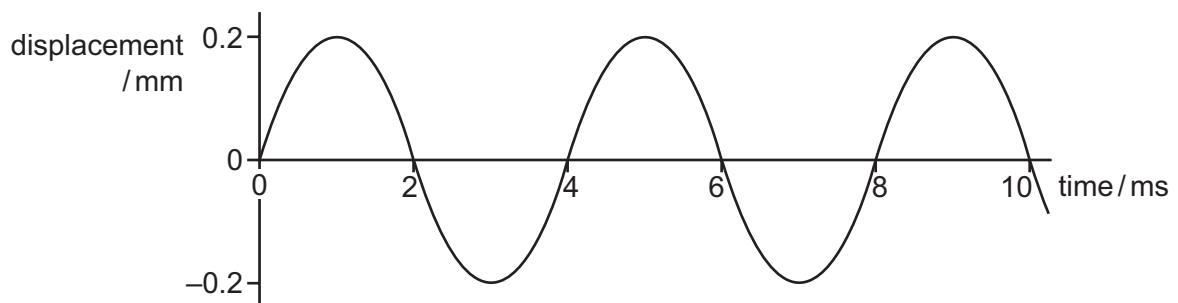
The speed of sound in the air column is v .

What is the lowest (fundamental) frequency of the note produced by the pipe?

- A $\frac{2v}{l}$
- B $\frac{v}{l}$
- C $\frac{v}{2l}$
- D $\frac{v}{4l}$

123 9702/11/M/J/15/Q24

A sound wave moves with a speed of 320 ms^{-1} through air. The variation with time of the displacement of an air particle due to this wave is shown in the graph.



Which statement about the sound wave is correct?

- A The frequency of the wave is 500 Hz.
- B The graph shows that sound is a transverse wave.
- C The intensity of the wave will be doubled if its amplitude is increased to 0.4 mm.
- D The wavelength of the sound wave is 1.28 m.

124 9702/1/O/N/02/Q27

A wave of amplitude a has an intensity of 3.0 W m^{-2} .

What is the intensity of a wave of the same frequency that has an amplitude $2a$?

- A 4.2 W m^{-2}
- B 6.0 W m^{-2}
- C 9.0 W m^{-2}
- D 12 W m^{-2}

125 9702/11/M/J/15/Q25

A wave of frequency 15 Hz travels at 24 m s^{-1} through a medium.

What is the phase difference between two points 2 m apart?

- A** There is no phase difference.
- B** They are out of phase by a quarter of a cycle.
- C** They are out of phase by half a cycle.
- D** They are out of phase by 0.8 of a cycle.

126 9702/11/M/J/15/Q27

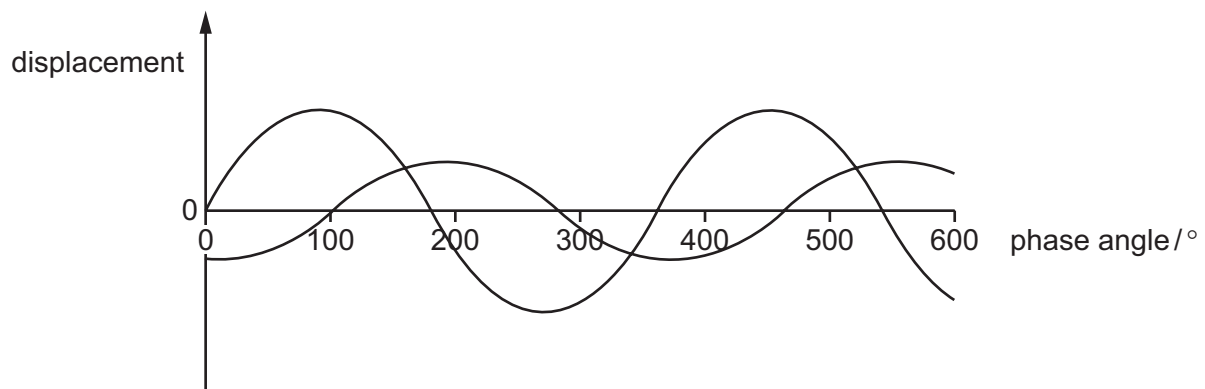
An electromagnetic wave has a wavelength that is numerically of the same order of magnitude as the diameter of a nucleus.

In which region of the electromagnetic spectrum does the wave occur?

- A** gamma ray
- B** X-ray
- C** visible light
- D** infra-red

127 9702/12/M/J/15/Q24

Two light waves of the same frequency are represented by the diagram.



What could be the phase difference between the two waves?

- A** 150°
- B** 220°
- C** 260°
- D** 330°

128 9702/12/M/J/15/Q25

A sound wave has a speed of 330 m s^{-1} and a frequency of 50 Hz .

What is a possible distance between two points on the wave that have a phase difference of 60° ?

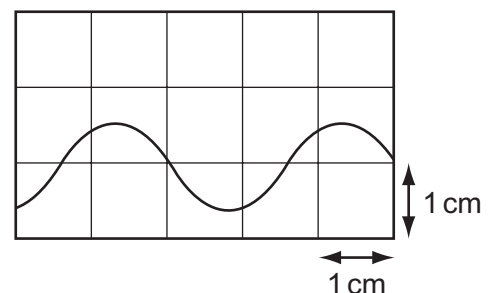
- A** 0.03 m
- B** 1.1 m
- C** 2.2 m
- D** 6.6 m

129 9702/13/M/J/15/Q25

A cathode-ray oscilloscope (c.r.o.) is used to display the trace from a sound wave. The time-base is set at $5 \mu\text{s mm}^{-1}$.

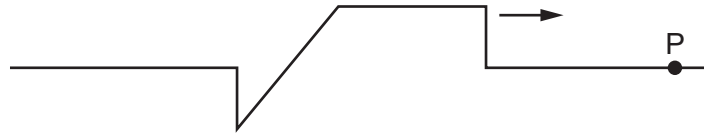
What is the frequency of the sound wave?

- A** 6.7 Hz
- B** 67 Hz
- C** 6.7 kHz
- D** 67 kHz

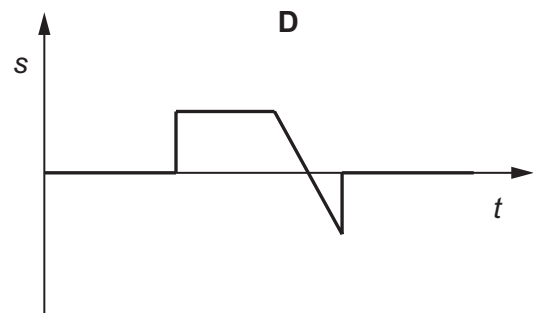
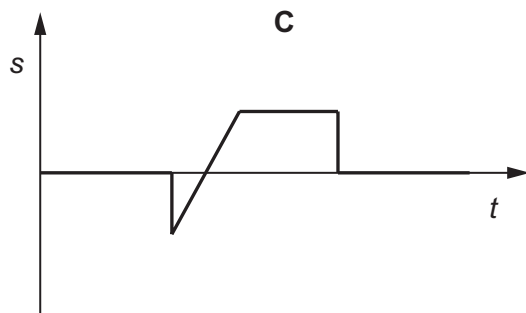
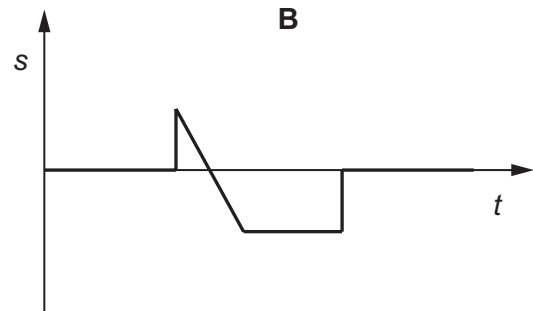
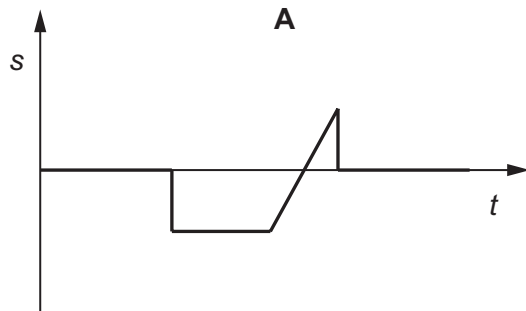


130 9702/13/M/J/15/Q26

A wave pulse moves along a stretched rope in the direction shown.



Which diagram correctly shows the variation with time t of the displacement s of the particle P in the rope?



131 9702/13/M/J/15/Q27

The table contains statements about stationary and progressive waves.

Which row is correct?

	stationary wave	progressive wave
A	all particles vibrate with the same amplitude	all particles vibrate with the same amplitude
B	energy is transferred along the wave	energy is transferred along the wave
C	particles in adjacent loops vibrate in antiphase	particles vibrate in phase with their immediate neighbours
D	particles one wavelength apart vibrate in phase	particles one wavelength apart vibrate in phase

132 9702/13/M/J/15/Q28

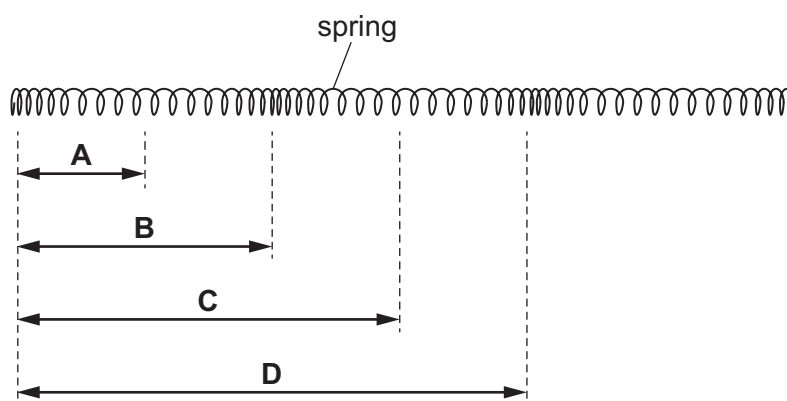
Which electromagnetic wave would cause the most significant diffraction effect for an atomic lattice of spacing around 10^{-10} m?

- A** infra-red **B** microwave **C** ultraviolet **D** X-ray

133 9702/11/M/J/17/Q22

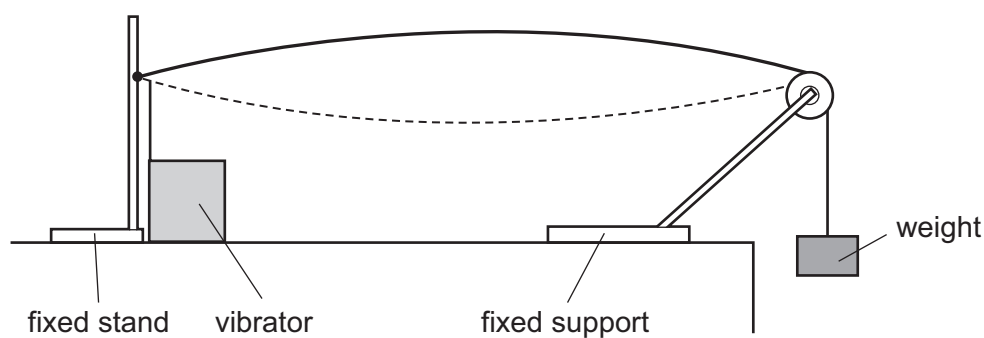
A longitudinal wave travels through a long spring. The spring is shown at one instant.

What is the wavelength of the wave?



134 9702/11/M/J/17/Q27

The diagram shows a steel wire clamped at one end. The other end is attached to a weight hanging over a pulley.



A vibrator is attached to the wire near the clamped end. A stationary wave with one loop is produced. The frequency of the vibrator is f .

Which frequency should be used to produce a stationary wave with two loops?

- A $\frac{f}{4}$ B $\frac{f}{2}$ C $2f$ D $4f$

135 9702/11/M/J/17/Q23

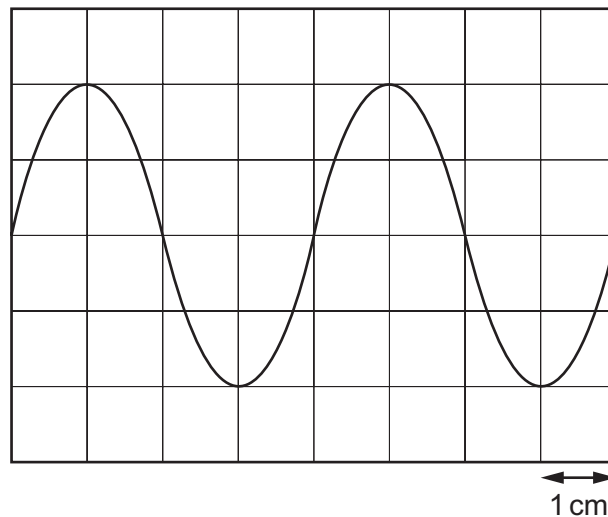
A sound wave has a frequency of 2500 Hz and a speed of 1500 m s^{-1} .

What is the shortest distance from a point of maximum pressure in the wave to a point of minimum pressure?

- A** 0.15 m **B** 0.30 m **C** 0.60 m **D** 1.20 m

136 9702/11/M/J/17/Q24

A sound wave is displayed on the screen of a cathode-ray oscilloscope (c.r.o.) as shown.



The time-base of the c.r.o. is set at 2.5 ms cm^{-1} .

What is the frequency of the sound wave?

- A** 50 Hz **B** 100 Hz **C** 200 Hz **D** 400 Hz

137 9702/11/M/J/17/Q25

A car travelling in a straight line at a speed of 30 m s^{-1} passes near a stationary observer while sounding its horn. The true frequency of sound from the horn is 400 Hz.

The speed of sound in air is 336 m s^{-1} .

What is the change in the frequency of the sound heard by the observer as the car passes?

- A** 39 Hz **B** 66 Hz **C** 72 Hz **D** 78 Hz

138 9702/11/M/J/17/Q26

Which list shows electromagnetic waves in order of increasing frequency?

- A** radio waves → gamma rays → ultraviolet → infra-red
B radio waves → infra-red → ultraviolet → gamma rays
C ultraviolet → gamma rays → radio waves → infra-red
D ultraviolet → infra-red → radio waves → gamma rays

139 9702/12/M/J/12/Q27

The period of an electromagnetic wave is 1.0 ns.

What are the frequency and wavelength of the wave?

	frequency / Hz	wavelength / m
A	1.0	3.0×10^8
B	1.0×10^6	300
C	1.0×10^9	0.30
D	1.0×10^{12}	3.0×10^{-4}

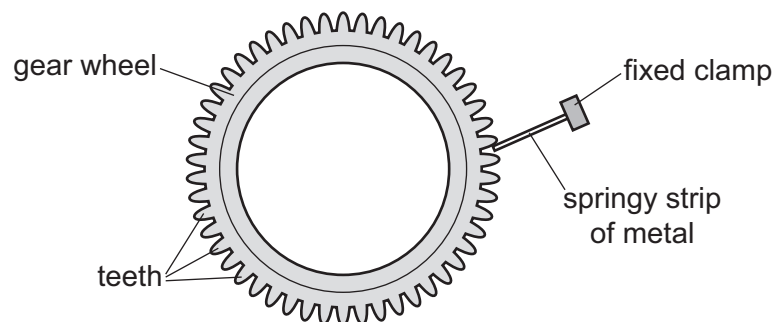
140 9702/12/M/J/17/Q23

Which statement about progressive longitudinal waves is **not** correct?

- A** The oscillations of the particles are parallel to the direction of travel of the wave energy.
- B** They have a series of nodes and antinodes.
- C** They need a medium through which to travel.
- D** They transfer energy.

141 9702/12/M/J/17/Q24

A bicycle gear wheel is a disc with 50 'teeth' equally spaced around its edge, as shown. The gear wheel is rotated 10 times each second. A springy strip of metal is vibrated by the rotating 'teeth'. The metal strip produces a sound of frequency that is equal to the frequency of vibration of the strip.



The speed of sound in air is 330 m s^{-1} .

What is the wavelength of the emitted sound?

- A** 0.66 m
- B** 1.5 m
- C** 6.6 m
- D** 500 m

142 9702/12/M/J/17/Q25

An ambulance travels along a straight road at a speed of 30.0 ms^{-1} . Its siren emits sound of frequency 2000 Hz . The speed of sound in the air is 340 ms^{-1} . The ambulance passes a man standing at the side of the road.

What is the frequency of the sound heard by the man as the ambulance moves towards him and as the ambulance moves away from him?

	frequency heard as ambulance moves towards man / Hz	frequency heard as ambulance moves away from man / Hz
A	1820	2180
B	1840	2190
C	2180	1820
D	2190	1840

143 9702/12/M/J/17/Q26

Three different electromagnetic waves P, Q and R have the frequencies shown.

	frequency / Hz
P	3×10^{10}
Q	3×10^{13}
R	6×10^{14}

Which row identifies P, Q and R?

	P	Q	R
A	infra-red	visible	ultraviolet
B	microwave	infra-red	visible
C	ultraviolet	X-ray	gamma ray
D	visible	ultraviolet	X-ray

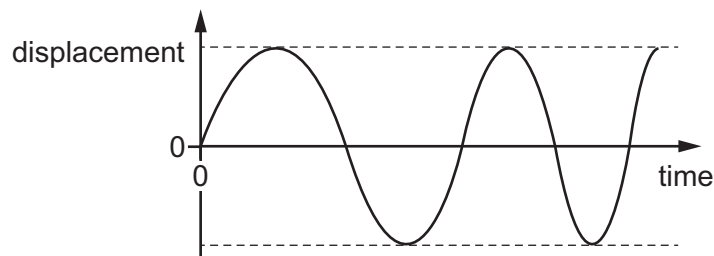
144 9702/12/M/J/17/Q27

Which row describes the oscillations of two moving particles in a stationary wave that are separated by a distance of half a wavelength?

	phase difference	amplitude
A	90°	different
B	90°	same
C	180°	different
D	180°	same

145 9702/11/O/N/17/Q22

The displacement-time graph for a layer of air in the path of a sound wave is shown.



Which wave quantity is increasing?

- A** amplitude **B** frequency **C** period **D** wavelength

146 9702/11/O/N/17/Q23

A loudspeaker emits a sound wave into a tube initially full of water.

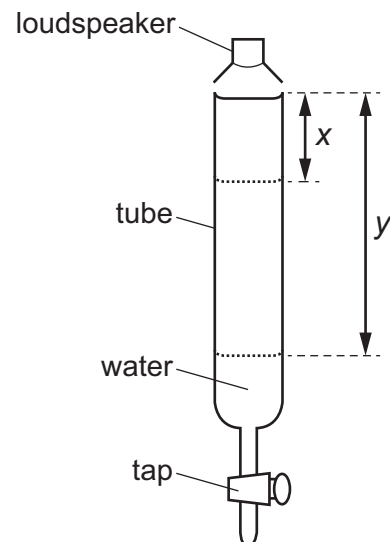
A tap at the bottom of the tube is opened so that water slowly leaves the tube. For some lengths of the air column in the tube, the sound heard is much louder.

The first loud sound is heard when the air column in the tube has length x .

The next time that a loud sound is heard is when the air column in the tube has length y .

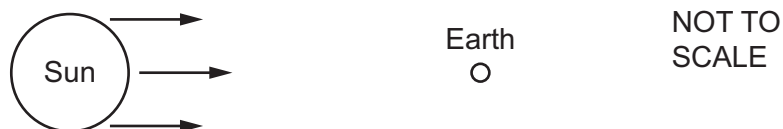
What is the wavelength of the sound wave from the loudspeaker?

- A** $2x$ **B** $4y$
C $2(y - x)$ **D** $4(y - x)$



147 9702/11/O/N/17/Q25

Light of a particular wavelength λ_s is emitted from the Sun. At any instant, a band of wavelengths ranging from less than λ_s to more than λ_s is observed on the Earth. This is caused by the Doppler effect.



What could be the explanation for this Doppler effect?

- A** The Sun is moving at right-angles to a line joining the Sun and the Earth.
B The Sun is moving away from the Earth.
C The Sun is moving towards the Earth.
D The Sun is rotating.

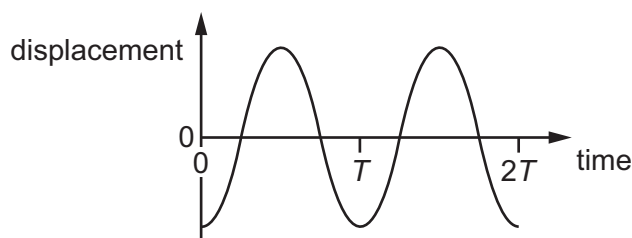
148 9702/11/O/N/17/Q26

What is the order of magnitude of the frequencies of electromagnetic waves in the visible spectrum?

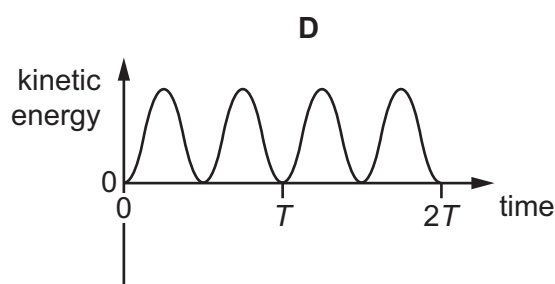
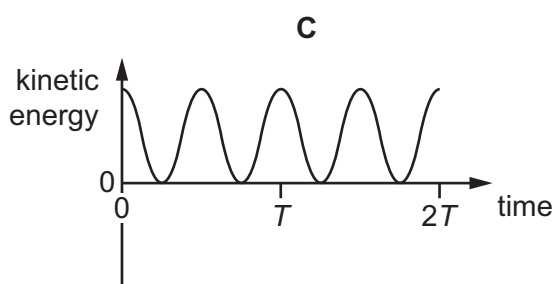
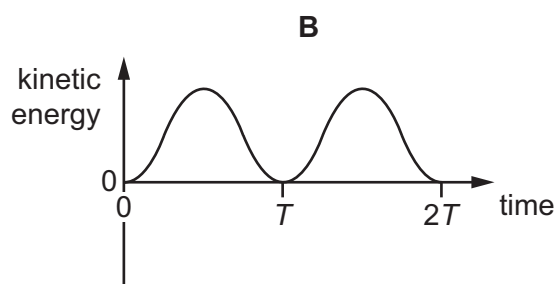
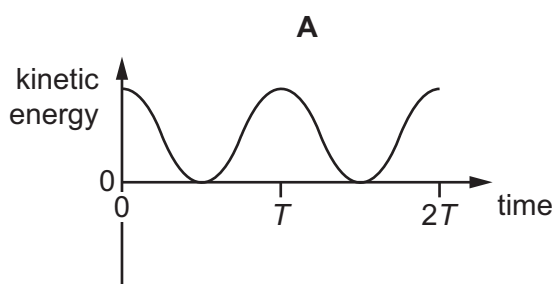
- A** 10^{10} Hz **B** 10^{12} Hz **C** 10^{14} Hz **D** 10^{16} Hz

149 9702/12/O/N/17/Q22

When sound travels through air, the air particles vibrate. A graph of displacement against time for a single air particle is shown.



Which graph best shows how the kinetic energy of the air particle varies with time?



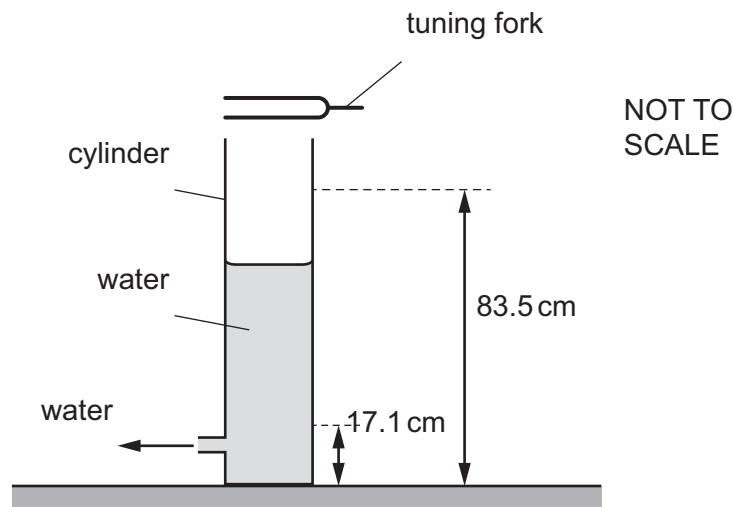
150 9702/12/O/N/17/Q23

Which wave is a longitudinal wave?

- A** a light wave travelling through air **C** a ripple on the surface of water
B a radio wave from a broadcasting station **D** a sound wave travelling through air

151 9702/12/O/N/17/Q24

A vibrating tuning fork is held above a glass cylinder filled to the top with water. The water level is steadily lowered. A loud sound is first heard when the water level is 83.5 cm above the bench. The next loud sound is heard when the water level is 17.1 cm above the bench.



The speed of sound in air is 340 m s^{-1} .

What is the frequency of the tuning fork?

- A** 128 Hz **B** 256 Hz **C** 384 Hz **D** 512 Hz

152 9702/12/O/N/17/Q25

A train that is moving in a straight line along a railway track has a whistle that continuously emits sound of frequency f .

A woman standing by the side of the track hears sound of frequency $0.85f$.

The speed of sound in the air is 340 m s^{-1} .

What is the velocity of the train?

- A** 51 m s^{-1} away from the woman **C** 60 m s^{-1} away from the woman
B 51 m s^{-1} towards the woman **D** 60 m s^{-1} towards the woman

153 9702/12/O/N/17/Q26

Orange light in a vacuum has a wavelength of 600 nm.

What is the frequency of this light?

- A** 180 Hz **B** $5.0 \times 10^5 \text{ Hz}$ **C** $1.8 \times 10^{11} \text{ Hz}$ **D** $5.0 \times 10^{14} \text{ Hz}$

154 9702/12/O/N/17/Q27

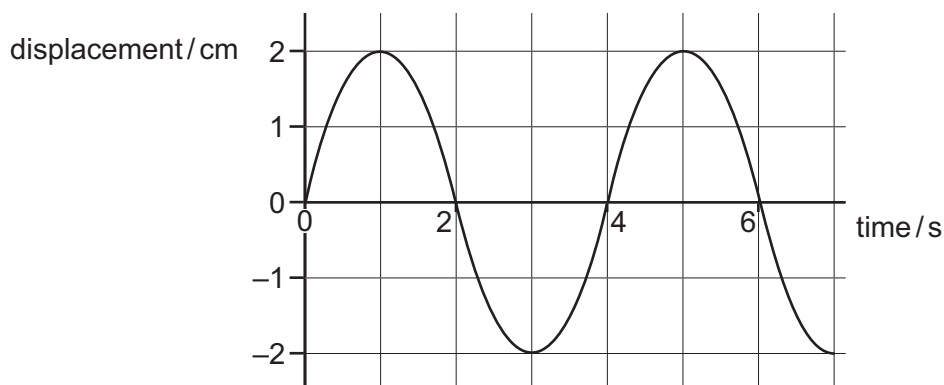
A stationary sound wave has a series of nodes. The distance between the first and the sixth node is 30.0 cm.

What is the wavelength of the sound wave?

- A** 5.0 cm **B** 6.0 cm **C** 10.0 cm **D** 12.0 cm

155 9702/13/O/N/17/Q22

The graph shows how the displacement of a particle in a wave varies with time.



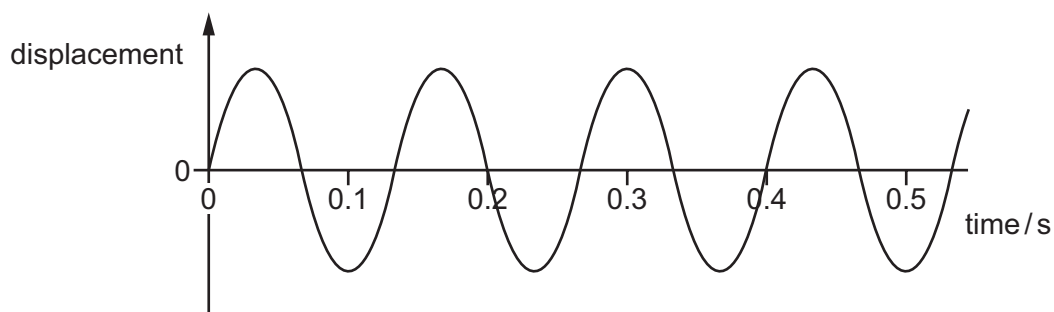
Which statement is correct?

- A** The wave has a period of 2 s and could be either transverse or longitudinal.
- B** The wave has a period of 2 s and must be transverse.
- C** The wave has a period of 4 s and could be either transverse or longitudinal.
- D** The wave has a period of 4 s and must be transverse.

156 9702/13/O/N/17/Q23

A wave travels along a coiled spring.

The graph shows the variation with time of the displacement of a point on the spring.



What is the frequency of the wave?

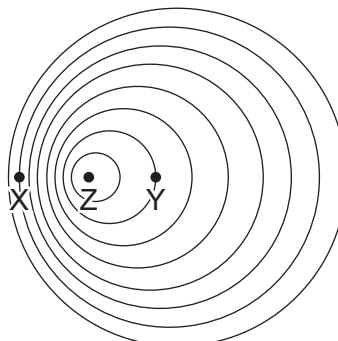
- A** 0.13 Hz
- B** 0.20 Hz
- C** 5.0 Hz
- D** 7.5 Hz

157 9702/13/O/N/17/Q25

A source of sound of frequency F at point Z is moving at a steady speed. The pattern of the emitted wavefronts is shown.

Which row describes the frequencies of the sound heard by stationary observers at X and Y?

	frequency heard at X	frequency heard at Y
A	$<F$	$<F$
B	$<F$	$>F$
C	$>F$	$<F$
D	$>F$	$>F$



158 9702/13/O/N/17/Q26

A car travelling at a steady speed in a straight line passes close to a stationary observer. The observer measures the frequency of the sound from the engine.

As the car approaches, the observed frequency is 220 Hz. When the car moves away, the observed frequency is 180 Hz.

The speed of sound in air is 340 m s^{-1} .

What is the speed of the car?

- A** 8.5 m s^{-1} **B** 31 m s^{-1} **C** 34 m s^{-1} **D** 38 m s^{-1}

159 9702/13/O/N/17/Q27

Which frequency of electromagnetic radiation could be ultraviolet?

- A** $1.0 \times 10^6 \text{ Hz}$ **C** $1.0 \times 10^{12} \text{ Hz}$
B $1.0 \times 10^9 \text{ Hz}$ **D** $1.0 \times 10^{15} \text{ Hz}$

160 9702/13/O/N/17/Q28

An electromagnetic wave travels in a straight line through a vacuum. The wave has a frequency of 6.0 THz.

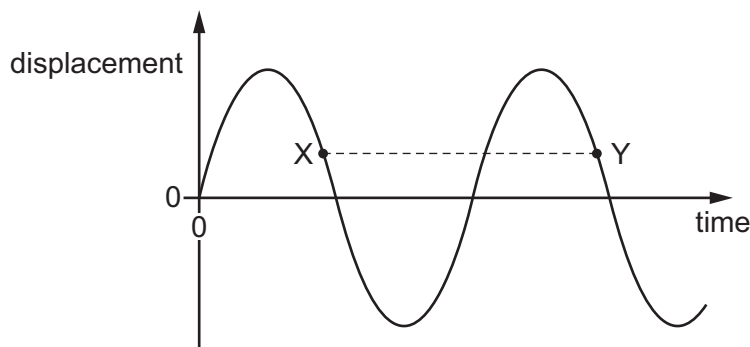
What is the number of wavelengths in a distance of 1.0 m along the wave?

- A** 5.0×10^{-5} **B** 2.0×10^1 **C** 2.0×10^4 **D** 5.0×10^7

161 9702/11/M/J/18/Q20

A transverse progressive wave is set up on a string.

The graph shows the variation with time of displacement for a point on this string.



The separation XY on the graph represents the1..... of the wave.

X and Y have equal2..... .

Which words correctly complete gaps 1 and 2?

	1	2
A	time period	amplitudes
B	time period	displacements
C	wavelength	amplitudes
D	wavelength	displacements

162 9702/11/M/J/18/Q21

Which region of the electromagnetic spectrum includes waves with a frequency of 10^7 MHz?

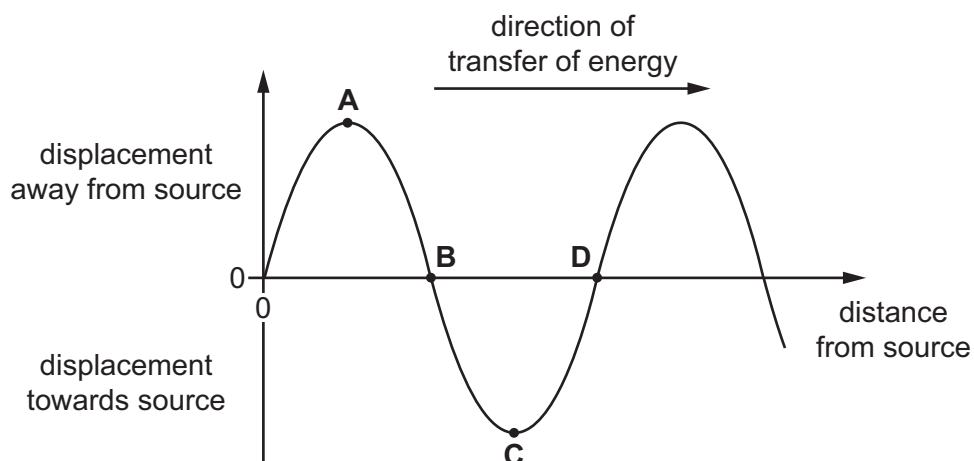
A infra-red waves **B** radio waves **C** ultraviolet waves **D** X-rays

163 9702/11/M/J/18/Q22

A longitudinal wave has vibrations parallel to the direction of transfer of energy by the wave.

The wave can be represented on a graph showing the variation of the displacement of the particles with distance from the source.

Which point on the graph is the centre of a compression?



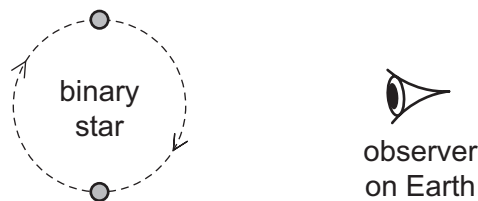
164 9702/11/M/J/18/Q23

What can be deduced from a table of wavelengths of the waves in the electromagnetic spectrum?

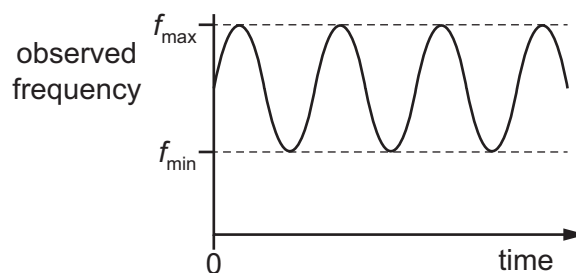
- A Green light has a shorter wavelength than X-rays.
- B Red light has a shorter wavelength than green light.
- C The wavelength range for radio waves is less than that for infra-red waves.
- D The wavelength range for X-rays is less than that for radio waves.

165 9702/11/M/J/18/Q24

A binary star consists of two stars rotating around a common centre. Light from one of the stars is observed on the Earth.



The observed frequency of the light varies between a minimum frequency $f_{\min}$ and a maximum frequency $f_{\max}$, as shown.



The rate of rotation of the binary star increases.

What is the change to $f_{\max}$ and the change to $f_{\min}$?

	$f_{\max}$	$f_{\min}$
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

166 9702/12/M/J/18/Q22

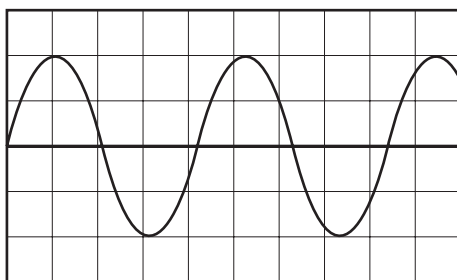
Two lasers emit light in a vacuum. One laser emits red light and the other emits green light.

Which property of the two laser beams **must** be different?

- A amplitude B frequency C intensity D speed

167 9702/12/M/J/18/Q23

The diagram shows the screen of a cathode-ray oscilloscope (c.r.o.) displaying a wave.



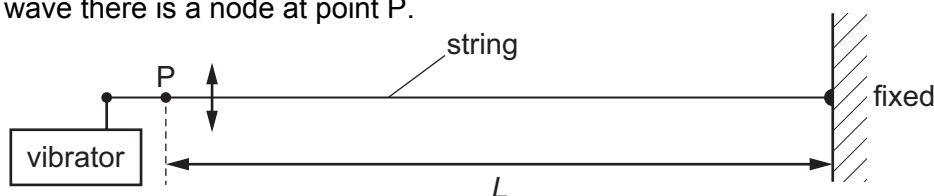
The time-base of the c.r.o. is set at 10 ms/division.

What is the frequency of the wave?

- A 0.24 Hz B 4.2 Hz C 12 Hz D 24 Hz

168 9702/12/M/J/18/Q24

A string is fixed at one end and the other end is attached to a vibrator. The frequency of the vibrator is slowly increased from zero. A series of stationary waves is formed. Assume that for stationary wave there is a node at point P.



What are the first five wavelengths of the stationary waves that could be formed?

- A $2\frac{L}{1}, 2\frac{L}{2}, 2\frac{L}{3}, 2\frac{L}{4}, 2\frac{L}{5}$
 B $2\frac{L}{2}, 2\frac{L}{3}, 2\frac{L}{4}, 2\frac{L}{5}, 2\frac{L}{6}$
 C $4\frac{L}{1}, 4\frac{L}{2}, 4\frac{L}{3}, 4\frac{L}{4}, 4\frac{L}{5}$
 D $4\frac{L}{1}, 4\frac{L}{3}, 4\frac{L}{5}, 4\frac{L}{7}, 4\frac{L}{9}$

169 9702/12/M/J/18/Q25

Which region of the electromagnetic spectrum has waves of wavelength 1000 times smaller than the wavelength of visible light?

- A infra-red B microwaves C ultraviolet D X-rays

170 9702/12/M/J/18/Q27

An astronomer observes the light from a star that is moving away from the Earth.

For the observed light, what has been increased due to the star's motion?

- A amplitude B frequency C speed D wavelength

171 9702/12/M/J/18/Q28

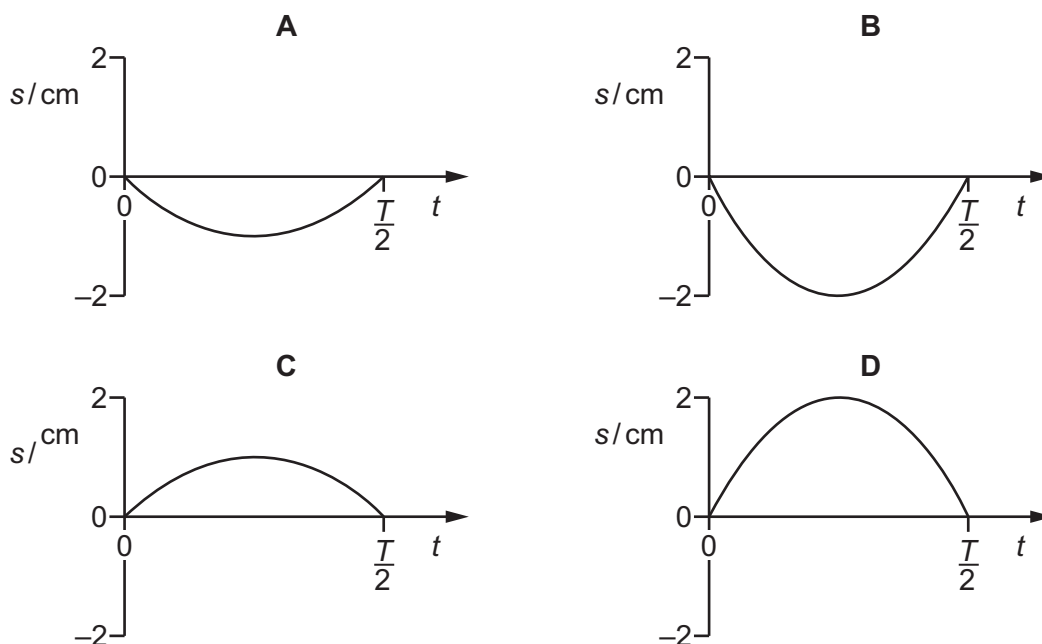
The diagram shows a stationary wave, at time $t = 0$, that has been set up on a string fixed between points P and S.



The nodes of the stationary wave occur on the string at P, Q, R and S. Point X is moving down at time $t = 0$. The points on the string vibrate with time period T and maximum amplitude 2 cm.

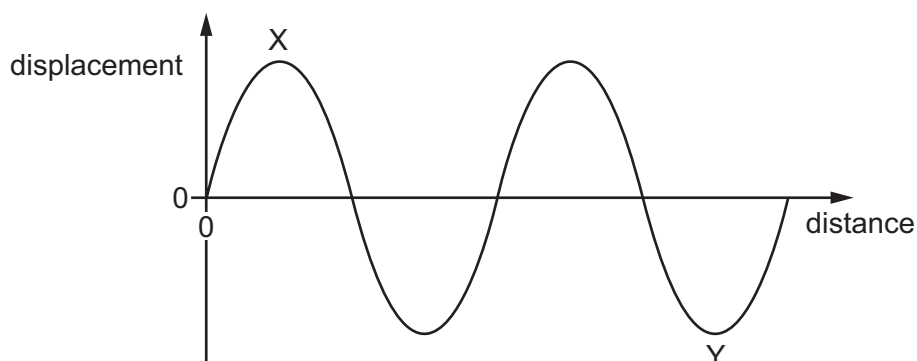
The displacement s is positive in the upward direction.

Which graph best shows the variation with t of the displacement s of point Y on the string?



172 9702/13/M/J/18/Q20

The displacement-distance graph for a transverse progressive wave is shown.



The phase difference between points X and Y can be expressed as $(180n)^\circ$.

What is the value of n ?

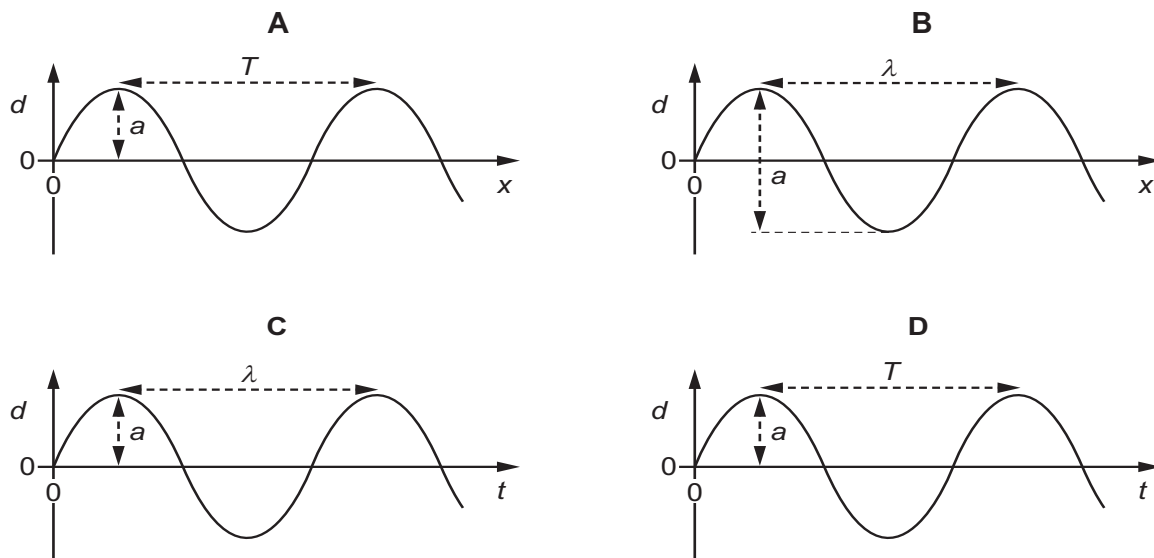
- A 1.5 B 2.5 C 3.0 D 6.0

173 9702/13/M/J/18/Q21

The four graphs represent a progressive wave on a stretched string. Graphs **A** and **B** show how the displacement d varies with distance x along the string at one instant. Graphs **C** and **D** show how the displacement d varies with time t at a particular value of x .

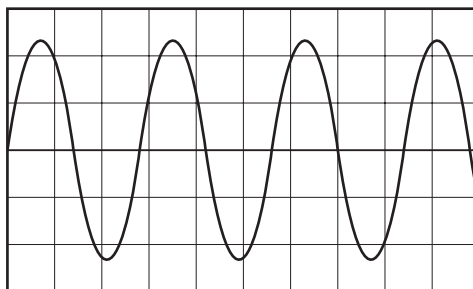
The labels on the graphs are intended to show the wavelength λ , the period T and the amplitude a of the wave, but only one graph is correctly labelled.

Which graph is correctly labelled?



174 9702/13/M/J/18/Q22

A cathode-ray oscilloscope (c.r.o.) is used to determine the frequency of a sound wave. The diagram shows the waveform on the screen.



The time-base setting is 5.0 ms/div .

What is the frequency of the sound wave?

- A** 57 Hz **B** 71 Hz **C** 114 Hz **D** 143 Hz

175 9702/13/M/J/18/Q23

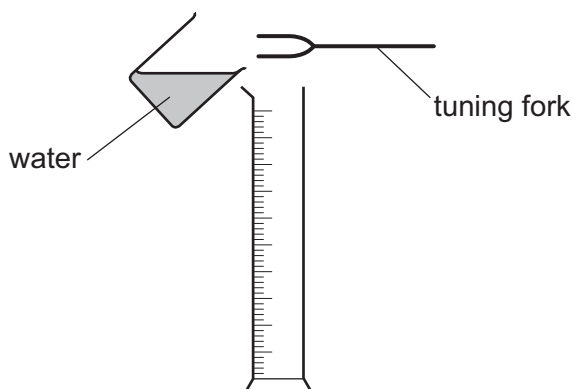
A police car travels at a velocity of 30.0 ms^{-1} directly towards a stationary observer. The horn of the car emits sound of frequency 2000 Hz . The speed of sound is 340 ms^{-1} .

What is the frequency of the sound heard by the observer?

- A** 1840 Hz **B** 2000 Hz **C** 2180 Hz **D** 2190 Hz

176 9702/13/M/J/18/Q24

A vibrating tuning fork is held over a measuring cylinder, as shown.



Water is then gradually poured into the measuring cylinder. A much louder sound is first heard when the water level is 2.9 cm above the base of the measuring cylinder. A second much louder sound is heard when the water level reaches a height of 67.3 cm above the base.

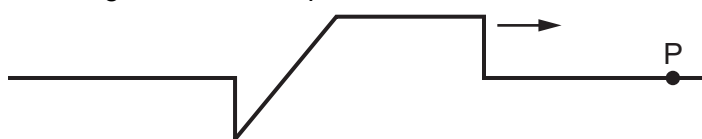
The speed of sound in air is 330 m s^{-1} .

What is the frequency of the tuning fork?

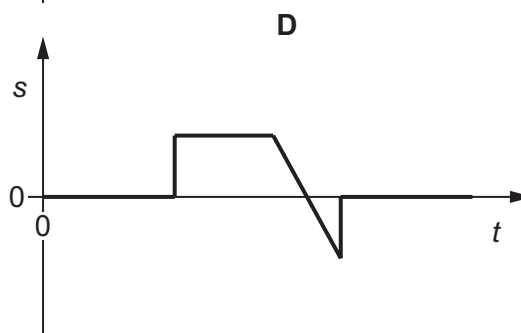
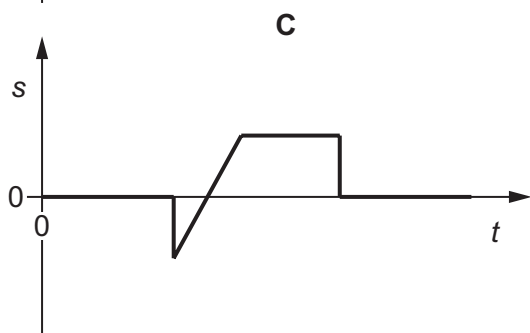
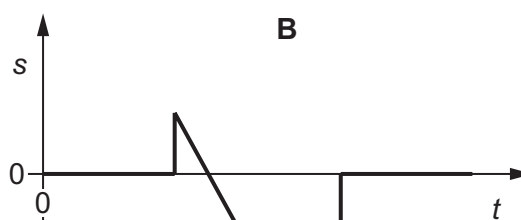
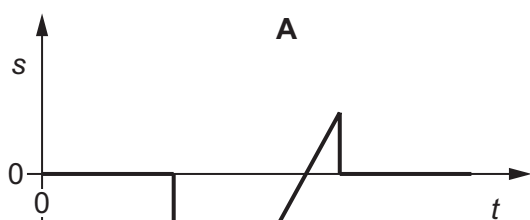
- A 128 Hz B 256 Hz C 512 Hz D 1024 Hz

177 9702/12/F/M/18/Q21

A wave pulse moves along a stretched rope in the direction shown.



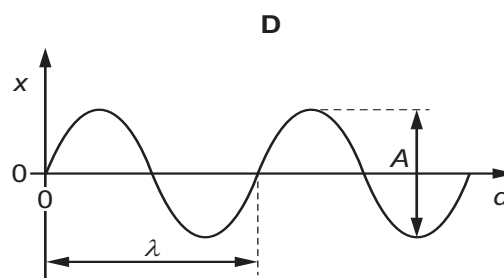
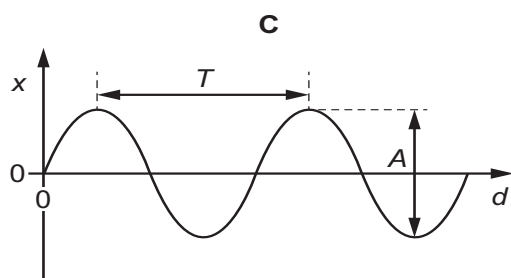
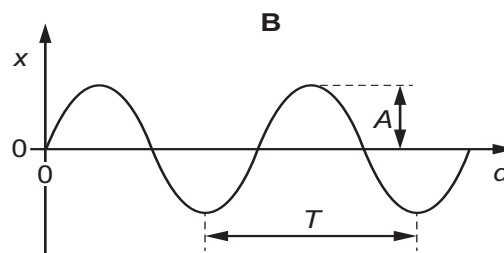
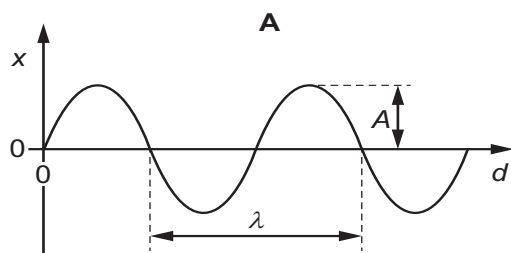
Which diagram shows the variation with time t of the displacement s of the particle P in the rope?



178 9702/12/F/M/18/Q22

A wave has period T , wavelength λ and amplitude A . The wave is shown on a graph of displacement x against distance d .

Which graph is correctly labelled?



179 9702/12/F/M/18/Q23

The table lists possible orders of magnitude of the wavelengths of some of the principal radiations of the electromagnetic spectrum.

Which row shows the correct orders of magnitude of the wavelengths?

	wavelength / m			
	microwaves	infra-red	ultraviolet	X-rays
A	10^{-6}	10^{-10}	10^{-12}	10^{-14}
B	10^{-4}	10^{-8}	10^{-10}	10^{-12}
C	10^{-2}	10^{-6}	10^{-8}	10^{-10}
D	10^2	10^{-4}	10^{-6}	10^{-8}

180 9702/12/F/M/18/Q24

A vehicle carries a microwave transmitter that emits microwaves of a constant frequency. A stationary observer has a microwave receiver.

The vehicle moves directly towards the observer at constant speed. The observer detects microwaves of frequency F_0 .

The vehicle then accelerates, still moving towards the observer, travels at higher steady speed for a time and then decelerates until it stops.

What is the variation in the frequency of the microwaves that are detected by the observer?

- A** The observed frequency will fall, then remain steady then return to the frequency F_0 .
- B** The observed frequency will fall, then remain steady then rise to a higher frequency than F_0 .
- C** The observed frequency will rise, then remain steady then fall to a lower frequency than F_0 .
- D** The observed frequency will rise, then remain steady then return to the frequency F_0 .

181 9702/12/F/M/18/Q26

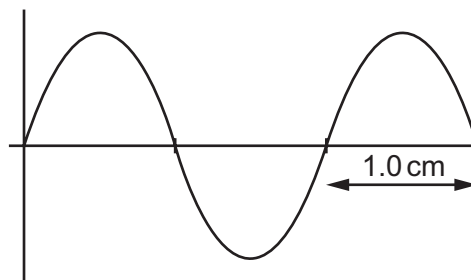
In a double-slit interference experiment, light of frequency $6.0 \times 10^{14} \text{ Hz}$ is incident on a pair of slits. Bright fringes that are 3.0 mm apart are observed on a screen some distance away.

What is the separation of the bright fringes when the frequency of the light is changed to $5.0 \times 10^{14} \text{ Hz}$?

- A** 1.8 mm **B** 2.5 mm **C** 3.0 mm **D** 3.6 mm

182 9702/12/F/M/18/Q25

The diagram shows a cathode-ray oscilloscope display of an electromagnetic wave.



The time base setting is $0.20 \mu\text{s cm}^{-1}$.

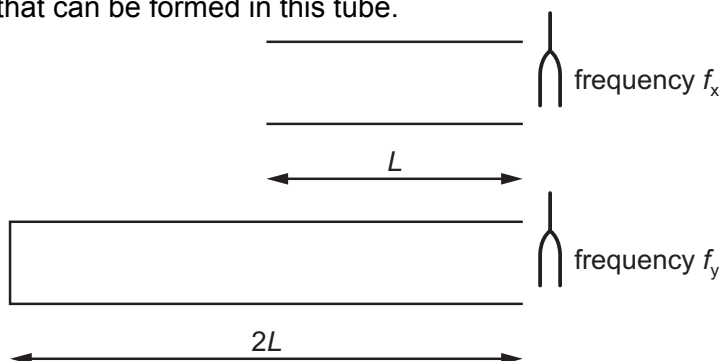
Which statement is correct?

- A** The frequency of the wave is 2.5 MHz and it lies in the microwave region of the electromagnetic spectrum.
- B** The frequency of the wave is 2.5 MHz and it lies in the radio-wave region of the electromagnetic spectrum.
- C** The frequency of the wave is 5.0 MHz and it lies in the microwave region of the electromagnetic spectrum.
- D** The frequency of the wave is 5.0 MHz and it lies in the radio-wave region of the electromagnetic spectrum.

183 9702/12/F/M/18/Q28

A tube of length L is open at both ends. A stationary wave is set up in this tube when a tuning fork vibrating with frequency f_x is held at one end. This is the lowest frequency of stationary wave that can be formed in this tube.

Another tube of length $2L$ is closed at one end. A stationary wave is set up in this tube when a tuning fork vibrating with frequency f_y is held at the open end. This is the lowest frequency of stationary wave that can be formed in this tube.



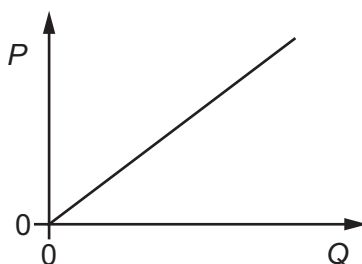
Assume the end correction for each tube is negligible.

Which equation is correct?

- A** $f_x = \frac{f_y}{4}$ **B** $f_x = \frac{f_y}{2}$ **C** $f_x = 2f_y$ **D** $f_x = 4f_y$

184 9702/11/O/N/18/Q21

The graph shows the variation of a quantity P with a quantity Q for a sound wave travelling in air.

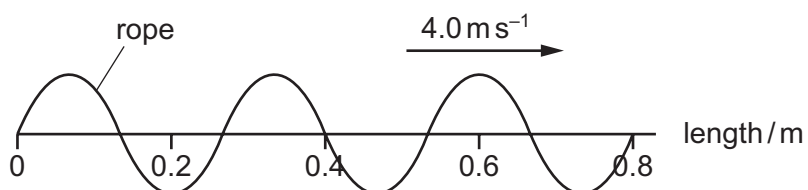


What could P and Q be?

	P	Q
A	amplitude	intensity
B	frequency	wavelength
C	speed	frequency
D	wavelength	period

185 9702/11/O/N/18/Q22

A vibration generator produces a progressive wave on a rope. The diagram shows the rope at one instant. The wave travels at a speed of 4.0 m s^{-1} .

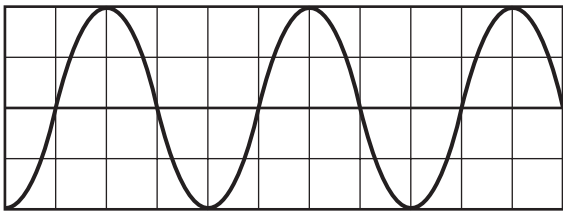


What are the wavelength and the frequency of the wave?

	wavelength / m	frequency / Hz
A	0.13	15
B	0.13	30
C	0.27	15
D	0.27	30

186 9702/11/O/N/18/Q23

The diagram shows the waveform of a signal displayed on a cathode-ray oscilloscope.



The time-base is set at 5.0 ms per division.

The Y-gain is set at 5.0 mV per division.

What are the amplitude and the frequency of the signal?

	amplitude / mV	frequency / Hz
A	10	50
B	10	100
C	20	50
D	20	100

187 9702/11/O/N/18/Q24

A jet aircraft travels at a speed of $0.8v$ where v is the speed of sound. The aircraft approaches a stationary observer. The frequency of sound emitted by the aircraft is 100 Hz.

Which frequency does the observer hear?

- A 56 Hz B 180 Hz C 400 Hz D 500 Hz

188 9702/11/O/N/18/Q25

A telescope detects and analyses some electromagnetic radiation of wavelength 2 cm.

Which type of telescope is it?

- A microwave telescope
B optical telescope
C radio telescope
D X-ray telescope

189 9702/12/O/N/18/Q22

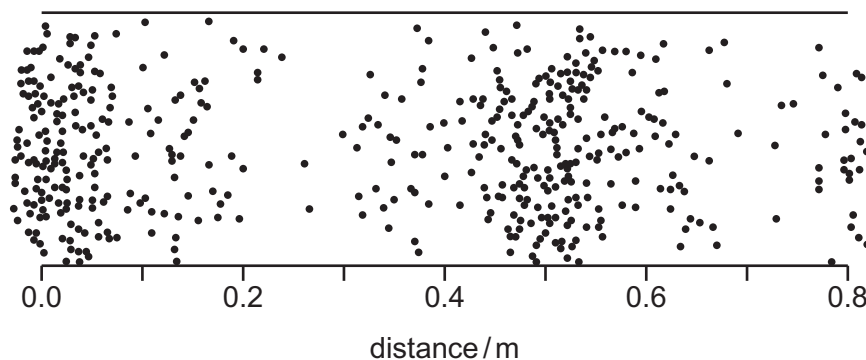
A progressive wave on a wire has a frequency of 10 Hz. Two points on the wire, separated by a distance of 0.25 m, have a phase difference of 22.5° .

What is the maximum speed of the wave?

- A** 2.5 ms^{-1} **B** 10 ms^{-1} **C** 20 ms^{-1} **D** 40 ms^{-1}

190 9702/12/O/N/18/Q23

When a guitar string is plucked, it causes a longitudinal sound wave in the air, as shown.



The speed of sound in the air is 340 ms^{-1} .

What is the approximate frequency of the sound wave shown?

- A** 430 Hz **B** 680 Hz **C** 1100 Hz **D** 1400 Hz

191 9702/12/O/N/18/Q25

A police car has a two-tone siren emitting sound of frequencies of 700 Hz and 1000 Hz.

The police car is travelling at a speed of 40.0 ms^{-1} towards a stationary observer. The speed of sound in the air is 340 ms^{-1} .

What is the difference between the two frequencies of the sound that is heard by the observer?

- A** 268 Hz **B** 300 Hz **C** 335 Hz **D** 340 Hz

192 9702/13/O/N/18/Q21

A progressive sound wave in air has amplitude x_0 and intensity I .

The amplitude of the wave increases to $3x_0$.

What is the new intensity of the wave?

- A** $\frac{I}{9}$ **B** $\frac{I}{3}$ **C** $3I$ **D** $9I$

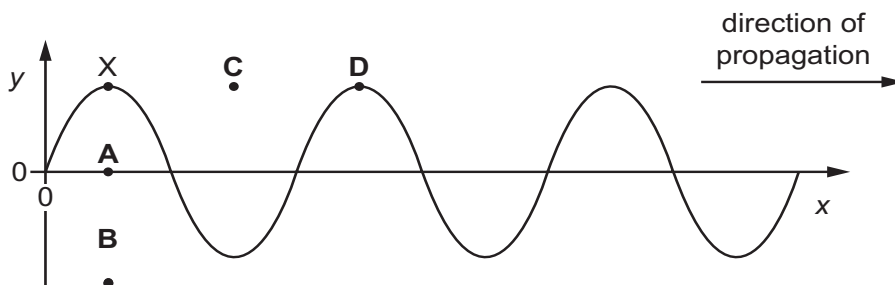
193 9702/13/O/N/18/Q22

The variation with distance x of the displacement y of a transverse wave on a rope is shown at time $t = 0$.

The wave has a frequency of 0.5 Hz.

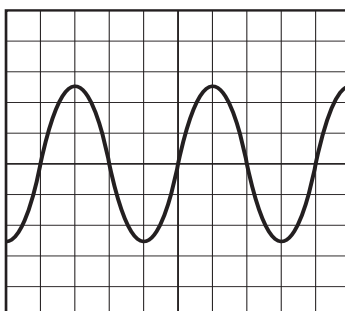
A point X on the rope is marked. The diagram shows the original position of X and four new positions.

What is the position of X at time $t = 1$ s?



194 9702/13/O/N/18/Q23

A sound wave is detected by a microphone. The output from the microphone is connected to the Y-input of a cathode-ray oscilloscope (c.r.o.). The trace on the c.r.o. is shown.



The time-base is set at 0.20 ms per division.

What is the frequency of the sound wave?

- A** 1000 Hz **B** 1250 Hz **C** 2000 Hz **D** 2500 Hz

195 9702/13/O/N/18/Q24

A bat flies directly towards a fixed ultrasound detector at a speed of 25.0 m s^{-1} emitting pulses of ultrasound of frequency 40.0 kHz.

The speed of sound in air is 330 m s^{-1} .

Which frequency does the ultrasound detector record?

- A** 37.0 kHz **B** 37.2 kHz **C** 43.0 kHz **D** 43.3 kHz

196 9702/13/O/N/18/Q25

An electromagnetic wave has a wavelength of $1.0 \times 10^{-7} \text{ m}$.

To which region of the electromagnetic spectrum does this wave belong?

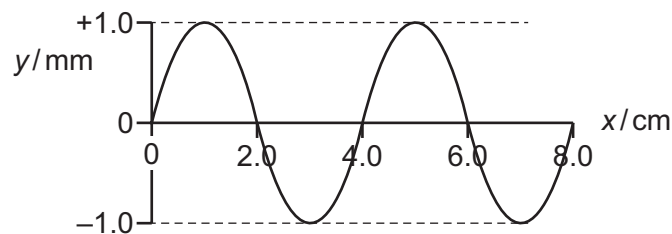
- | | |
|----------------------|------------------|
| A infra-red | C visible |
| B ultraviolet | D X-ray |

197 9702/11/M/J/19/Q22

A transverse wave in a medium has the waveform shown, where

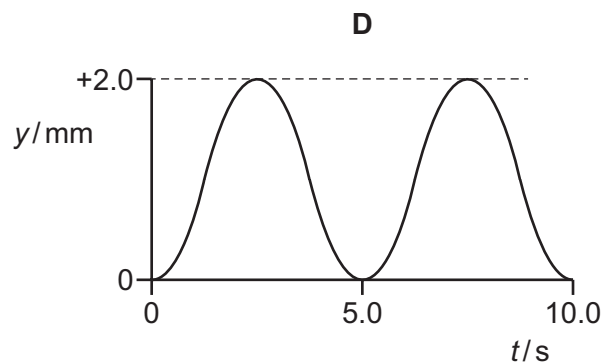
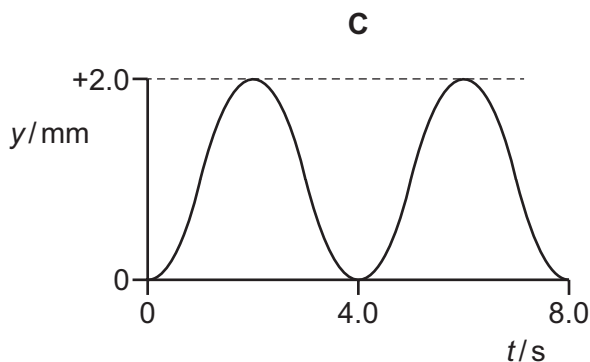
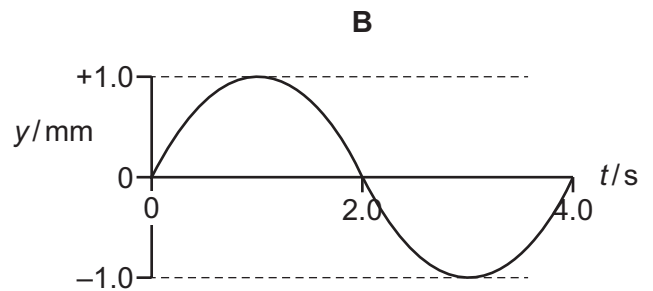
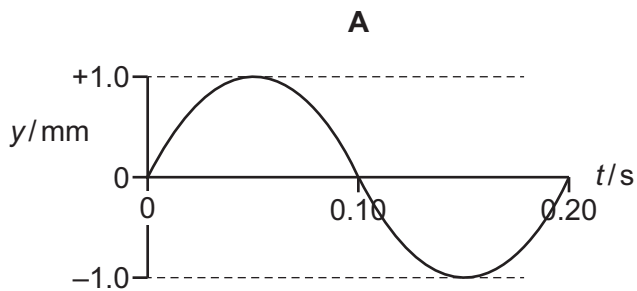
y = vertical displacement and x = horizontal distance.

The speed of the wave is 20.0 cm s^{-1} .



A particle of the medium oscillates vertically.

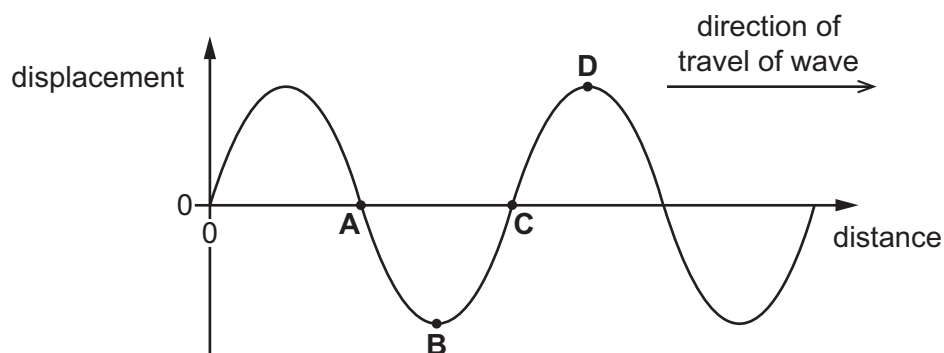
Which graph of vertical displacement y against time t best represents the motion of this particle?



198 9702/11/M/J/19/Q23

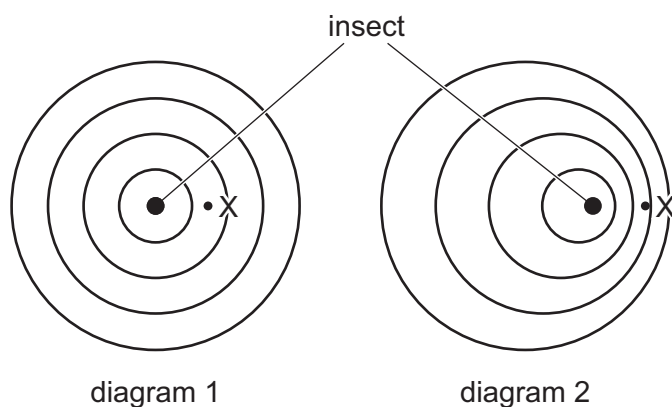
The graph shows the variation of the displacement of particles with distance along a transverse wave at an instant in time. The wave is moving to the right.

Which position along the wave corresponds to a point where particles in the wave are travelling the fastest upwards?



199 9702/11/M/J/19/Q25

A stationary insect on the surface of water creates circular waves with its legs, as shown in diagram 1. The insect begins to travel to the right as shown in diagram 2.



Which row describes the change to the waves at X caused by the movement of the insect?

	frequency	wave speed
A	decreases	increases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

200 9702/11/M/J/19/Q26

A toy motorboat moving with constant velocity v vibrates up and down on the surface of a pond. This causes the boat to act as a source of circular water waves of frequency 2.0 Hz. The speed of the waves is 1.5 m s^{-1} .

A man, standing at the edge of the pond, observes that the waves from the boat approach him with a frequency of 3.0 Hz.

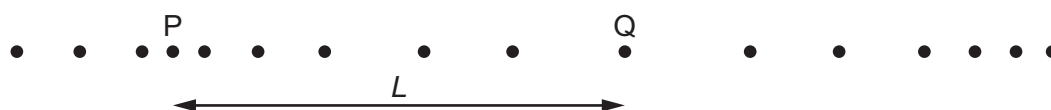
The formula for Doppler effect calculations with sound waves may also be used for water waves.

What is a possible value of v ?

	speed / m s^{-1}	direction
A	0.50	directly away from the man
B	0.50	directly towards the man
C	0.75	directly away from the man
D	0.75	directly towards the man

201 9702/12/M/J/19/Q23

The diagram illustrates the position of particles in a progressive sound wave at one instant in time.



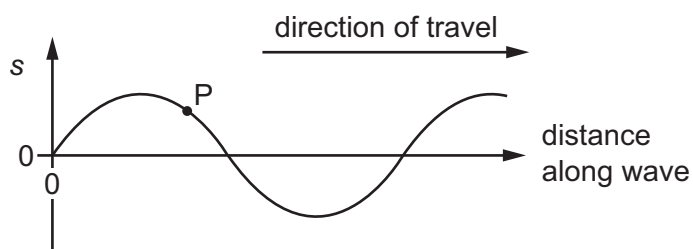
The speed of the wave is v . P and Q are two points in the wave a distance L apart.

What is an expression for the frequency of the wave?

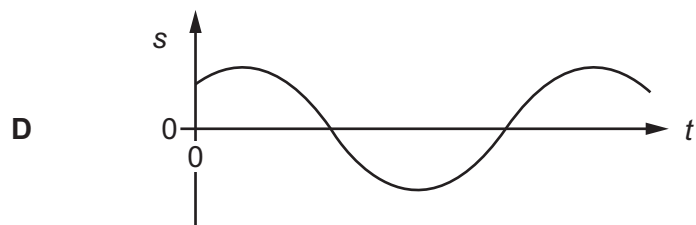
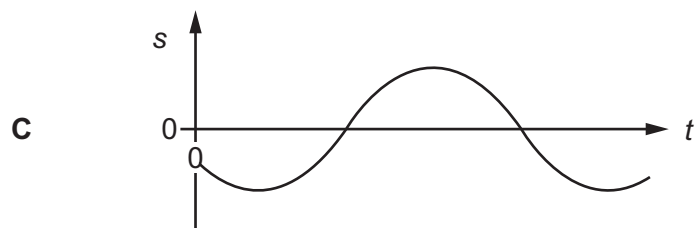
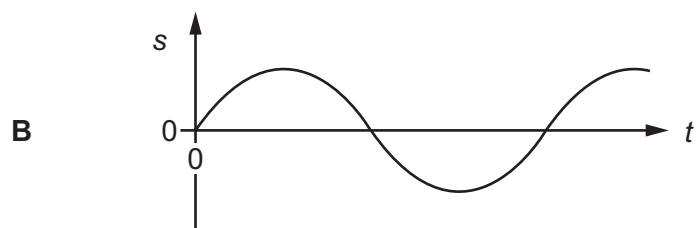
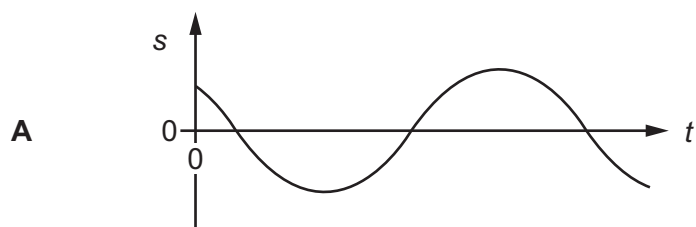
- A** $\frac{v}{2L}$ **B** $\frac{v}{L}$ **C** $\frac{2v}{L}$ **D** $\frac{L}{v}$

A wave moves along the surface of water.

The diagram shows the variation of displacement s with distance along the wave at time $t = 0$.



Which graph best shows the variation with time t of the displacement s of the point P on the wave?



203 9702/12/M/J/19/Q27

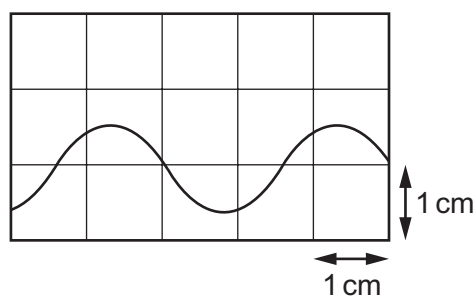
The electromagnetic spectrum consists of waves with different wavelengths.

Which row correctly identifies regions of the electromagnetic spectrum?

	10^{-10} m	10^{-8} m	10^{-5} m	10^{-2} m
A	microwaves	X-rays	ultraviolet	infrared
B	infrared	microwaves	X-rays	ultraviolet
C	microwaves	infrared	ultraviolet	X-rays
D	X-rays	ultraviolet	infrared	microwaves

204 9702/12/M/J/19/Q28

A cathode-ray oscilloscope (CRO) is used to display the trace from a sound wave. The time-base is set at $5 \mu\text{s mm}^{-1}$.

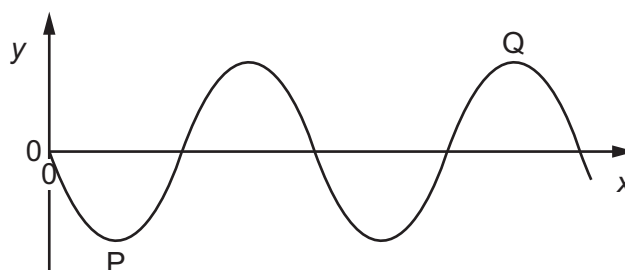


What is the frequency of the sound wave?

- A** 6.7 Hz **B** 67 Hz **C** 6.7 kHz **D** 67 kHz

205 9702/13/M/J/19/Q21

The graph shows the variation of displacement y with distance x along a progressive wave at one instant in time.



What is the phase difference between points P and Q on the wave?

- A** 90° **B** 270° **C** 540° **D** 630°

206 9702/13/M/J/19/Q22

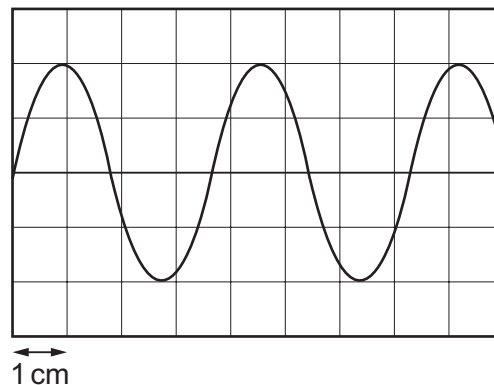
Wave-power generators take advantage of the energy that is transferred by the motion of waves across the surface of the oceans. The energy of a wave depends on its amplitude.

What is the correct definition of amplitude?

- A** the average amount of energy possessed by a wave
- B** the difference in displacement between a peak and a trough
- C** the maximum displacement of a point on the wave from equilibrium
- D** the number of oscillations of a wave that occur per second

207 9702/13/M/J/19/Q23

A sound wave of frequency 270 Hz is recorded by a cathode-ray oscilloscope (CRO). The waveform on the CRO is shown.



What is the time-base setting on the CRO?

- A** 0.1 ms cm⁻¹
- B** 1 ms cm⁻¹
- C** 10 ms cm⁻¹
- D** 100 ms cm⁻¹

208 9702/13/M/J/19/Q24

A motor boat vibrates in the water so that it produces water waves of frequency 0.20 Hz. The speed of these waves in the water is 20 ms⁻¹. The motor boat moves with a speed of 5.0 ms⁻¹ directly towards a stationary sailing boat.

The Doppler effect equation for sound waves also applies to water waves.

What is the frequency with which the waves hit the stationary sailing boat?

- A** 0.15 Hz
- B** 0.16 Hz
- C** 0.25 Hz
- D** 0.27 Hz

209 9702/13/M/J/19/Q25

Infrared laser light is used for the transmission of data along optic fibres.

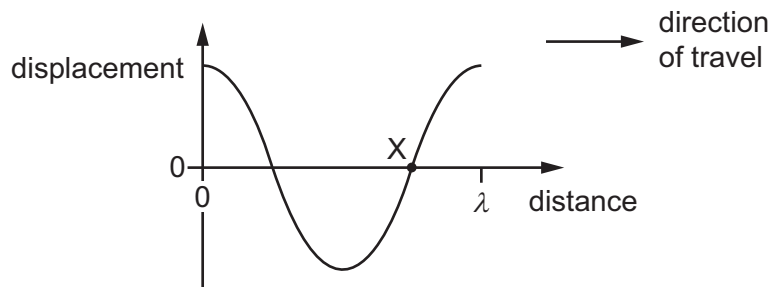
What is a typical wavelength of infrared radiation?

- A** 5 × 10⁻⁵ m
- B** 5 × 10⁻⁷ m
- C** 2 × 10⁻⁹ m
- D** 2 × 10⁻¹¹ m

210 9702/11/O/N/19/Q21

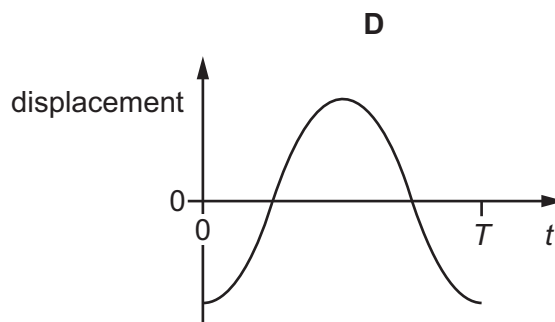
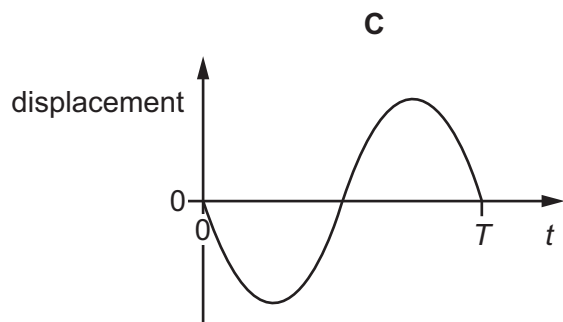
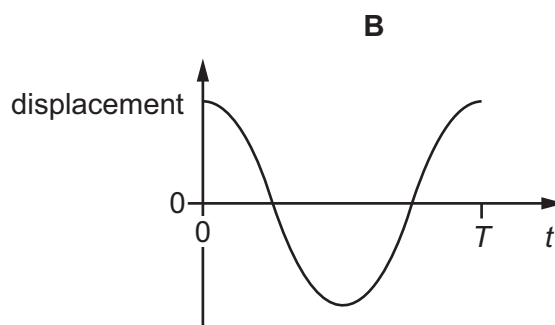
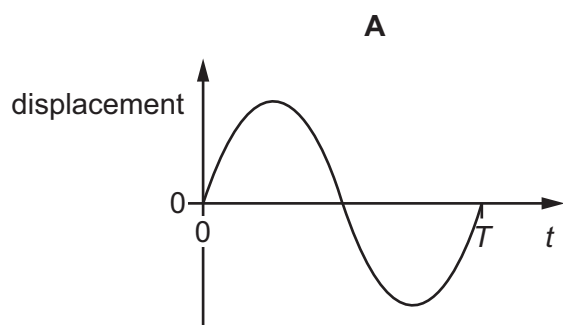
A transverse wave on a rope has wavelength λ and period T .

The graph shows the variation of the displacement of the particles of the rope with distance in the direction of travel of the wave at time $t = 0$.



A particle X is labelled.

Which graph shows the variation of the displacement of particle X with time t ?

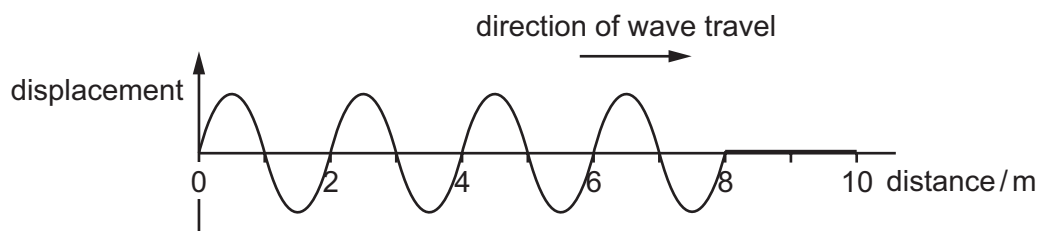


211 9702/11/O/N/19/Q22

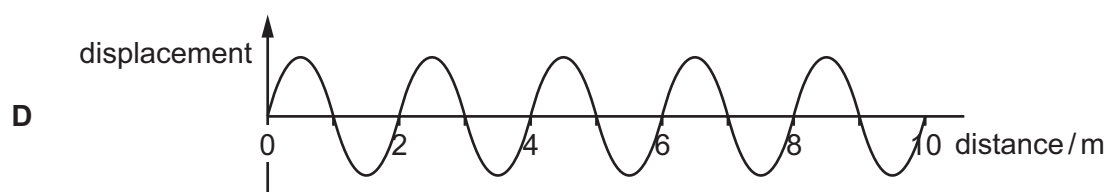
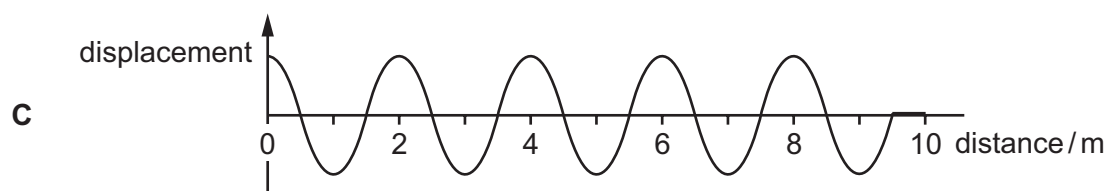
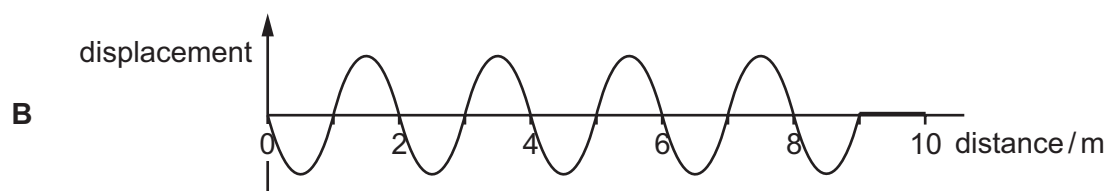
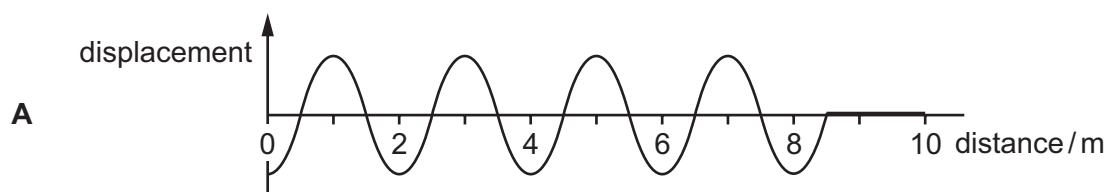
Which statement about all types of transverse waves is correct?

- A** They all have the same speed.
- B** They all have vibrations that are parallel to the direction of propagation of energy.
- C** They can all form stationary waves.
- D** They can all travel through a vacuum.

A transverse wave is travelling along a rope. The frequency of the wave is 2.0 Hz . The graph shows the variation with distance of the displacement of the wave at time $t = 0$.



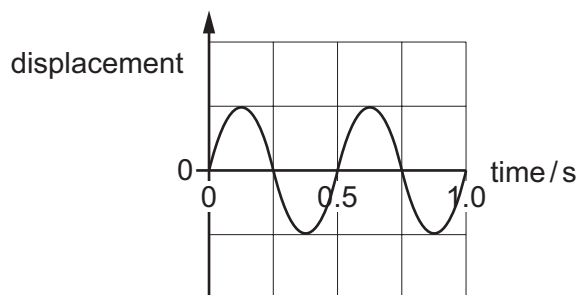
Which diagram shows the position of the wave at time $t = 0.5\text{ s}$?



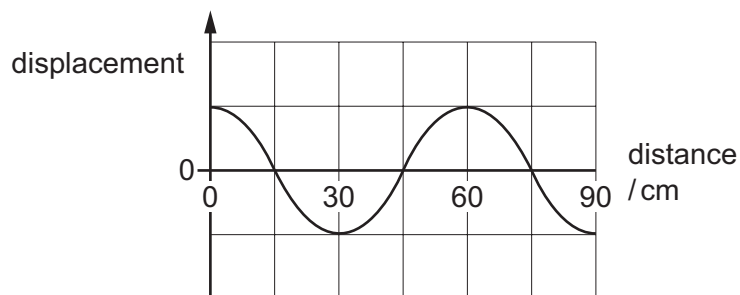
213 9702/12/O/N/19/Q21

The two graphs represent the same wave.

Graph 1 shows the variation with time of the displacement at a particular distance. Graph 2 shows the variation with distance of the displacement at one instant.



graph 1



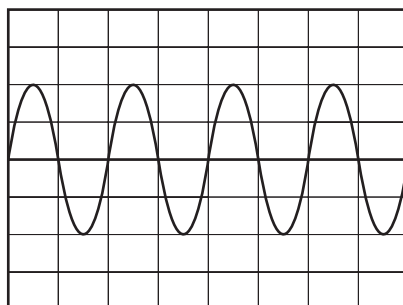
graph 2

What is the speed of the wave?

- A** 22.5 cm s^{-1} **B** 30.0 cm s^{-1} **C** 90.0 cm s^{-1} **D** 120 cm s^{-1}

214 9702/12/O/N/19/Q22

A microphone is connected to a cathode-ray oscilloscope (CRO). When a tuning fork is struck and then held next to the microphone, the following waveform is shown on the display of the CRO.



The time-base setting on the CRO is 2.00 ms per division.

What is the best estimate of the frequency of the sound produced by the tuning fork?

- A** 63 Hz **B** 170 Hz **C** 250 Hz **D** 500 Hz

215 9702/12/O/N/19/Q24

Two electromagnetic waves have wavelengths of $5.0 \times 10^{-7} \text{ m}$ and $5.0 \times 10^{-2} \text{ m}$.

Which row identifies the regions of the electromagnetic spectrum to which the waves belong?

	wavelength $5.0 \times 10^{-7} \text{ m}$	wavelength $5.0 \times 10^{-2} \text{ m}$
A	ultraviolet	infrared
B	visible	microwave
C	ultraviolet	microwave
D	visible	infrared

216 9702/12/O/N/19/Q26

Which statement about a light wave and a sound wave is correct?

- A** Both can travel through free space.
- B** Both have a frequency inversely proportional to their wavelength.
- C** Both have an intensity proportional to their amplitude.
- D** Both have oscillations perpendicular to the direction of energy transfer.

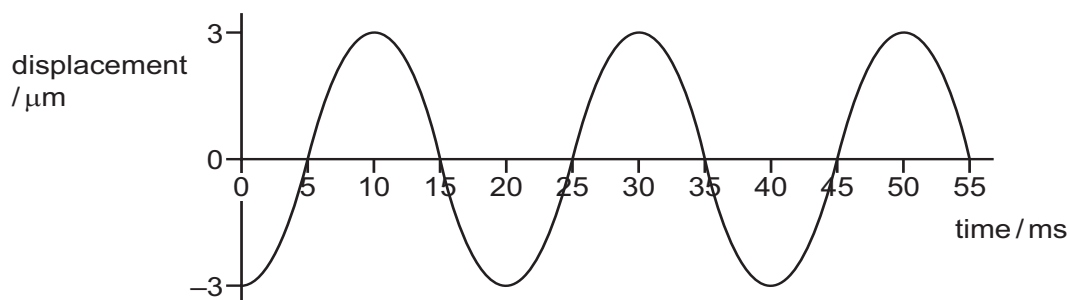
217 9702/13/O/N/19/Q21

Which statement about light waves and sound waves is **not** correct?

- A** Light waves and sound waves can both demonstrate the Doppler effect.
- B** Light waves are transverse waves and sound waves are longitudinal waves.
- C** Light waves can be diffracted but sound waves cannot.
- D** Light waves can travel in a vacuum but sound waves cannot.

218 9702/13/O/N/19/Q22

The graph represents a sound wave.

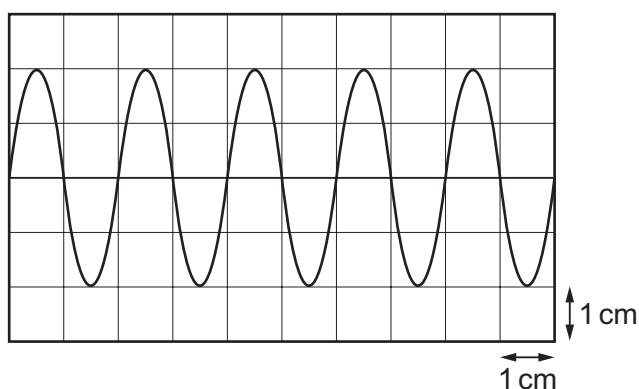


Which statement is correct?

- A** The wave is longitudinal and has a period of 25 ms.
- B** The wave is longitudinal and has a frequency of 50 Hz.
- C** The wave is transverse and has an amplitude of $3\ \mu\text{m}$.
- D** The wave is transverse and has a wavelength of 20 ms.

219 9702/13/O/N/19/Q23

A cathode-ray oscilloscope (CRO) is used to display a wave of frequency $5.0\ \text{kHz}$. The display is shown.

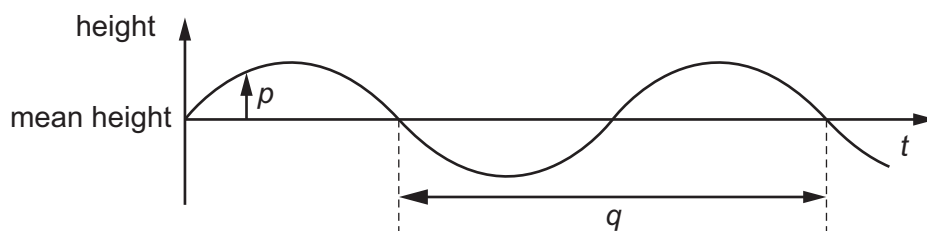


What is the time-base setting of the CRO?

- A** $10\ \mu\text{s cm}^{-1}$
- B** $100\ \mu\text{s cm}^{-1}$
- C** $10\ \text{ms cm}^{-1}$
- D** $100\ \text{ms cm}^{-1}$

220 9702/13/O/N/19/Q27

The graph shows how the height of the water surface at a point in a harbour varies with time t as waves pass the point.



What are p and q ?

	p	q
A	displacement	period
B	displacement	wavelength
C	amplitude	period
D	amplitude	wavelength

221. 9702/12/F/M/20/Q21

A wave of frequency 15 Hz travels at 24 m s^{-1} through a medium.

What is the phase difference between two points 2.0 m apart?

- A There is no phase difference.
- B They are out of phase by a quarter of a cycle.
- C They are out of phase by half a cycle.
- D They are out of phase by 0.80 of a cycle.

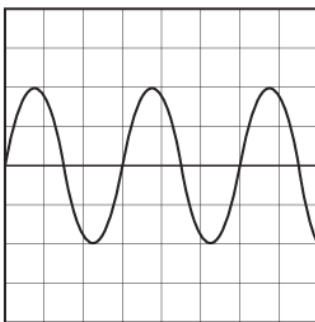
222. 9702/12/F/M/20/Q22

Which row describes a longitudinal wave and a medium through which it can travel?

	direction of oscillation of the medium compared with the direction of propagation of wave energy	medium
A	parallel	air
B	parallel	vacuum
C	perpendicular	air
D	perpendicular	vacuum

223. 9702/12/F/M/20/Q23

A sound wave is displayed on the screen of a cathode-ray oscilloscope, as shown.



The time-base setting is 0.50 ms per division.

What is the frequency of the sound wave?

- A** 500 Hz **B** 670 Hz **C** 1000 Hz **D** 1300 Hz

224. 9702/12/F/M/20/Q24

An observer is situated at the top of a tall tower. An aeroplane emitting sound at a frequency of 1000 Hz approaches the observer at a speed of 165 m s^{-1} .

The speed of sound is 330 m s^{-1} .

What is the frequency of the sound received by the observer?

- A** 330 Hz **B** 667 Hz **C** 1000 Hz **D** 2000 Hz

225. 9702/12/F/M/20/Q25

What is the order of magnitude of the wavelengths of microwaves and X-rays?

	wavelength of microwaves / m	wavelength of X-rays / m
A	10^{-6}	10^3
B	10^{-2}	10^3
C	10^{-6}	10^{-10}
D	10^{-2}	10^{-10}

226. 9702/11/M/J/20/Q22

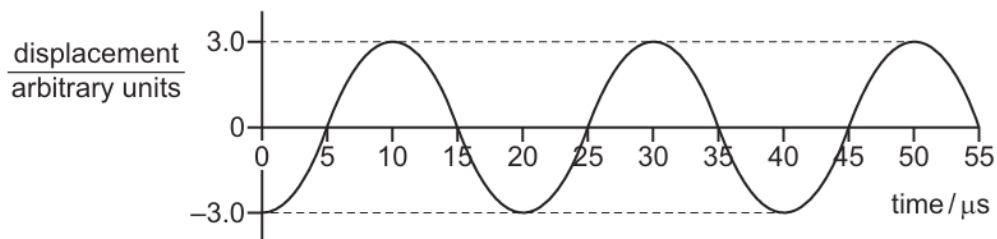
A sound wave reduces in intensity but maintains a constant frequency as it travels through air.

Which statement is correct?

- A** The maximum displacement of the particles changes between one particle and the next particle.
- B** The phase difference between adjacent particles is zero.
- C** The wavelength is the distance between two particles that have a phase difference of 180° .
- D** Two particles that have a phase difference of 360° have the same maximum displacement.

227. 9702/11/M/J/20/Q23

The graph shows the variation with time of the displacement of an electromagnetic wave at a point.



The wave is travelling in a vacuum.

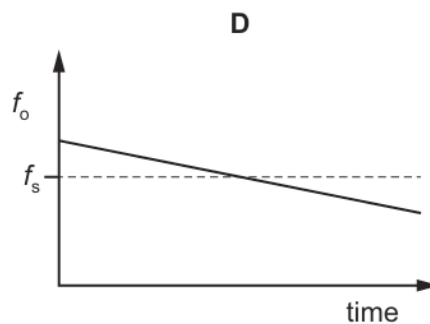
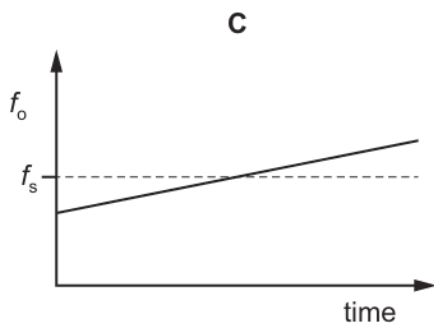
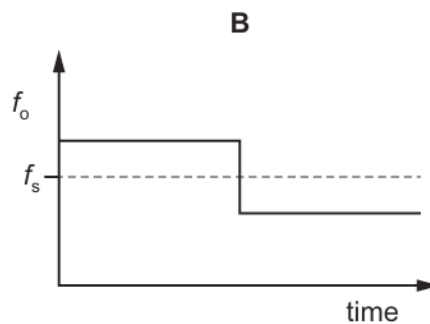
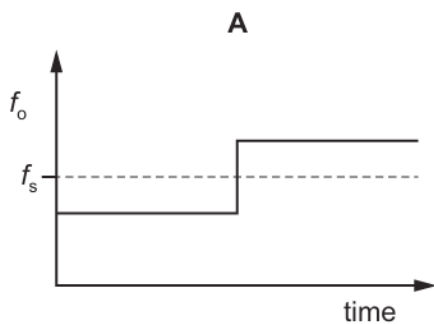
What is the amplitude and what is the wavelength of the wave?

	amplitude /arbitrary units	wavelength /m
A	3.0	6000
B	6.0	6000
C	3.0	7500
D	6.0	7500

228. 9702/11/M/J/20/Q25

A source emitting sound of a single frequency f_s travels at constant speed directly towards an observer. The source then passes the observer and continues to move directly away from the observer. The velocity of the source remains constant.

Which graph represents the variation with time of the frequency f_o of the sound heard by the observer?



229. 9702/11/M/J/20/Q26

What are the names of the electromagnetic waves that have wavelengths in a vacuum of 100 pm and of $100\text{ }\mu\text{m}$?

	wavelength 100 pm	wavelength $100\text{ }\mu\text{m}$
A	γ -rays	infrared
B	γ -rays	red light
C	X-rays	infrared
D	X-rays	red light

230. 9702/12/M/J/20/Q21

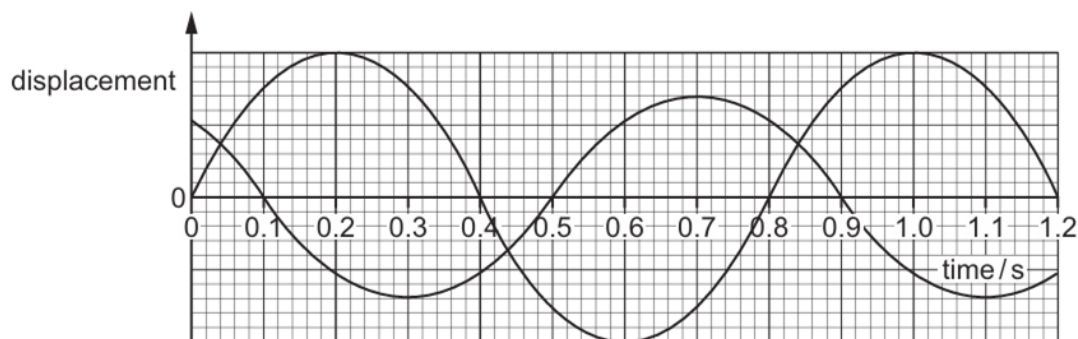
A transverse wave is moving along a rope. Two points X and Y on the rope are a quarter of a wavelength apart from each other.

Which statement is **not** possible for the two points X and Y at any instant?

- A** They are both stationary.
- B** They are displaced in opposite directions from their equilibrium position.
- C** They are moving in opposite directions.
- D** They both have displacements of the same magnitude from their equilibrium positions.

231. 9702/12/M/J/20/Q22

Two progressive waves meet at a fixed point P. The variation with time of the displacement of each wave at point P is shown in the graph.

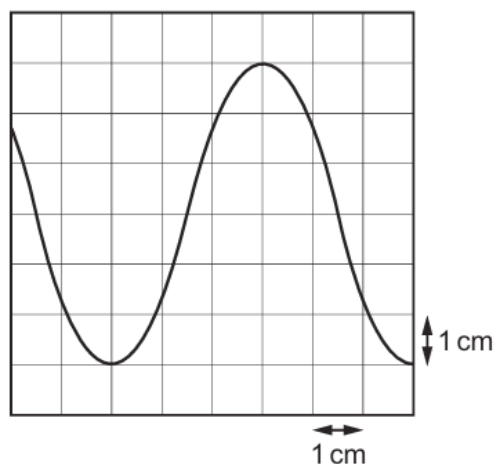


What is the phase difference between the two waves at point P?

- A** 45°
- B** 90°
- C** 135°
- D** 180°

232. 9702/12/M/J/20/Q23

A microphone connected to the Y-plates of a cathode-ray oscilloscope (CRO) is placed in front of a loudspeaker. The trace on the screen of the CRO is shown.



The time-base setting is 0.5 ms cm^{-1} and the Y-plate sensitivity is 0.2 mV cm^{-1} .

What is the frequency of the sound from the loudspeaker and what is the amplitude of the trace on the CRO?

	frequency /Hz	amplitude /mV
A	330	0.6
B	330	1.2
C	670	0.6
D	670	1.2

233. 9702/12/M/J/20/Q25

A stationary person measures the speed and wavelength of the sound from a horn on a stationary vehicle. The person then repeats the measurements when the vehicle is approaching at a constant speed.

Which row describes the measured wavelength and the measured speed of the sound wave from the moving vehicle when compared with the sound wave from the stationary vehicle?

	wavelength of the sound wave	speed of the sound wave
A	longer	greater
B	shorter	greater
C	longer	same
D	shorter	same

234. 9702/12/M/J/20/Q26

The table shows the wavelengths of five electromagnetic waves.

Which row correctly identifies the principal radiation for each of these wavelengths?

	10^{-14} m	10^{-10} m	10^{-6} m	10^{-2} m	10^2 m
A	gamma-ray	X-ray	infrared	microwave	radio wave
B	radio wave	microwave	infrared	X-ray	gamma-ray
C	radio wave	microwave	ultraviolet	infrared	X-ray
D	X-ray	infrared	ultraviolet	microwave	radio wave

235. 9702/13/M/J/20/Q22

A wave of amplitude A has an intensity I .

After passing through a certain medium, the wave has a new intensity of $\frac{I}{4}$.

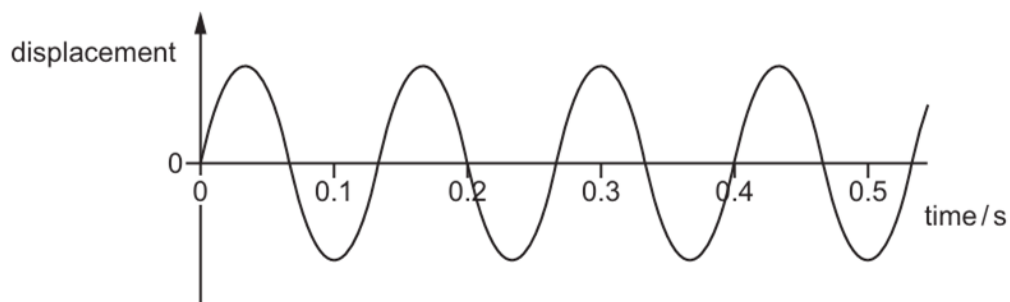
What is the new amplitude of the wave?

- A** $2A$ **B** $\frac{A}{2}$ **C** $\frac{A}{4}$ **D** $\frac{A}{16}$

236. 9702/13/M/J/20/Q23

A wave travels along a coiled spring.

The graph shows the variation with time of the displacement of a point on the spring.



What is the frequency of the wave?

- A** 0.13 Hz **B** 0.20 Hz **C** 5.0 Hz **D** 7.5 Hz

237. 9702/13/M/J/20/Q25

A stationary source S emits a sound wave of frequency f .

The source now moves away from a stationary observer.

Which statement is correct?

- A** The frequency of the source S and the observed frequency are now both higher than f .
- B** The frequency of the source S and the observed frequency are now both lower than f .
- C** The frequency of the source S is now lower than f .
- D** The observed frequency is now lower than f .

238. 9702/13/M/J/20/Q26

The table lists possible wavelengths of four different electromagnetic waves.

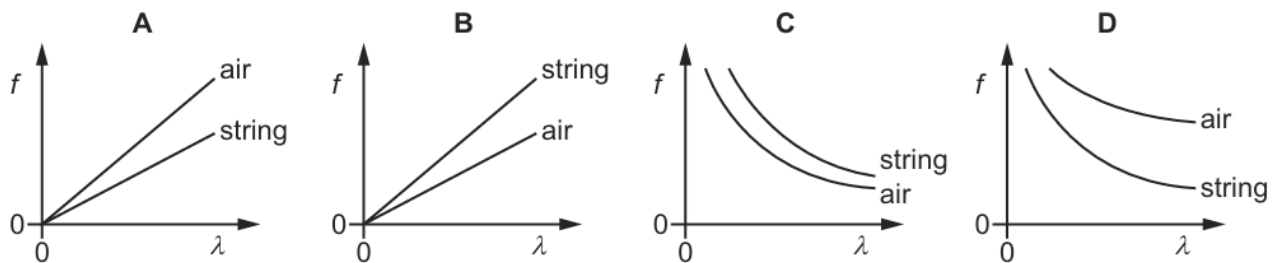
Which row is correct?

	type of wave	approximate wavelength / m
A	infrared	10^{-5}
B	radio	10^{-3}
C	ultraviolet	10^{-12}
D	X-rays	10^{-7}

239. 9702/11/O/N/20/Q21

A guitar string vibrates to create a sound. The speed of the wave in the guitar string is always 440 m s^{-1} . The vibrating string creates a sound wave that moves in the air with a speed of 330 m s^{-1} .

Which graph shows the variation of frequency f with the wavelength λ for the waves in the string and in the air?

**240. 9702/11/O/N/20/Q24**

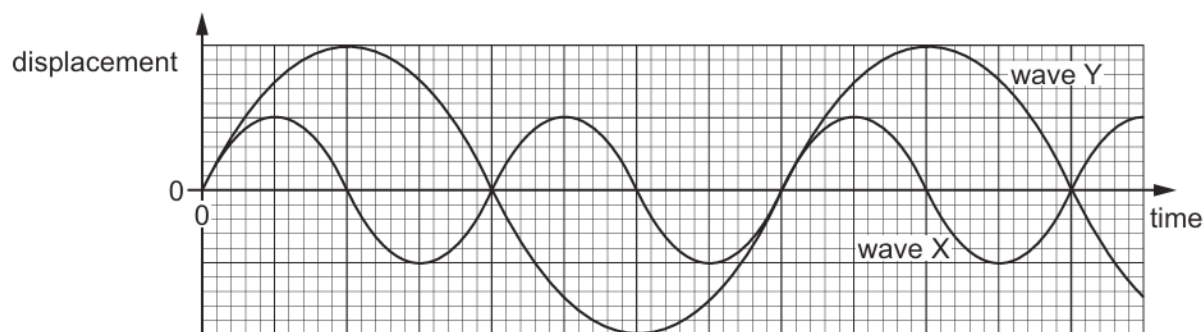
A source of sound of frequency 1000 Hz directly approaches a stationary observer. The observer measures the frequency of the received sound to be 1500 Hz . The speed of sound in still air is 330 m s^{-1} .

What is the speed of the source of sound?

- A** 110 m s^{-1}
- B** 165 m s^{-1}
- C** 220 m s^{-1}
- D** 330 m s^{-1}

241. 9702/11/O/N/20/Q22

The graph shows the variation with time of displacement for two different waves X and Y.



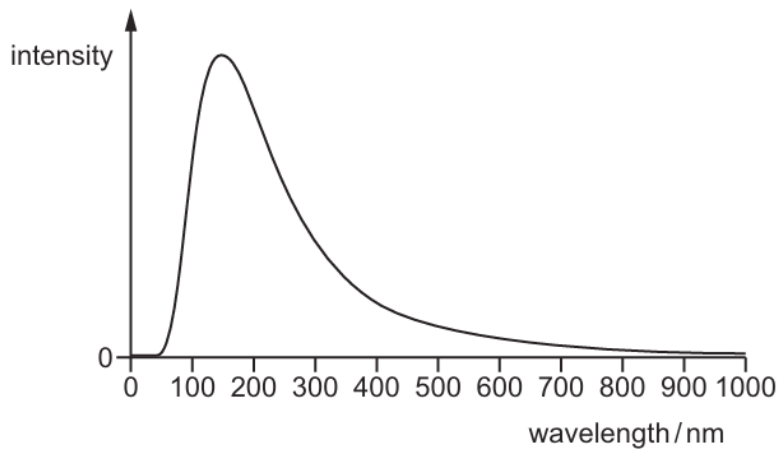
Wave X has frequency f and amplitude A .

What is the frequency and what is the amplitude of wave Y?

	frequency	amplitude
A	$\frac{1}{2}f$	$\frac{1}{2}A$
B	$\frac{1}{2}f$	$2A$
C	$2f$	$\frac{1}{2}A$
D	$2f$	$2A$

242. 9702/11/O/N/20/Q25

The graph shows how the intensity of electromagnetic radiation emitted from a distant star varies with wavelength.



In which region of the electromagnetic spectrum is the radiation of greatest intensity?

- A** infrared
- B** visible light
- C** ultraviolet
- D** X-ray

243. 9702/12/O/N/20/Q22

The speed v of waves in deep water is given by the equation

$$v^2 = \frac{g\lambda}{2\pi}$$

where λ is the wavelength of the waves and g is the acceleration of free fall.

A student measures the wavelength λ and the frequency f of a number of these waves.

Which graph should he plot to give a straight line through the origin?

A f^2 against λ

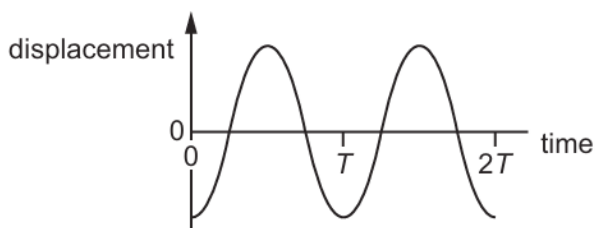
B f against λ^2

C f against $\frac{1}{\lambda}$

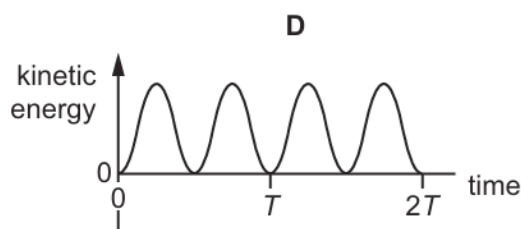
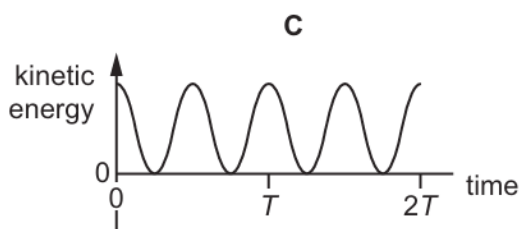
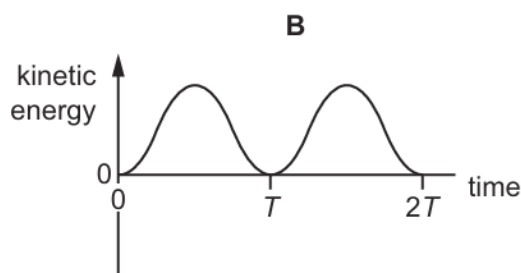
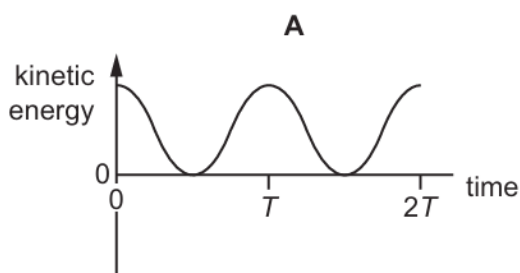
D f^2 against $\frac{1}{\lambda}$

244. 9702/12/O/N/20/Q23

When sound travels through air, the air particles vibrate. A graph of displacement against time for a single air particle is shown.

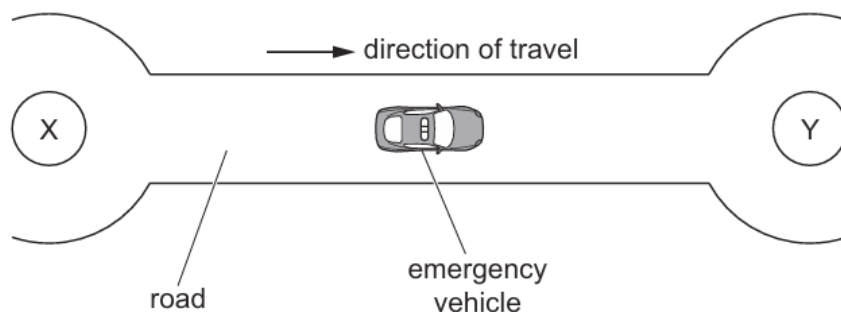


Which graph shows how the kinetic energy of the air particle varies with time?



245. 9702/12/O/N/20/Q25

An emergency vehicle sounds its siren as it accelerates along a straight road between two points X and Y, as shown in the diagram.



The frequency of the sound emitted by the siren is 750 Hz.

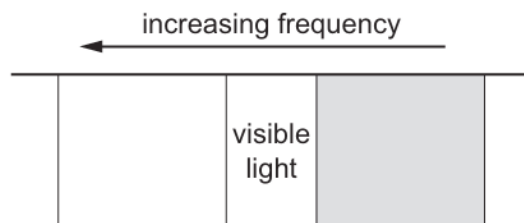
A person stands at X and another person stands at Y.

What describes the sounds heard by the people at X and at Y as the vehicle accelerates?

	sound heard by person at X	sound heard by person at Y
A	higher than 750 Hz, increasing in frequency	lower than 750 Hz, decreasing in frequency
B	higher than 750 Hz, decreasing in frequency	lower than 750 Hz, increasing in frequency
C	lower than 750 Hz, decreasing in frequency	higher than 750 Hz, increasing in frequency
D	lower than 750 Hz, increasing in frequency	higher than 750 Hz, decreasing in frequency

246. 9702/12/O/N/20/Q26

Part of the electromagnetic spectrum is shown.



What is the name of the shaded region and what is the order of magnitude of a wavelength of a wave from this region?

	name	wavelength /m
A	infrared	10^{-5}
B	infrared	10^{-8}
C	ultraviolet	10^{-5}
D	ultraviolet	10^{-8}

247. 9702/13/O/N/20/Q22

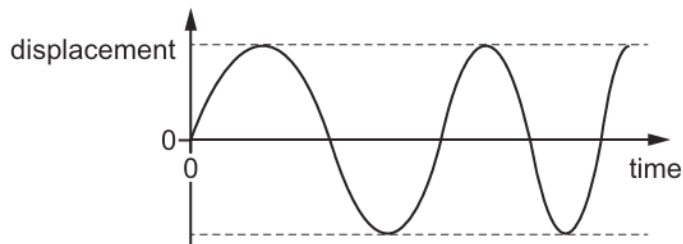
Two waves X and Y have the same frequency. The amplitude of X is $1.5A_0$ and the amplitude of Y is $2.5A_0$. The waves meet at a point and superpose to form a resultant wave.

For the resultant wave, what is the ratio $\frac{\text{maximum possible intensity}}{\text{minimum possible intensity}}$?

- A** 1.7 **B** 2.8 **C** 4.0 **D** 16

248. 9702/13/O/N/20/Q23

The displacement–time graph for an air particle in the path of a sound wave is shown.

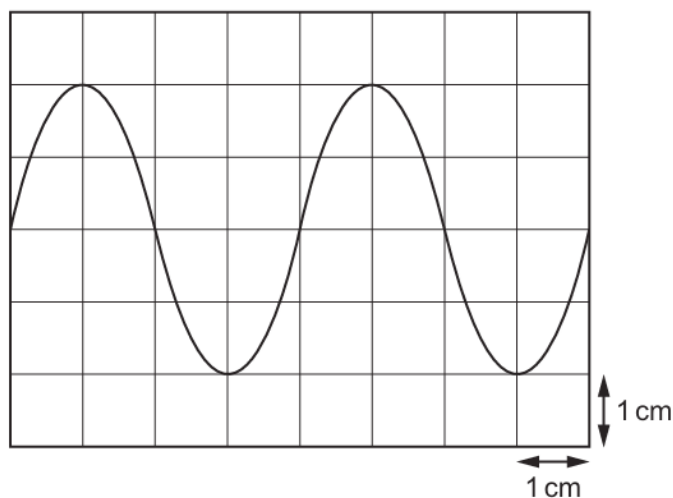


Which property of the sound wave must be increasing?

- A** amplitude
B frequency
C period
D speed

249. 9702/13/O/N/20/Q24

A sound wave is displayed on the screen of a cathode-ray oscilloscope (CRO) as shown.



The time-base of the CRO is set at 2.5 ms cm^{-1} .

What is the frequency of the sound wave?

- A** 50 Hz **B** 100 Hz **C** 200 Hz **D** 400 Hz

250. 9702/13/O/N/20/Q25

The horn of a train emits sound of frequency f_1 . While the horn is sounding, the train moves directly towards a stationary person. The speed of the train is $0.20v$, where v is the speed of sound.

The frequency of the sound heard by the person is f_2 .

What is the ratio $\frac{f_1}{f_2}$?

- A** $\frac{0.80}{1}$ **B** $\frac{1}{1.2}$ **C** $\frac{1.2}{1}$ **D** $\frac{1}{0.80}$

251. 9702/13/O/N/20/Q26

Two satellites in orbit around the Earth are at a constant distance of 100 km apart from each other.

Satellite X transmits a microwave pulse towards satellite Y. The pulse takes time T to reach Y.

Satellite Y then transmits a pulse of visible light towards satellite X.

What is the time taken for the pulse of light to reach X?

- A** $10^{-5}T$ **B** $10^{-3}T$ **C** $10^{-2}T$ **D** T

252. 9702/12/F/M/21/Q21

A progressive radio wave in a vacuum has a frequency of 75 MHz.

What is the phase difference between two points on the wave that are 50 cm apart from each other?

- A** 23° **B** 45° **C** 90° **D** 180°

253. 9702/12/F/M/21/Q22

Which statement is correct for longitudinal waves but **not** correct for transverse waves?

- A** They can form stationary waves.
B They can only travel through a medium.
C They can transfer energy in the direction of travel.
D They consist of peaks and troughs.

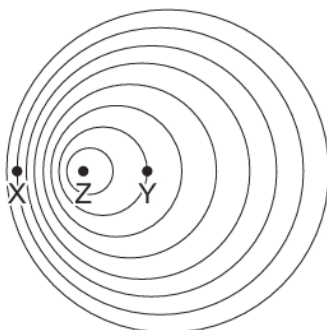
254. 9702/12/F/M/21/Q24

What is **not** a possible value for the wavelength of the named electromagnetic waves when it is travelling in a vacuum?

	electromagnetic wave	wavelength / m
A	γ -rays	3×10^{-13}
B	X-rays	3×10^{-10}
C	infrared	3×10^{-6}
D	microwaves	3×10^{-5}

255. 9702/12/F/M/21/Q25

A source of sound of frequency F at point Z is moving at a steady speed. The pattern of the emitted wavefronts is shown.



Which row describes the frequencies of the sound heard by stationary observers at X and Y?

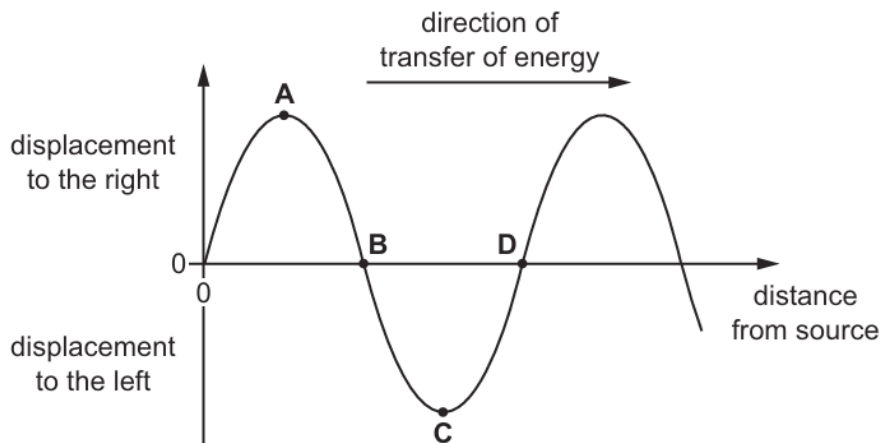
	frequency heard at X	frequency heard at Y
A	$<F$	$<F$
B	$<F$	$>F$
C	$>F$	$<F$
D	$>F$	$>F$

256. 9702/11/M/J/21/Q22

A longitudinal wave travelling from left to right has vibrations parallel to the direction of transfer of energy by the wave.

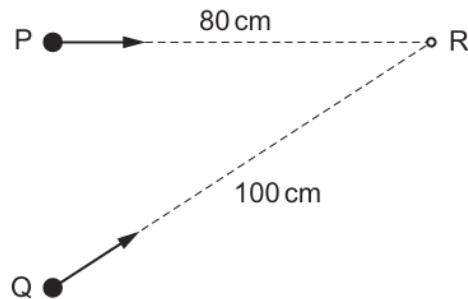
The wave can be represented on a graph showing the variation with distance of the displacement of the particles from their equilibrium positions at one instant.

Which point on the graph is the centre of a compression?



257. 9702/11/M/J/21/Q21

Two identical waves are produced by sources at points P and Q. The waves travel along different paths to reach point R, as shown.



Both waves have a wavelength of 6.0 cm. The waves are in phase at point R.

What is the phase difference between the waves as they leave points P and Q?

- A** 0° **B** 60° **C** 90° **D** 120°

258. 9702/11/M/J/21/Q24

An ambulance has a siren that emits sound of a constant frequency. The ambulance is moving directly towards a stationary observer.

The ambulance decelerates as it is approaching the observer and then accelerates after it has passed the observer.

How does the frequency of the sound heard by the observer change as the ambulance is approaching and as it is moving away from the observer?

	approaching observer	moving away from observer
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

259. 9702/11/M/J/21/Q25

Microwaves in a vacuum travel at speed X and have wavelength of order of magnitude Y .

What are the speed and a possible order of magnitude of wavelength of X-rays in a vacuum?

	speed	wavelength
A	X	$10^{-8} Y$
B	X	$10^{-4} Y$
C	$10^4 X$	Y
D	$10^8 X$	Y

260. 9702/12/M/J/21/Q21

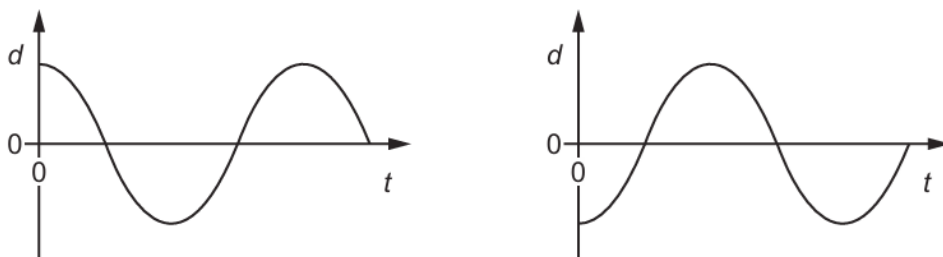
Two lasers emit light in a vacuum. One laser emits red light and the other emits green light.

Which property of the light from the two lasers **must** be different?

- A amplitude
- B frequency
- C intensity
- D speed

261. 9702/12/M/J/21/Q22

Two particles in a progressive wave are a distance 10 cm apart. The two graphs show the variation with time t of the displacement d of the two particles.



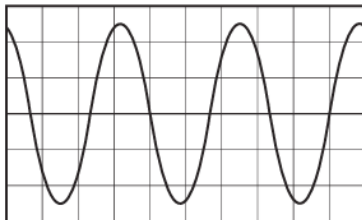
What could be represented by the two graphs?

- A particles in a longitudinal wave with a compression and the nearest rarefaction separated by 10 cm
- B particles in a longitudinal wave with a compression and the nearest rarefaction separated by 20 cm
- C particles in a transverse wave with a peak and the nearest trough separated by 20 cm
- D particles in a transverse wave with two adjacent peaks separated by 10 cm

262. 9702/12/M/J/21/Q23

A sound wave is detected by a microphone that is connected to a cathode-ray oscilloscope (CRO).

The screen of the CRO displays a waveform, as shown.



The time-base is set to $20 \mu\text{s div}^{-1}$.

What is the frequency of the sound wave?

- A 15 Hz
- B 15 000 Hz
- C 20 000 Hz
- D 30 000 Hz

263. 9702/12/M/J/21/Q24

A person stands at the side of a straight railway track. A train moves towards the person and emits sound from its whistle. The person hears a sound of frequency 1690 Hz as the train approaches him.

The person then hears sound of frequency 1500 Hz as the train moves away from him. The speed of sound in air is 340 m s^{-1} .

What is the speed of the train?

- A** 20 m s^{-1} **B** 38 m s^{-1} **C** 41 m s^{-1} **D** 43 m s^{-1}

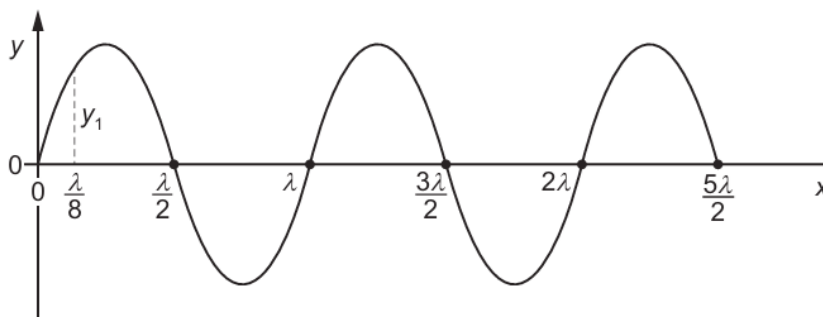
264. 9702/12/M/J/21/Q25

Which list shows electromagnetic waves in order of decreasing frequency?

- A** gamma-rays $\rightarrow$ infrared $\rightarrow$ ultraviolet $\rightarrow$ radio waves
B gamma-rays $\rightarrow$ ultraviolet $\rightarrow$ infrared $\rightarrow$ radio waves
C radio waves $\rightarrow$ infrared $\rightarrow$ ultraviolet $\rightarrow$ gamma-rays
D radio waves $\rightarrow$ ultraviolet $\rightarrow$ infrared $\rightarrow$ gamma-rays

265. 9702/13/M/J/21/Q22

A transverse progressive wave of wavelength λ is set up on a stretched string. The graph shows the variation of displacement y with distance x at a particular instant of time. The wave has displacement $+y_1$ at distance $x = \frac{\lambda}{8}$.

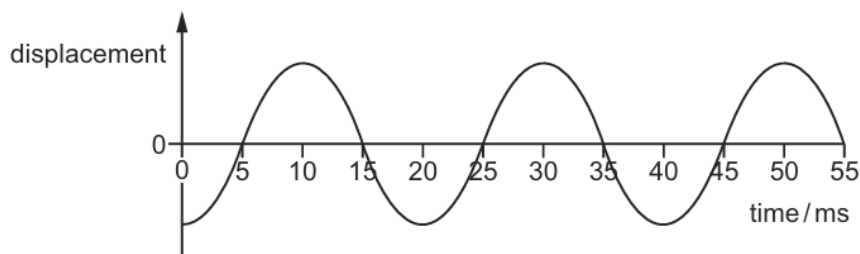


What are the next two values of x where the displacement y is again equal to $+y_1$?

- A** $\frac{3\lambda}{8}$ and $\frac{5\lambda}{8}$
B $\frac{3\lambda}{8}$ and $\frac{9\lambda}{8}$
C $\frac{5\lambda}{8}$ and $\frac{9\lambda}{8}$
D $\frac{9\lambda}{8}$ and $\frac{17\lambda}{8}$

266. 9702/13/M/J/21/Q23

The graph shows the variation with time of the displacement of an air particle as a progressive sound wave passes through the air. The speed of sound in air is 330 m s^{-1} .

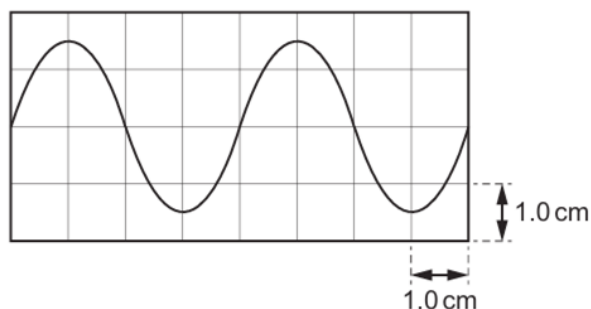


What is the wavelength of the wave?

- A** 6.6 m **B** 8.3 m **C** 20 m **D** 25 m

267. 9702/13/M/J/21/Q24

A microphone is connected to a cathode-ray oscilloscope (CRO). A sound wave of constant frequency is detected by the microphone. The screen of the CRO is shown.



The time-base is set at 1.0 ms cm^{-1} .

What is the frequency of the sound wave?

- A** 250 Hz **B** 500 Hz **C** 670 Hz **D** 4000 Hz

268. 9702/13/M/J/21/Q25

A train's whistle is emitting sound of frequency 500 Hz as the train moves with a speed of 20 m s^{-1} along a straight track. The train moves directly towards a stationary observer standing next to the track and then passes the observer.

The speed of sound in air is 330 m s^{-1} .

What is the difference between the frequencies of the sound heard by the observer before and after the train has passed the observer?

- A** 29 Hz **B** 32 Hz **C** 40 Hz **D** 61 Hz

269. 9702/13/M/J/21/Q26

Gamma-rays, microwaves, visible light and X-rays are four regions of the electromagnetic spectrum.

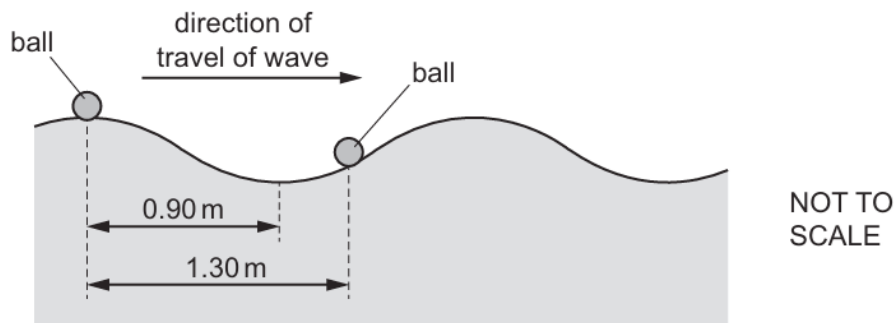
Which list shows these four regions in order of increasing wavelength?

- A microwaves $\rightarrow$ visible light $\rightarrow$ X-rays $\rightarrow$ gamma-rays
- B gamma-rays $\rightarrow$ X-rays $\rightarrow$ visible light $\rightarrow$ microwaves
- C X-rays $\rightarrow$ gamma-rays $\rightarrow$ microwaves $\rightarrow$ visible light
- D microwaves $\rightarrow$ visible light $\rightarrow$ gamma-rays $\rightarrow$ X-rays

270. 9702/11/O/N/21/Q21

Two balls float on the surface of the sea. The balls are separated by a distance of 1.30 m.

A wave travels on the surface of the sea so that the balls move vertically up and down.



The distance between a crest and an adjacent trough of the wave is 0.90 m.

What is the phase difference between the two balls?

- A 55°
- B 110°
- C 160°
- D 260°

271. 9702/11/O/N/21/Q22

Which statement about transverse or longitudinal waves is **not** correct?

- A Longitudinal waves can be used to demonstrate diffraction.
- B Longitudinal waves can travel in a vacuum.
- C Transverse waves can form stationary waves.
- D Transverse waves can transfer energy.

272. 9702/11/O/N/21/Q24

With which waves can the Doppler effect be observed?

- A all waves including sound and light
- B light waves only
- C sound and light waves only
- D sound waves only

273. 9702/11/O/N/21/Q25

Which radiation could consist of waves of wavelength 0.5 nm?

- A γ -rays
- B ultraviolet
- C visible light
- D X-rays

274. 9702/12/O/N/21/Q23

The table contains descriptions and examples of waves.

Which row is correct?

	description of wave	example
A	oscillations are parallel to the direction of energy transfer	gamma-rays
B	oscillations are parallel to the direction of energy transfer	ultraviolet waves
C	oscillations are perpendicular to the direction of energy transfer	sound waves
D	oscillations are perpendicular to the direction of energy transfer	X-rays

275. 9702/12/O/N/21/Q25

A train travels in a straight line at a constant speed of 30 m s^{-1} . The train's horn continuously emits sound of frequency 2400 Hz.

A stationary observer stands next to the train track. The train approaches the stationary observer, passes him and then moves away.

The speed of sound is 340 m s^{-1} .

What is the maximum difference in the frequencies of the sound heard by the stationary observer?

- A 190 Hz B 230 Hz C 430 Hz D 460 Hz

276. 9702/12/O/N/21/Q26

Electromagnetic waves of frequency 30 THz are in a vacuum.

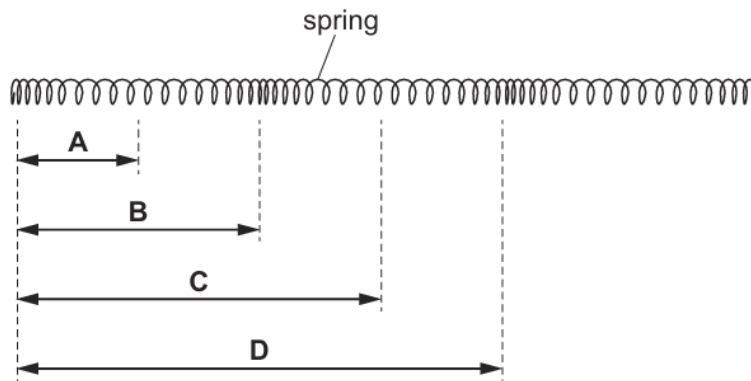
In which region of the electromagnetic spectrum are the waves?

- A infrared
- B microwave
- C ultraviolet
- D visible light

277. 9702/13/O/N/21/Q22

A longitudinal wave travels through a long spring. The spring is shown at one instant.

What is the wavelength of the wave?



278. 9702/13/O/N/21/Q23

Which statement about waves is correct?

- A** Both longitudinal and transverse waves can travel through a vacuum.
- B** Both longitudinal and transverse waves transfer matter.
- C** Longitudinal progressive waves consist of alternate nodes and antinodes.
- D** The particles of a transverse wave vibrate perpendicular to the direction of energy propagation.

279. 9702/13/O/N/21/Q25

A train travels at constant speed along a straight track. The train's horn emits sound of frequency 500 Hz.

A person standing by the side of the track hears sound of frequency 450 Hz.

The speed of sound in air is 340 m s^{-1} .

What is the speed of the train and in which direction is it travelling relative to the person?

	speed / m s^{-1}	direction
A	34	away from the person
B	34	towards the person
C	38	away from the person
D	38	towards the person

280. 9702/13/O/N/21/Q26

A smooth surface has bumps on the surface that are smaller than the wavelength of visible light.

What is the approximate maximum size of the largest bumps on the surface?

- A** 20 nm **B** 350 nm **C** 720 nm **D** $5.0 \mu\text{m}$

281. 9702/12/F/M/22/Q20

Which statement about progressive waves is correct?

- A They are always transverse waves.
- B They can exist in solids but not liquids.
- C They decrease in frequency as their speed increases.
- D They transfer energy away from their source.

282. 9702/12/F/M/22/Q22

The warning signal on an ambulance has a frequency of 600 Hz. The speed of sound is 330 m s^{-1} . The ambulance is travelling with a constant velocity of 25 m s^{-1} towards an observer. The ambulance passes, and then moves away from the observer with no change in velocity.



Which overall change in observed frequency takes place between the times at which the ambulance is a long way behind the observer and when it is a long way in front of the observer?

- A 49 Hz
- B 84 Hz
- C 91 Hz
- D 98 Hz

283. 9702/12/F/M/22/Q23

Brief pulses of red, blue and green light are emitted from the Sun at the same time.

The pulses travel the same distance to reach Mars. Assume that the pulses travel in a vacuum for the full duration of their journey.

In which order would these pulses of light arrive at Mars?

- A all arrive at the same time
- B blue first, then green, then red
- C red first, then blue, then green
- D red first, then green, then blue

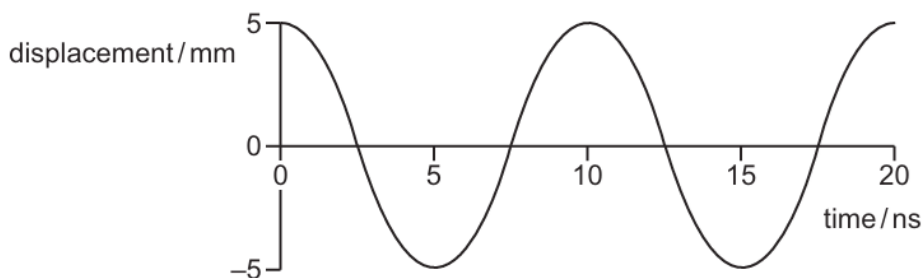
284. 9702/11/M/J/22/Q23

Which statement is correct?

- A Gases cannot transmit longitudinal waves.
- B Longitudinal sound waves cannot form stationary waves.
- C Solids can transmit both transverse and longitudinal waves.
- D Transverse waves cannot pass through a vacuum.

285. 9702/11/M/J/22/Q21

The graph shows the variation with time of the displacement of a particle as a progressive wave passes.



What are the frequency and the amplitude of the wave?

	frequency / MHz	amplitude / mm
A	100	5
B	200	5
C	100	10
D	200	10

286. 9702/11/M/J/22/Q24

A car is travelling at a constant velocity directly towards a man standing in the middle of the road.

The driver sounds the car's horn as a warning. The horn emits a sound wave of constant frequency.

The frequency of the sound heard by the man is different from the frequency of the sound emitted by the horn.

Which statement is correct?

- A** The frequency of the sound emitted by the horn is greater than the frequency of the sound heard by the man.
- B** The frequency of the sound heard by the man depends on the distance between the car and the man.
- C** The sound waves continually accelerate as they move from the horn to the man.
- D** The wavelength of the sound heard by the man is less than the wavelength of the sound emitted by the horn.

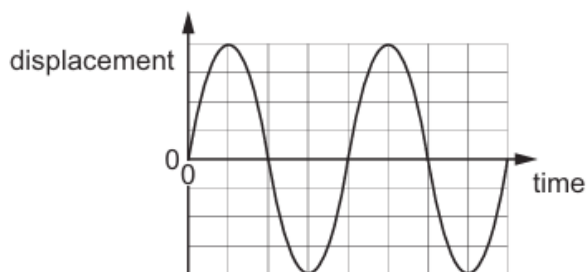
287. 9702/11/M/J/22/Q25

Which statement about electromagnetic waves is correct?

- A** A wave of wavelength 5.0×10^{-6} m is invisible to the human eye.
- B** They can all travel at different speeds in free space.
- C** They cannot be polarised.
- D** They consist of vibrating atoms.

288. 9702/11/M/J/22/Q22

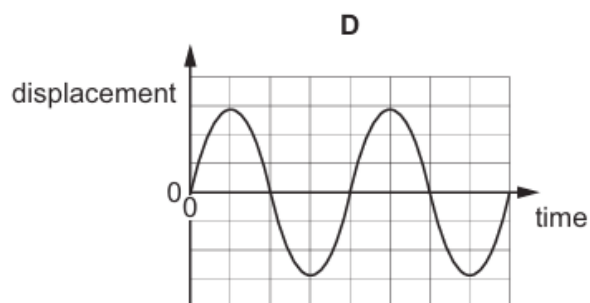
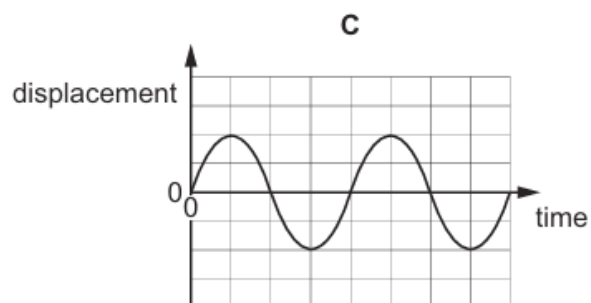
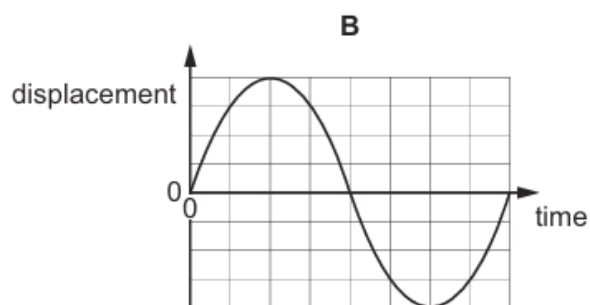
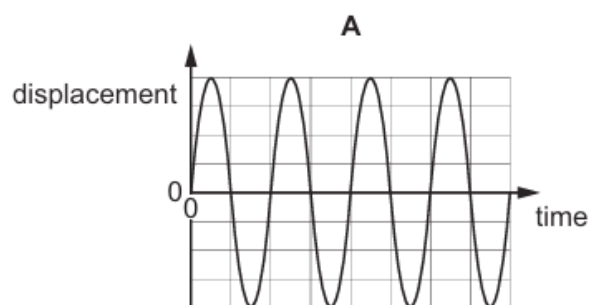
The graph shows the variation of the displacement of an air particle with time as a sound wave passes through air.



The intensity of the sound is halved while the frequency remains constant.

The four graphs below are drawn to the same scale as the graph above.

Which graph shows the displacement of the air particle?



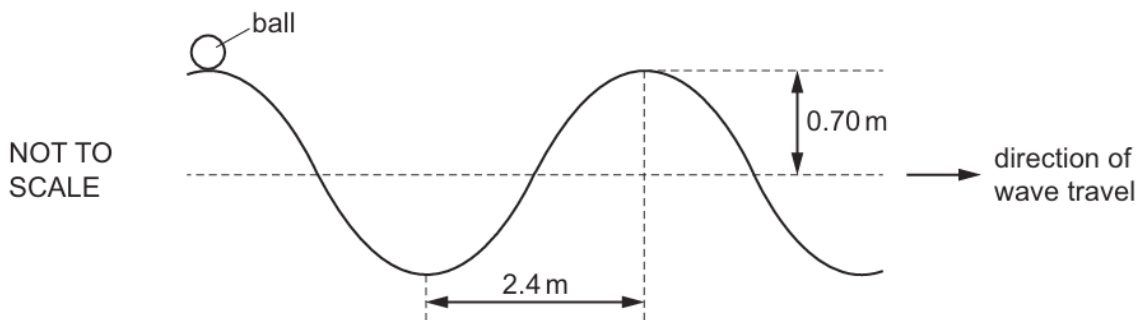
289. 9702/12/M/J/22/Q21

Which statement is correct for **all** types of progressive wave?

- A** The distance from a peak to the next trough is equal to a wavelength.
- B** They can be demonstrated in ripple tanks.
- C** They consist of vibrating atoms.
- D** They transfer energy from one position to another.

290. 9702/12/M/J/22/Q22

A transverse water wave is moving along the surface of some water. This causes a ball to move vertically without moving horizontally as it floats upon the surface. At one instant, the ball is at the position shown.



The wave has a frequency of 0.20 Hz and an amplitude of 0.70 m. The distance between a trough and an adjacent peak is 2.4 m.

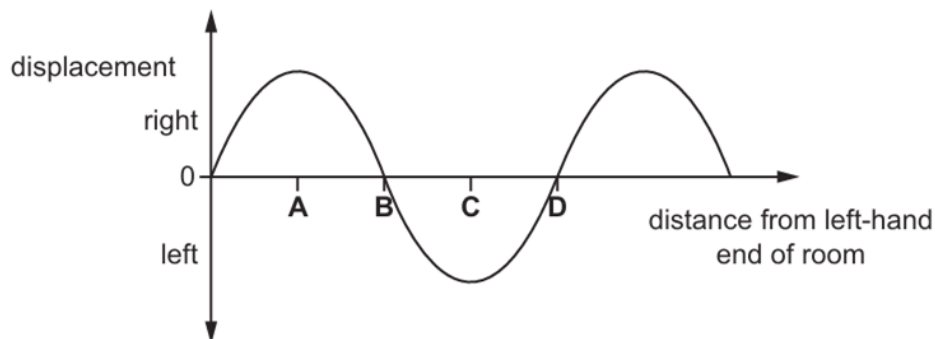
What is the distance travelled by the ball in a time of 20 s?

- A** 5.6 m **B** 9.6 m **C** 11.2 m **D** 19.2 m

291. 9702/12/M/J/22/Q23

A sound wave travels from left to right across a room. The variation with distance across the room of the displacement of the air molecules at one instant is shown.

At which distance will the air pressure be lowest?



292. 9702/12/M/J/22/Q24

A source emits a sound wave of a single frequency. The Doppler effect causes a different frequency of sound to be heard by a stationary observer.

What is a requirement for the Doppler effect to occur?

- A** a source that is moving as it produces the sound wave
B a source that produces a polarised sound wave
C a source that produces a sound wave of changing amplitude
D a source that produces a sound wave of changing frequency

293. 9702/12/M/J/22/Q25

An electromagnetic wave in free space has a frequency of 2.5×10^{14} Hz.

Which region of the electromagnetic spectrum includes this frequency?

- A infrared
- B microwave
- C ultraviolet
- D X-ray

294. 9702/12/M/J/22/O26

Two polarising filters are placed next to each other so that their planes are parallel.

The first polarising filter has its transmission axis at an angle of 50° to the vertical.

The second polarising filter has its transmission axis at an angle of 20° to the vertical. The angle between the transmission axes of the two polarising filters is 30° .

A beam of vertically polarised light of intensity 8.0 W m^{-2} is incident normally on the first polarising filter.

What is the intensity of the light that is transmitted from the second polarising filter?

- A zero B 2.5 W m^{-2} C 2.9 W m^{-2} D 6.0 W m^{-2}

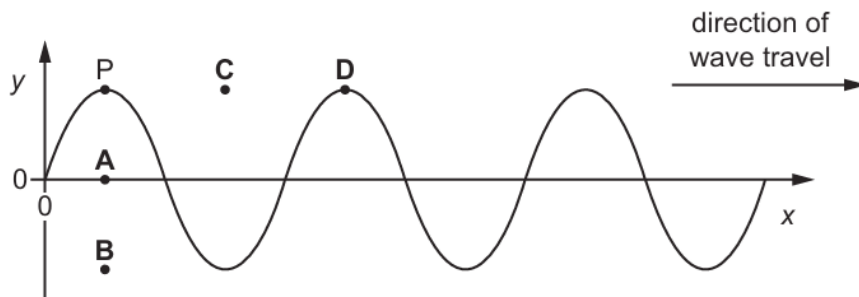
295. 9702/13/M/J/22/Q22

The variation with distance x of the displacement y of a transverse wave on a rope is shown at time $t = 0$.

The wave has a frequency of 0.5 Hz.

A point P on the rope is marked. The diagram shows the original position of P and four new positions.

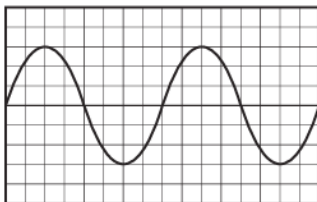
What is the position of P at time $t = 1 \text{ s}$?



296. 9702/13/M/J/22/Q23

A cathode-ray oscilloscope (CRO) is connected to a microphone which detects sound of constant frequency.

The trace on the screen of the CRO is shown.



Which property of the sound wave is measured using **only** information from the CRO?

- A amplitude
- B period
- C speed
- D wavelength

297. 9702/13/M/J/22/Q24

A transverse wave and a longitudinal wave both travel in the same direction down a long stretched spring.

Which statement is **not** correct for these two forms of wave?

- A The displacement measurements for the particles of the two waves are made at right angles to each other.
- B The energy transferred by the two waves is in the same direction.
- C The velocities of the two waves are in the same direction.
- D The wavelength measurements for the two waves are made at right angles to each other.

298. 9702/13/M/J/22/Q25

A man standing next to a stationary train hears sound of frequency 400 Hz emitted from the train's horn. The train then moves directly away from the man and sounds its horn when it has a speed of 50 m s^{-1} . The speed of sound in the air is 340 m s^{-1} .

What is the difference in frequency of the sound heard by the man on the two occasions?

- A 51 Hz B 69 Hz C 350 Hz D 470 Hz

299. 9702/13/M/J/22/Q26

Which list of regions of the electromagnetic spectrum is in order of increasing wavelength from left to right?

- A gamma-ray $\rightarrow$ ultraviolet $\rightarrow$ infrared
- B infrared $\rightarrow$ microwave $\rightarrow$ ultraviolet
- C microwave $\rightarrow$ X-ray $\rightarrow$ infrared
- D X-ray $\rightarrow$ ultraviolet $\rightarrow$ gamma-ray

300. 9702/11/O/N/22/Q21

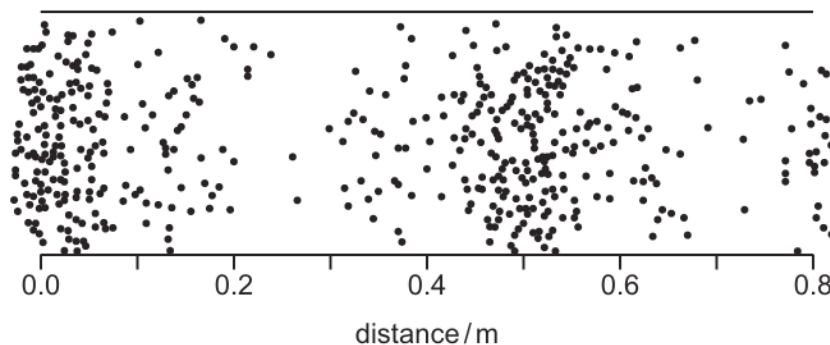
A signal generator, amplifier and loudspeaker are used to produce different sound waves in the air of a room. The relationships between the properties of these waves are investigated.

Which relationship is **not** correct?

- A Amplitude is proportional to wavelength.
- B Frequency is inversely proportional to wavelength.
- C Intensity is proportional to amplitude squared at a given frequency.
- D Period is equal to the reciprocal of frequency.

301. 9702/11/O/N/22/Q22

When a guitar string is plucked, it causes a longitudinal sound wave in the air, as shown.



The speed of sound in the air is 340 m s^{-1} .

What is the approximate frequency of the sound wave shown?

- A 430 Hz B 680 Hz C 1100 Hz D 1400 Hz

302. 9702/11/O/N/22/Q24

A jet aircraft travels at a speed of $0.80v$, where v is the speed of sound. The aircraft directly approaches a stationary observer. The frequency of sound emitted by the aircraft is 100 Hz.

Which frequency does the observer hear?

- A 56 Hz B 180 Hz C 400 Hz D 500 Hz

303. 9702/11/O/N/22/Q25

A telescope detects and analyses some electromagnetic radiation of wavelength 2 cm.

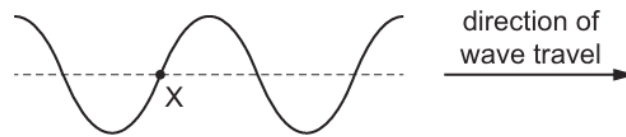
Which type of telescope is it?

- A microwave telescope
- B optical telescope
- C radio telescope
- D X-ray telescope

304. 9702/11/O/N/22/Q23

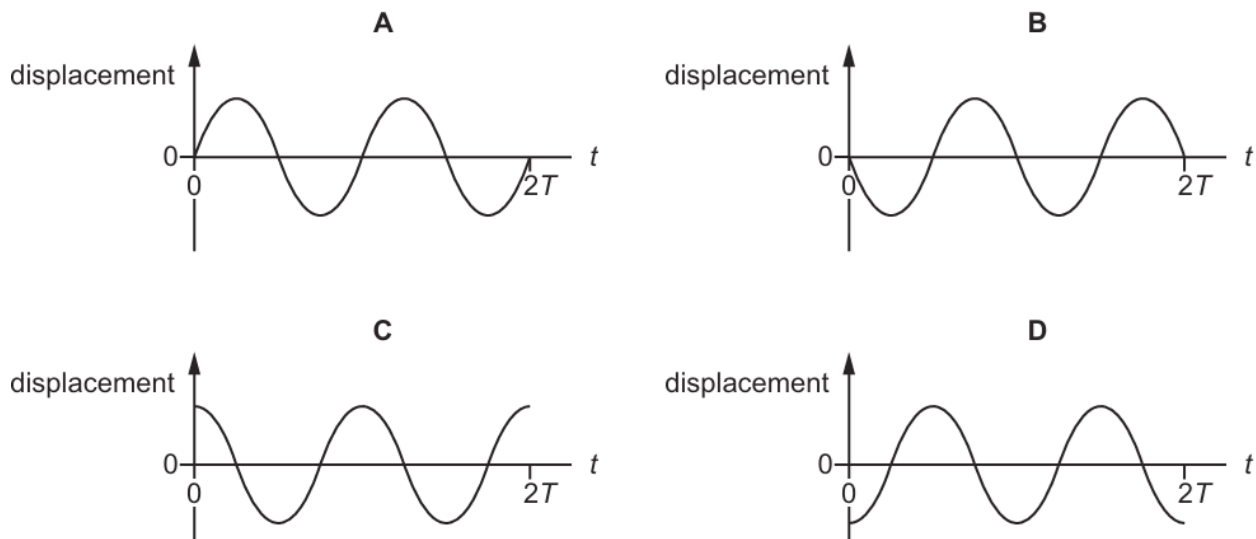
A transverse wave travels along a rope. The diagram shows the rope at time $t = 0$.

The wave is travelling from left to right. The period of the wave is T .



One particle of the rope is labelled X.

Which graph shows the variation with time of the displacement of particle X between $t = 0$ and $t = 2T$?

**305. 9702/12/O/N/22/Q23**

Which statement about progressive transverse and longitudinal waves is correct?

- A** Particles in a transverse wave have fixed equilibrium positions but those in longitudinal waves do not.
- B** Transverse waves can be polarised but longitudinal waves cannot.
- C** Transverse waves transfer energy but longitudinal waves do not.
- D** Two-source interference can be demonstrated with transverse waves but not with longitudinal waves.

306. 9702/12/O/N/22/Q24

A miniature loudspeaker, initially at rest, falls vertically from a window in a high building. When the speaker has fallen a distance of 10.0 m, it emits a very short pulse of sound of constant frequency 256 Hz in all directions. The pulse of sound, travelling at a speed of 330 m s^{-1} , is heard by a person leaning out of the window.

Air resistance is negligible.

What is the frequency of the pulse of sound heard by the person?

- A** 246 Hz
- B** 249 Hz
- C** 267 Hz
- D** 313 Hz

307. 9702/12/O/N/22/Q25

Two electromagnetic waves have wavelengths of $5.0 \times 10^{-7} \text{ m}$ and $5.0 \times 10^{-2} \text{ m}$ in a vacuum.

Which row identifies the regions of the electromagnetic spectrum to which the waves belong?

	wavelength $5.0 \times 10^{-7} \text{ m}$	wavelength $5.0 \times 10^{-2} \text{ m}$
A	ultraviolet	infrared
B	visible	microwave
C	ultraviolet	microwave
D	visible	infrared

308. 9702/13/O/N/22/Q22

The period of an electromagnetic wave in a vacuum is 1.0 ns.

What are the frequency and wavelength of the wave?

	frequency / Hz	wavelength / m
A	1.0	3.0×10^8
B	1.0×10^6	300
C	1.0×10^9	0.30
D	1.0×10^{12}	3.0×10^{-4}

309. 9702/13/O/N/22/Q23

An observer is standing on a railway platform. A train passes the observer at constant speed while emitting sound of constant frequency f from its whistle.

What does the observer hear?

- A** sound of a decreasing frequency as the train approaches and of an increasing frequency as it moves away
- B** sound of a higher frequency than f as the train approaches and of a lower frequency than f as the train moves away
- C** sound of a lower frequency than f as the train approaches and of a higher frequency than f as the train moves away
- D** sound of an increasing frequency as the train approaches and as it moves away

310. 9702/13/O/N/22/Q24

Which microorganisms have a length that is equal to the wavelength in free space of an electromagnetic wave that is visible to the human eye?

- A** algae of length $0.5 \mu\text{m}$
- B** bacteria of length $5.0 \mu\text{m}$
- C** fungi of length $50 \mu\text{m}$
- D** protozoa of length $100 \mu\text{m}$

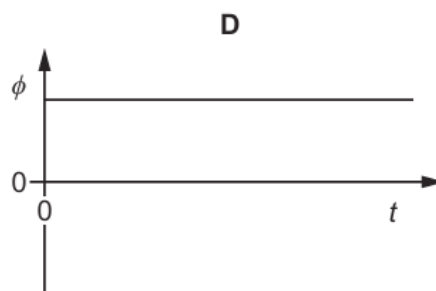
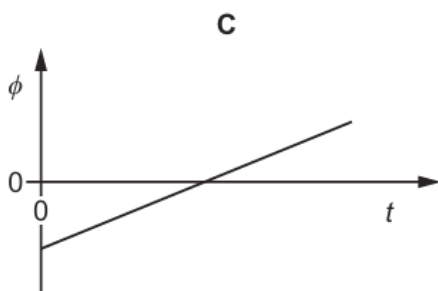
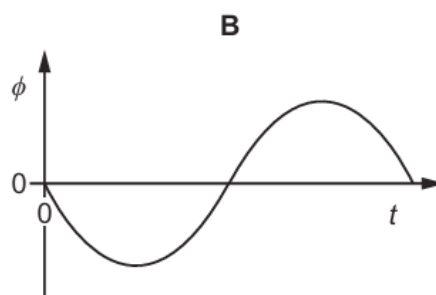
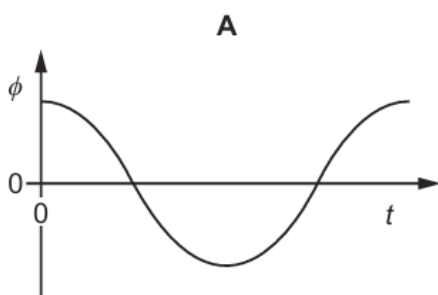
311. 9702/12/F/M/23/Q20

In a progressive water wave, two particles, P and Q, on the surface of the water, are a fixed horizontal distance apart. P and Q oscillate vertically.

At time $t = 0$, the wave is as shown.



Which graph best represents the variation with time t of the phase difference ϕ between the oscillation of the water particle P and the oscillation of the water particle Q?



312. 9702/12/F/M/23/Q21

Which statement about longitudinal waves and transverse waves is **not** correct?

- A** Both waves can be polarised.
- B** Both waves can form stationary waves.
- C** Both waves can transfer energy as progressive waves.
- D** Both waves obey the equation $v = f\lambda$.

313. 9702/12/F/M/23/Q23

What is the approximate range of frequencies of electromagnetic radiation visible to the human eye?

- A** (430–750) kHz
- B** (430–750) MHz
- C** (430–750) GHz
- D** (430–750) THz

314. 9702/12/F/M/23/Q22

An observer hears a sound wave emitted from a moving source.

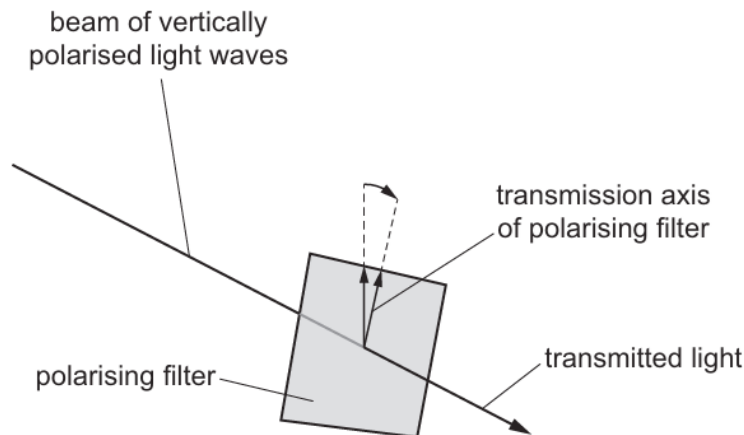
The observed frequency is less than the frequency of sound emitted from the source.

What could be the reason for this?

- A** The source is moving away from the observer.
- B** The source is moving towards the observer.
- C** The speed of the sound wave in air decreases due to the movement of the source.
- D** The speed of the sound wave in air increases due to the movement of the source.

315. 9702/12/F/M/23/Q24

A beam of vertically polarised light is incident normally on a polarising filter. The filter can be rotated so that it is always in a plane perpendicular to the beam. The transmission axis of the filter is initially vertical.



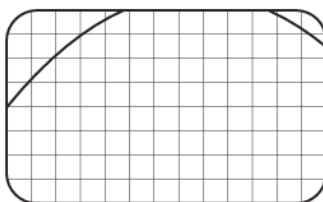
The filter is first rotated clockwise by an angle of 30° so that the transmitted light waves have intensity I_{30} . The filter is then rotated clockwise by a further angle of 30° .

What is the new intensity of the transmitted light waves?

- A** $0.25 I_{30}$
- B** $0.33 I_{30}$
- C** $0.75 I_{30}$
- D** $0.87 I_{30}$

316. 9702/11/M/J/23/20

In an experiment, a student uses a microphone and a cathode-ray oscilloscope (CRO) to analyse a sound wave. The diagram shows the trace on the screen of the CRO.



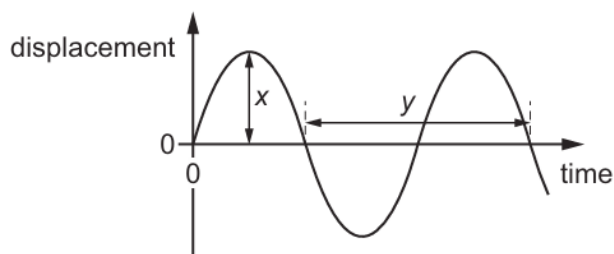
The student is expecting a sinusoidal waveform to be shown on the screen.

Which changes should the student make to the time-base and the y-gain of the CRO so that the screen shows a continuous trace for one complete cycle of the waveform?

	time-base	y-gain
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

317. 9702/11/M/J/23/21

The graph shows the variation with time of the displacement of a particle in a progressive wave.



Two measurements, x and y , are labelled on the graph.

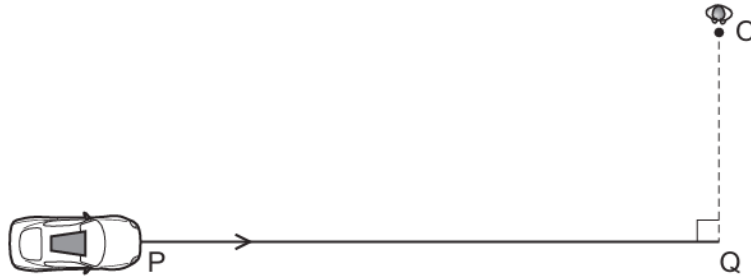
What do x and y represent?

	x	y
A	amplitude	period
B	frequency	period
C	amplitude	wavelength
D	frequency	wavelength

318. 9702/11/M/J/23/22

A car travels at a constant speed along a straight line PQ.

A loudspeaker attached to the car emits sound of constant frequency f . A stationary observer is at point O.



What does the observer hear as the car moves from P towards Q?

- A** a frequency less than f that decreases as the car moves from P towards Q
- B** a frequency less than f that increases as the car moves from P towards Q
- C** a frequency more than f that decreases as the car moves from P towards Q
- D** a frequency more than f that increases as the car moves from P towards Q

319. 9702/11/M/J/23/23

Some sources of electromagnetic waves in free space are listed.

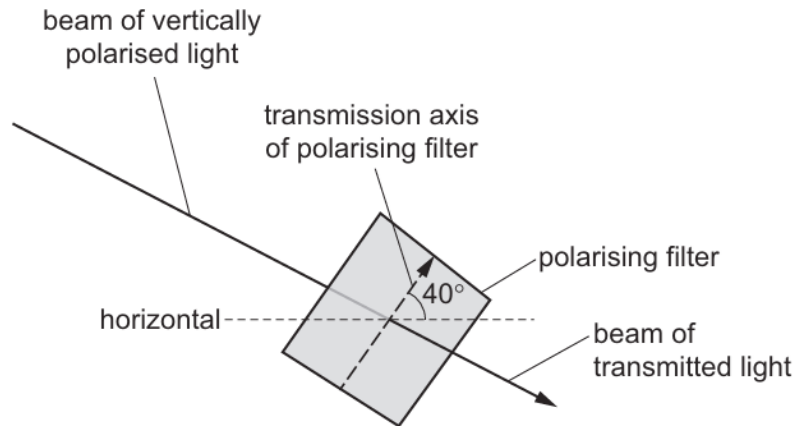
- 1 a radio wave transmitter
- 2 a source of X-rays
- 3 a 30 mm wavelength radar transmitter
- 4 a light-emitting diode that emits red light

Which list gives the sources in order of increasing wavelength, from left to right, of the waves emitted by the sources?

- A** 1 → 3 → 4 → 2
- B** 2 → 4 → 1 → 3
- C** 2 → 4 → 3 → 1
- D** 3 → 1 → 4 → 2

320. 9702/11/M/J/23/24

A vertically polarised beam of light is incident normally on a polarising filter. The transmission axis of the filter is at an angle of 40° to the horizontal.



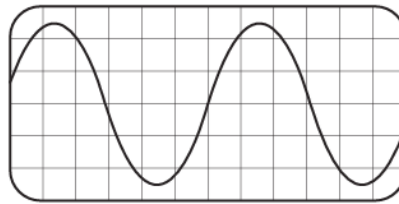
What is the ratio $\frac{\text{amplitude of transmitted beam}}{\text{amplitude of incident beam}}$?

- A** 0.41 **B** 0.59 **C** 0.64 **D** 0.77

321. 9702/12/M/J/23/Q20

A microphone detects a sound wave. The microphone is connected to a cathode-ray oscilloscope (CRO).

The shape of the trace on the screen of the CRO is shown.



Which property of the sound wave can be determined by using only the measurement of a horizontal distance on the screen and the value of a control setting of the CRO?

- A** amplitude
B frequency
C speed
D wavelength

322. 9702/12/M/J/23/Q23

A beam of visible light is in a vacuum.

What could be the frequency of the light?

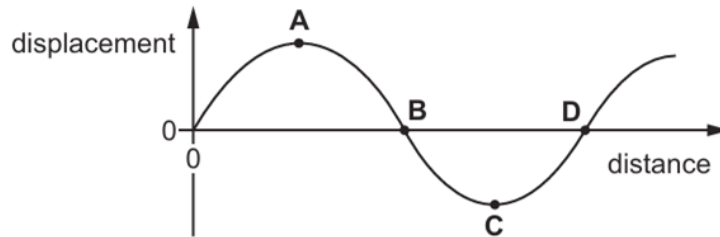
- A** $5.0 \times 10^5 \text{ Hz}$ **B** $5.0 \times 10^8 \text{ Hz}$ **C** $5.0 \times 10^{11} \text{ Hz}$ **D** $5.0 \times 10^{14} \text{ Hz}$

323. 9702/12/M/J/23/Q21

A longitudinal wave is travelling from left to right. The graph shows the variation of the displacement of the particles with distance along the wave at one instant in time.

Displacements to the right are positive; displacements to the left are negative.

Which labelled point represents a compression?



324. 9702/12/M/J/23/Q22

A source X emits a sound wave of constant frequency f .

The wave is subsequently received at a stationary detector Y.

The frequency of the wave that is detected by Y is less than f .

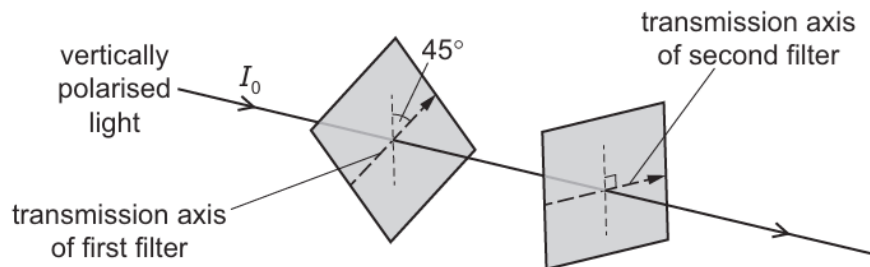
What could be the reason for this?

- A** Between X and Y, the wave undergoes diffraction.
- B** Between X and Y, the wave undergoes reflection.
- C** X is moving away from Y.
- D** X is moving towards Y.

325. 9702/12/M/J/23/Q24

A vertically polarised beam of light of intensity I_0 is incident normally on a polarising filter.

The transmission axis of the filter is at 45° to the vertical. The beam of light transmitted by this filter is then incident normally on a second filter. The transmission axis of the second filter is horizontal.



What is the intensity of the beam of light after transmission through the second filter?

- A** 0
- B** $\frac{1}{4} I_0$
- C** $\frac{1}{2} I_0$
- D** I_0

326. 9702/13/M/J/23/Q20

A progressive wave of frequency 1.5 kHz travels in a medium at a speed of 340 m s^{-1} .

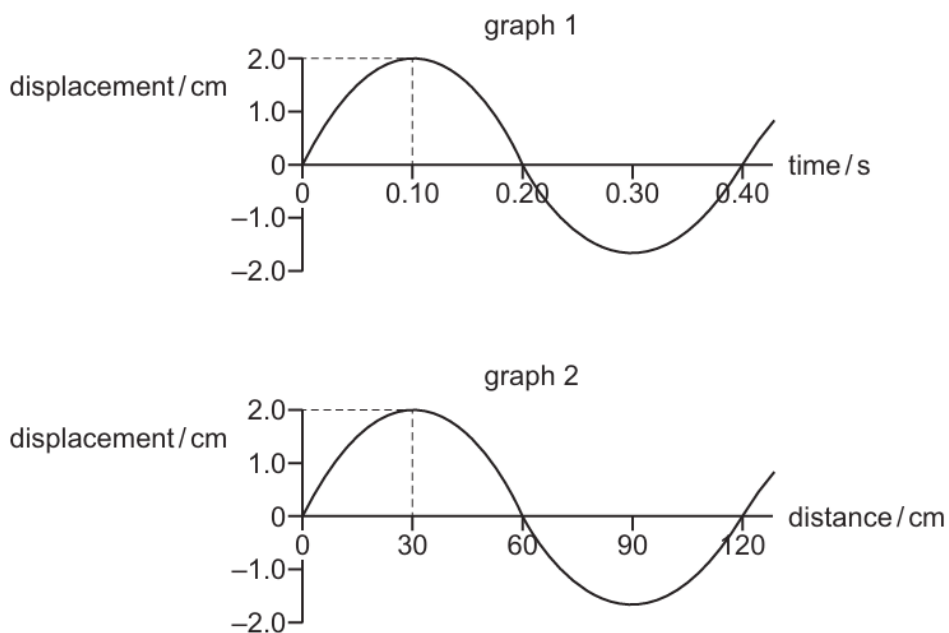
What is the minimum distance between two points on the wave that have a phase difference of 70° ?

- A** 4.4 cm **B** 8.8 cm **C** 18 cm **D** 23 cm

327. 9702/13/M/J/23/Q21

Graph 1 shows the variation with time of displacement at a fixed distance along a progressive wave.

Graph 2 represents the same wave and shows the variation with distance of displacement at an instant in time.



What is the speed of the wave?

- A** 5.0 cm s^{-1} **B** 48 cm s^{-1} **C** 150 cm s^{-1} **D** 300 cm s^{-1}

328. 9702/13/M/J/23/Q22

A vehicle moves with constant velocity along a road directly towards an observer. The observed frequency of the sound from the vehicle changes as the vehicle moves past the observer.

Which phenomenon explains the change in frequency?

- A** diffraction
B interference
C polarisation
D the Doppler effect

329. 9702/13/M/J/23/Q23

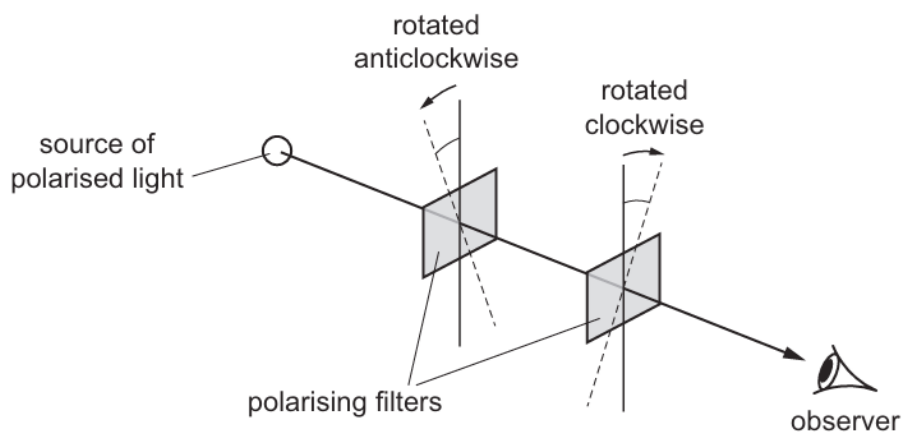
An electromagnetic wave has a wavelength of 2.1 cm in a vacuum.

Which region of the electromagnetic spectrum contains this wave?

- A** infrared
- B** microwave
- C** radio wave
- D** visible light

330. 9702/13/M/J/23/Q24

A source of plane polarised light is observed through two polarising filters.



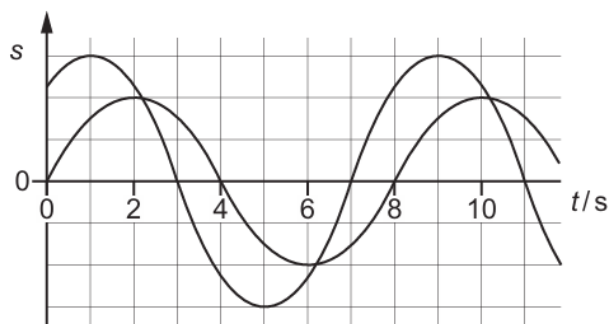
The filters are positioned so that the source appears at its brightest. One of the filters is then rotated clockwise and the other filter is rotated anticlockwise through the same angle.

How does the source appear when both filters have been rotated 90° and 180° from their initial positions?

	90°	180°
A	brightest	brightest
B	brightest	darkest
C	darkest	brightest
D	darkest	darkest

331. 9702/11/O/N/23/Q20

Two waves pass through a point P. The graph shows the variation with time t of the displacement s of the two waves at point P.



What is the phase difference between the two waves at point P?

- A** 0° **B** 45° **C** 90° **D** 180°

332. 9702/11/O/N/23/Q21

Which row is correct for both progressive transverse waves and progressive longitudinal waves?

	transverse waves	longitudinal waves
A	contain compressions and rarefactions	some can travel in a vacuum
B	can be polarised	contain compressions and rarefactions
C	vibrations are perpendicular to the direction of travel of the wave energy	can be polarised
D	some can travel in a vacuum	vibrations are perpendicular to the direction of travel of the wave energy

333. 9702/11/O/N/23/Q23

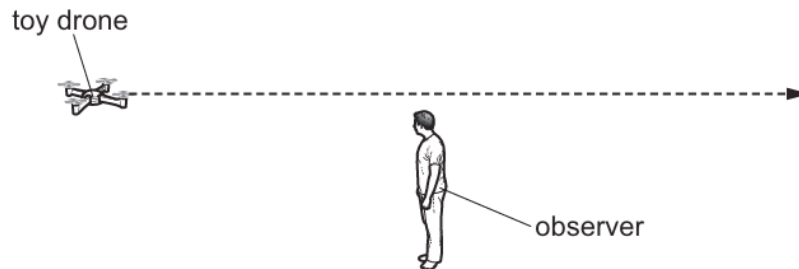
Which statement about electromagnetic waves in a vacuum is correct?

- A** Infrared waves have shorter wavelengths than visible light waves.
B Microwaves have longer wavelengths than radio waves.
C Ultraviolet waves have higher frequencies than visible light waves.
D γ -rays have lower frequencies than X-rays.

334. 9702/11/O/N/23/Q22

A toy drone emits a sound of constant frequency 800 Hz. The speed of the sound in the air is 330 m s^{-1} .

The drone moves along a straight path directly towards an observer and then continues in a straight line directly away from the observer. The speed of the drone is constant.



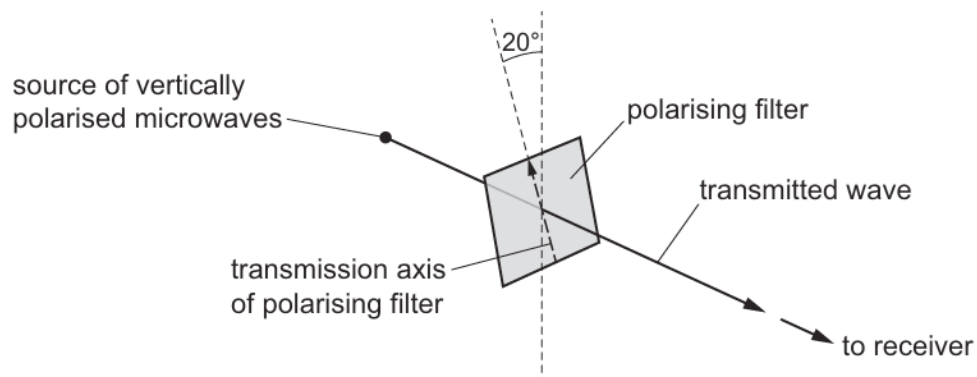
What is the velocity of the drone when the frequency of the sound heard by the observer is 850 Hz?

	magnitude of velocity / m s^{-1}	direction of velocity
A	19	away from the observer
B	21	away from the observer
C	19	towards the observer
D	21	towards the observer

335. 9702/11/O/N/23/Q24

Vertically polarised microwaves are emitted from a source. The microwaves are detected by a receiver that is connected to a cathode-ray oscilloscope (CRO). The waveform displayed on the screen of the CRO has an amplitude of 2.6 cm.

A metal wire grid that acts as a polarising filter is now placed between the source and the receiver. The filter is orientated so that the plane of polarisation of the transmitted wave is at an angle of 20° to the vertical.



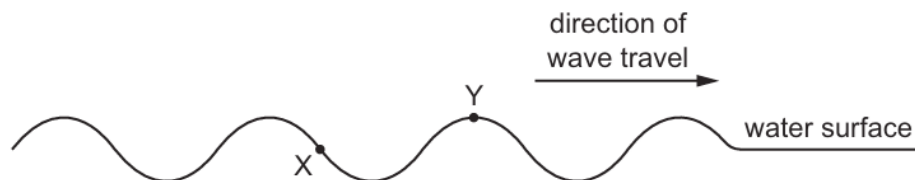
The distance between the source and receiver is unchanged. The settings on the CRO are also unchanged.

What is now the amplitude of the waveform displayed on the screen of the CRO?

- A** 0.30 cm **B** 0.89 cm **C** 2.3 cm **D** 2.4 cm

336. 9702/12/O/N/23/Q20

X and Y are two points on the surface of water in a ripple tank. A source of constant frequency generates a wave which travels past X and Y, causing them to oscillate vertically.



What is the phase difference between X and Y?

- A** 45° **B** 135° **C** 180° **D** 270°

337. 9702/12/O/N/23/Q22

A source of sound waves is moving at a constant speed directly towards a stationary observer.

The sound waves have a speed of 340 ms^{-1} and a frequency of 480 Hz. The observer hears sound waves of frequency 650 Hz.

What is the speed of the source?

- A** 89 ms^{-1} **B** 120 ms^{-1} **C** 250 ms^{-1} **D** 340 ms^{-1}

338. 9702/12/O/N/23/Q23

A student is investigating two electromagnetic waves, X and Y, in a vacuum.

Wave X has a wavelength of $5.2 \times 10^{-7} \text{ m}$. Wave Y has a frequency of 9.4 GHz.

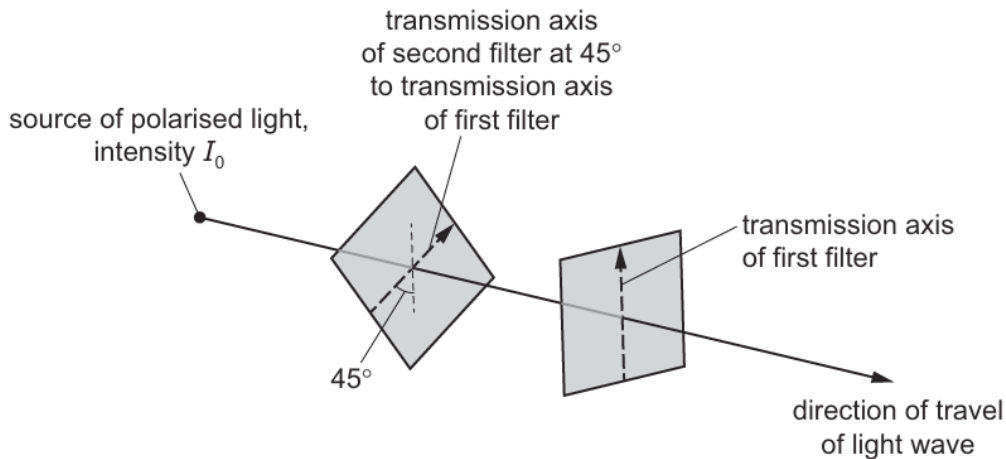
Which principal regions of the electromagnetic spectrum contain waves X and Y?

	X	Y
A	radio wave	ultraviolet
B	ultraviolet	visible
C	visible	microwave
D	microwave	radio wave

339. 9702/12/O/N/23/Q24

A plane polarised light wave of intensity I_0 is incident normally on a polarising filter. The initial intensity of the transmitted wave is 0.

A second polarising filter is then inserted between the source and the first filter. Its transmission axis is at 45° to the transmission axis of the first filter, as shown.



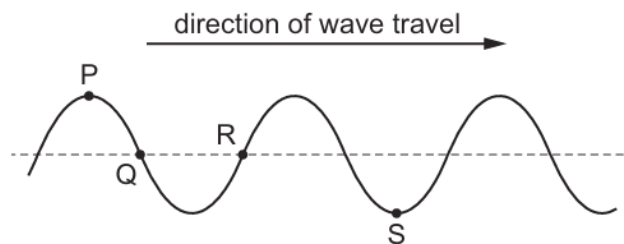
What is the intensity of the transmitted wave from the filter combination?

- A 0 B $\frac{I_0}{8}$ C $\frac{I_0}{4}$ D $\frac{I_0}{2}$

340. 9702/13/O/N/23/Q21

A wave travels on the surface of water. P, Q, R and S are four particles of water on the surface.

The diagram shows the positions of the particles at one instant. The direction of travel of the wave is from left to right.



Which two particles are about to move upwards?

- A P and R B P and S C R and S D Q and S

341. 9702/13/O/N/23/Q22

A loudspeaker emits sound of frequency f_s . The loudspeaker is attached to a car that moves with increasing speed directly towards a stationary observer.

Which statement describes the frequency of the sound heard by the observer?

- A a frequency greater than f_s and increasing
 B a frequency greater than f_s but decreasing
 C a frequency less than f_s and decreasing
 D a frequency less than f_s but increasing

342. 9702/13/O/N/23/Q23

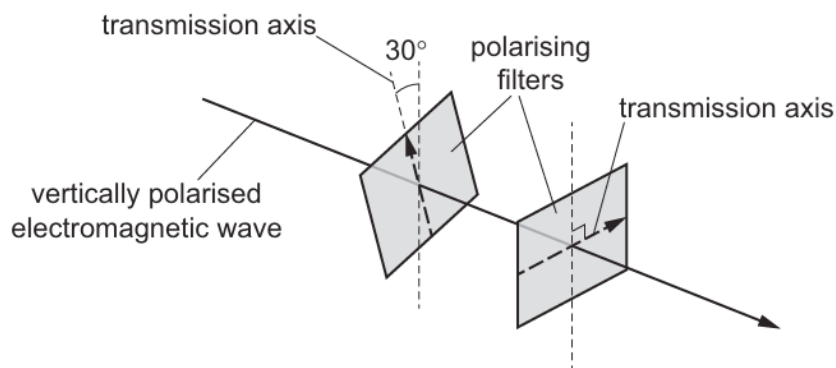
Which statement about electromagnetic waves in a vacuum is correct?

- A** Amplitude is inversely proportional to velocity.
- B** Frequency is inversely proportional to wavelength.
- C** Intensity is proportional to amplitude.
- D** Velocity is proportional to wavelength.

343. 9702/13/O/N/23/Q24

A vertically polarised electromagnetic wave of intensity I_0 is incident normally on a polarising filter. The transmission axis of the filter is at an angle of 30° to the vertical.

The transmitted wave from the first filter is then incident normally on a second polarising filter. The transmission axis of this filter is at an angle of 90° to the vertical.



What is the intensity of the wave after passing through the second filter?

- A** 0
- B** $0.063 I_0$
- C** $0.19 I_0$
- D** $0.56 I_0$

ANSWERS

1. C	48. D	95. D	142. D	189. D	234. A
2. B	49. D	96. B	143. B	190. B	235. B
3. B	50. B	97. A	144. D	191. D	236. D
4. A	51. A	98. C	145. B	192. D	237. D
5. D	52. D	99. C	146. C	193. B	238. A
6. B	53. C	100. C	147. D	194. B	239. C
7. D	54. B	101. B	148. C	195. D	240. A
8. D	55. B	102. D	149. D	196. B	241. B
9. C	56. B	103. C	150. D	197. A	242. C
10. B	57. B	104. C	151. B	198. A	243. D
11. A	58. C	105. A	152. C	199. D	244. D
12. D	59. C	106. D	153. D	200. B	245. C
13. D	60. A	107. B	154. D	201. A	246. A
14. A	61. C	108. D	155. C	202. D	247. D
15. B	62. A	109. B	156. D	203. D	248. B
16. A	63. A	110. C	157. C	204. C	249. B
17. B	64. C	111. C	158. C	205. C	250. A
18. D	65. C	112. C	159. D	206. C	251. D
19. C	66. A	113. C	160. C	207. B	252. B
20. C	67. D	114. B	161. B	208. D	253. B
21. C	68. D	115. B	162. A	209. A	254. C
22. A	69. B	116. D	163. B	210. C	255. D
23. B	70. A	117. D	164. D	211. C	256. B
24. B	71. B	118. A	165. C	212. D	257. D
25. D	72. D	119. A	166. B	213. D	258. A
26. B	73. C	120. D	167. D	214. C	259. A
27. B	74. D	121. C	168. A	215. B	260. B
28. B	75. A	122. C	169. D	216. B	261. A
29. C	76. A	123. D	170. D	217. C	262. B
30. A	77. B	124. D	171. A	218. B	263. A
31. B	78. B	125. B	172. C	219. B	264. B
32. B	79. D	126. A	173. D	220. A	265. B
33. C	80. B	127. C	174. B	221. B	266. A
34. C	81. B	128. B	175. D	222. A	267. A
35. B	82. C	129. C	176. B	223. B	268. D
36. C	83. C	130. D	177. D	224. D	269. B
37. D	84. B	131. D	178. A	225. D	270. D
38. A	85. C	132. D	179. C	226. A	271. B
39. C	86. B	133. B	180. C	227. A	272. A
40. A	87. C	134. C	181. A	228. B	273. D
41. D	88. A	135. B	182. B	229. C	274. D
42. D	89. A	136. B	183. D	230. A	275. C
43. B	90. B	137. C	184. D	231. C	276. A
44. B	91. B	138. B	185. C	232. A	277. B
45. B	92. C	139. C	186. A	233. D	278. D
46. B	93. C	140. B	187. D		
47. D	94. A	141. A	188. A		

279. C	326. A
280. B	327. D
281. D	328. D
282. C	329. B
283. A	330. C
284. C	331. B
285. A	332. B
286. D	333. C
287. A	334. C
288. D	335. D
289. D	336. D
290. C	337. A
291. D	338. C
292. A	339. C
293. A	340. D
294. B	341. A
295. B	342. B
296. B	343. C
297. D	
298. A	
299. A	
300. A	
301. B	
302. D	
303. A	
304. B	
305. B	
306. A	
307. B	
308. C	
309. B	
310. A	
311. D	
312. A	
313. D	
314. A	
315. B	
316. D	
317. A	
318. C	
319. C	
320. C	
321. B	
322. D	
323. B	
324. C	
325. B	

1 (a) State what is meant by a *progressive wave*.

9702/21/O/N/09/Q5

.....
..... [2]

(b) The variation with distance x along a progressive wave of a quantity y , at a particular time, is shown in Fig. 5.1.

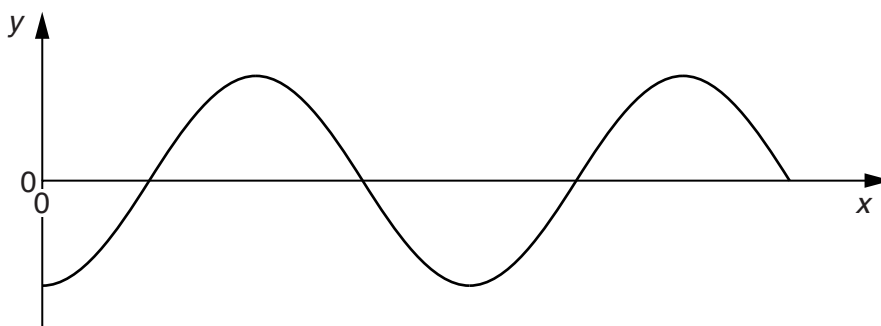


Fig. 5.1

(i) State what the quantity y could represent.

.....
..... [1]

(ii) Distinguish between the quantity y for

1. a transverse wave,

..... [1]

2. a longitudinal wave.

..... [1]

(c) The wave nature of light may be demonstrated using the phenomena of diffraction and interference.

Outline how diffraction and how interference may be demonstrated using light.

In each case, draw a fully labelled diagram of the apparatus that is used and describe what is observed.

diffraction

.....
.....

interference

.....
..... [6]

- 2 A uniform string is held between a fixed point P and a variable-frequency oscillator, as shown in Fig. 5.1.

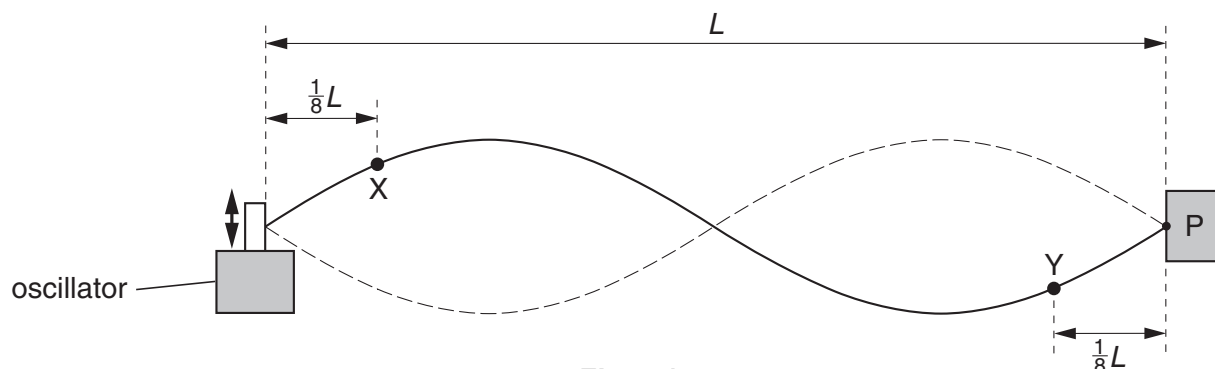


Fig. 5.1

The distance between point P and the oscillator is L .

The frequency of the oscillator is adjusted so that the stationary wave shown in Fig. 5.1 is formed.

Points X and Y are two points on the string.

Point X is a distance $\frac{1}{8}L$ from the end of the string attached to the oscillator. It vibrates with frequency f and amplitude A .

Point Y is a distance $\frac{1}{8}L$ from the end P of the string.

- (a) For the vibrations of point Y, state

- (i) the frequency (in terms of f),

frequency = [1]

- (ii) the amplitude (in terms of A).

amplitude = [1]

- (b) State the phase difference between the vibrations of point X and point Y.

phase difference = [1]

- (c) (i) State, in terms of f and L , the speed of the wave on the string.

speed = [1]

- (ii) The wave on the string is a stationary wave.

Explain, by reference to the formation of a stationary wave, what is meant by the speed stated in (i).

.....

 [3]

- 3 (a) State two features of a stationary wave that distinguish it from a progressive wave.

1.

.....

2.

..... [2]

- (b) A long tube is open at one end. It is closed at the other end by means of a piston that can be moved along the tube, as shown in Fig. 4.1.

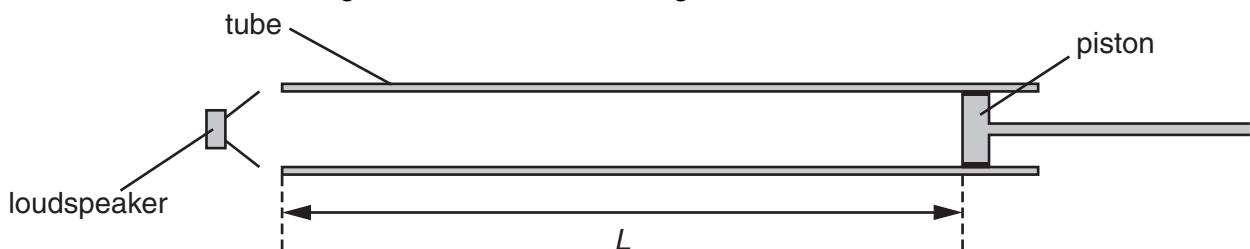


Fig. 4.1

A loudspeaker producing sound of frequency 550 Hz is held near the open end of the tube.

The piston is moved along the tube and a loud sound is heard when the distance L between the piston and the open end of the tube is 45 cm.

The speed of sound in the tube is 330 m s^{-1} .

- (i) Show that the wavelength of the sound in the tube is 60 cm.

[1]

- (ii) On Fig. 4.1, mark all the positions along the tube of

1. the displacement nodes (label these with the letter N),
2. the displacement antinodes (label these with the letter A).

[3]

- (c) The frequency of the sound produced by the loudspeaker in (b) is gradually reduced.

Determine the lowest frequency at which a loud sound will be produced in the tube of length $L = 45 \text{ cm}$.

frequency = Hz [3]

- 4 A student is studying a water wave in which all the wavefronts are parallel to one another. The variation with time t of the displacement x of a particular particle in the wave is shown in Fig. 5.1.

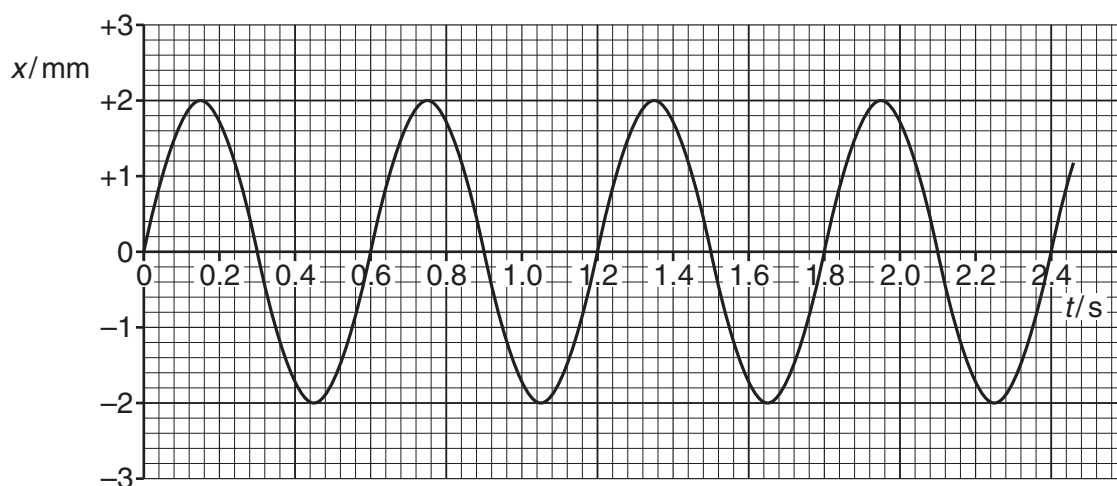


Fig. 5.1

The distance d of the oscillating particles from the source of the waves is measured. At a particular time, the variation of the displacement x with this distance d is shown in Fig. 5.2.

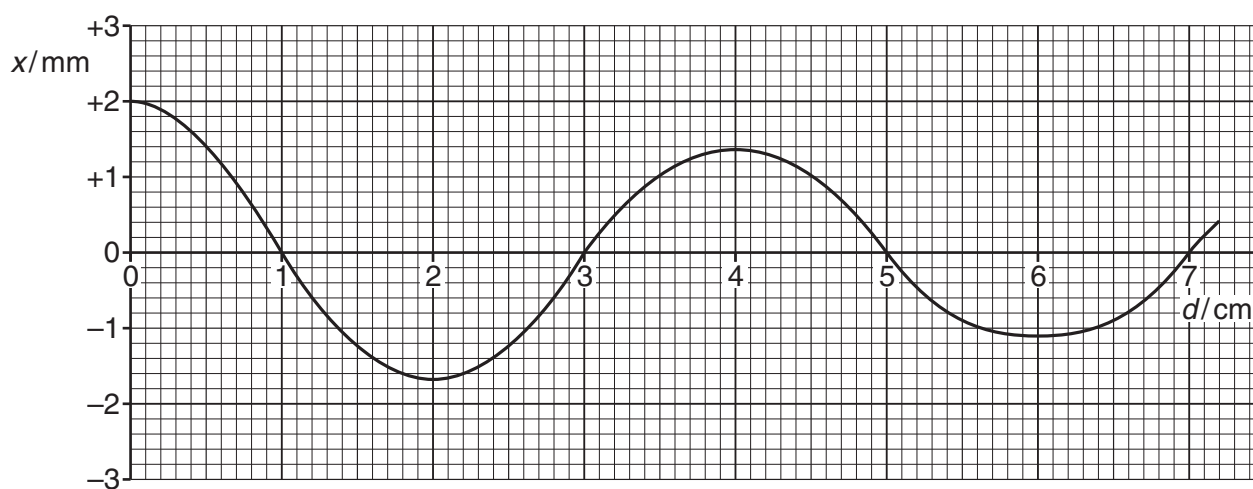


Fig. 5.2

- (a) Define, for a wave, what is meant by

(i) *displacement*,

.....[1]

(ii) *wavelength*.

.....[1]

(b) Use Figs. 5.1 and 5.2 to determine, for the water wave,

(i) the period T of vibration,

$T = \dots\dots\dots$ s [1]

(ii) the wavelength λ ,

$\lambda = \dots\dots\dots$ cm [1]

(iii) the speed v .

$v = \dots\dots\dots$ cm s⁻¹ [2]

(c) (i) Use Figs. 5.1 and 5.2 to state and explain whether the wave is losing power as it moves away from the source.

.....
.....
.....[2]

(ii) Determine the ratio

$$\frac{\text{intensity of wave at source}}{\text{intensity of wave 6.0 cm from source}} .$$

ratio = [3]

(b) Describe an experiment to determine the wavelength of sound in air using stationary waves. Include a diagram of the apparatus in your answer.

..... [3]

- (c) The variation with distance x of the intensity I of a stationary sound wave is shown in Fig. 6.1.

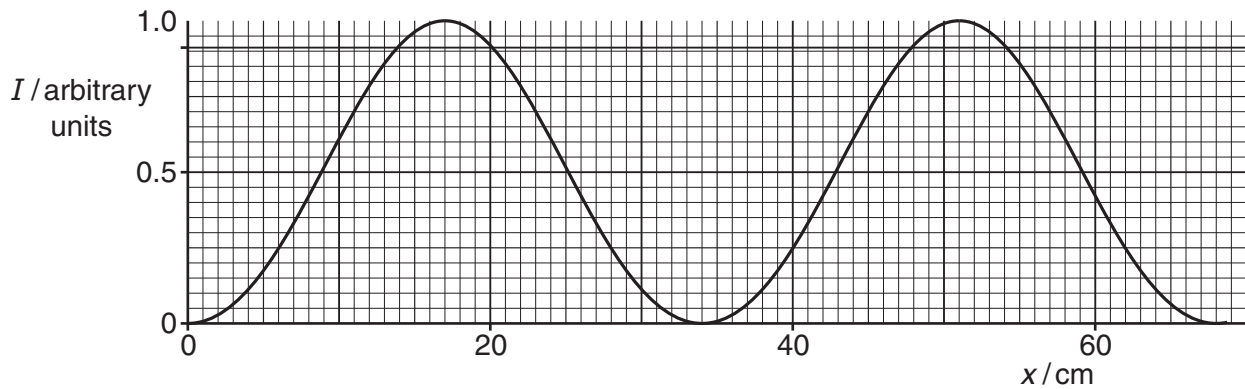


Fig. 6.1

- (i) On the x -axis of Fig. 6.1, indicate the positions of all the nodes and antinodes of the stationary wave. Label the nodes **N** and the antinodes **A**. [1]
- (ii) The speed of sound in air is 340 m s^{-1} .
Use Fig. 6.1 to determine the frequency of the sound wave.

frequency = Hz [3]

- 6 Fig. 4.1 shows an arrangement for producing stationary waves in a tube that is closed at one end.

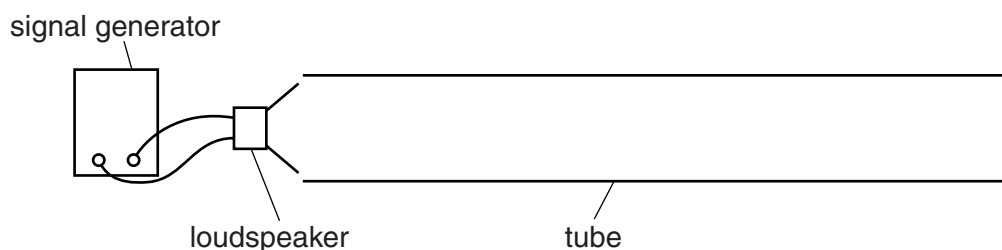


Fig. 4.1

- (a) Explain how waves from the loudspeaker produce stationary waves in the tube.

.....

.....

.....

.....

..... [3]

- (b) One of the stationary waves that may be formed in the tube is represented in Fig. 4.2.

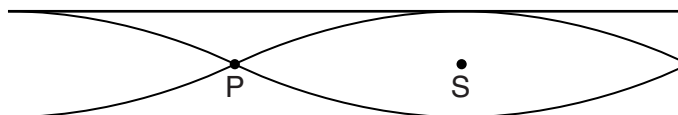


Fig. 4.2

- (i) Describe the motion of the air particles in the tube at

1. point P,

..... [1]

2. point S.

..... [1]

- (ii) The speed of sound in the tube is 330 ms^{-1} and the frequency of the waves from the loudspeaker is 880 Hz . Calculate the length of the tube.

length = m [3]

- 7 (a) State one property of electromagnetic waves that is **not** common to other transverse waves.

..... [1]

- (b) The seven regions of the electromagnetic spectrum are represented by blocks labelled A to G in Fig. 5.1.

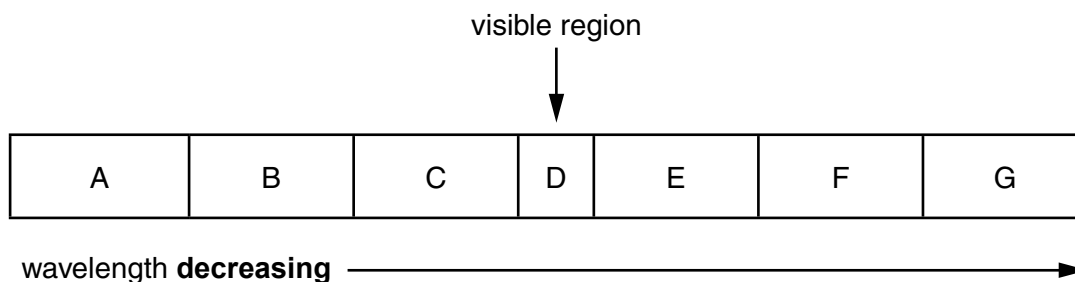


Fig. 5.1

A typical wavelength for the visible region D is 500 nm.

- (i) Name the principal radiations and give a typical wavelength for each of the regions B, E and F.

B: name: wavelength: m

E: name: wavelength: m

F: name: wavelength: m
[3]

- (ii) Calculate the frequency corresponding to a wavelength of 500 nm.

frequency = Hz [2]

- (c) All the waves in the spectrum shown in Fig. 5.1 can be polarised. Explain the meaning of the term *polarised*.

.....

 [2]

8 Fig. 5.1 shows a string stretched between two fixed points P and Q.

9702/22/M/J/13/Q5

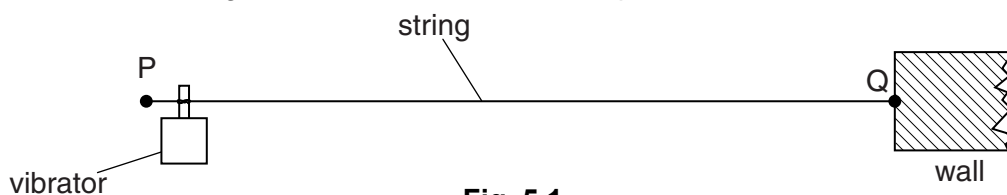


Fig. 5.1

A vibrator is attached near end P of the string. End Q is fixed to a wall. The vibrator has a frequency of 50 Hz and causes a transverse wave to travel along the string at a speed of 40 m s^{-1} .

(a) (i) Calculate the wavelength of the transverse wave on the string.

wavelength = m [2]

(ii) Explain how this arrangement may produce a stationary wave on the string.

.....

 [2]

(b) The stationary wave produced on PQ at one instant of time t is shown on Fig. 5.2. Each point on the string is at its maximum displacement.

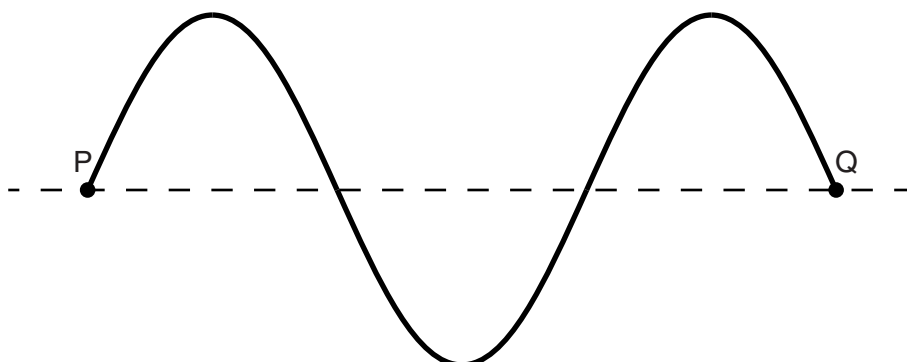


Fig. 5.2 (not to scale)

(i) On Fig. 5.2, label all the nodes with the letter **N** and all the antinodes with the letter **A**. [2]

(ii) Use your answer in (a)(i) to calculate the length of string PQ.

length = m [1]

(iii) On Fig. 5.2, draw the stationary wave at time $(t + 5.0 \text{ ms})$. Explain your answer.

..... [3]

- 9 A long rope is held under tension between two points A and B. Point A is made to vibrate vertically and a wave is sent down the rope towards B as shown in Fig. 5.1.

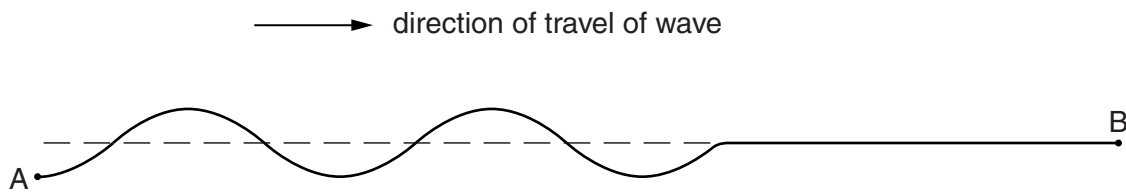


Fig. 5.1 (not to scale)

The time for one oscillation of point A on the rope is 0.20 s. The point A moves a distance of 80 mm during one oscillation. The wave on the rope has a wavelength of 1.5 m.

- (a) (i) Explain the term *displacement* for the wave on the rope.

.....
 [1]

- (ii) Calculate, for the wave on the rope,

1. the amplitude,

amplitude = mm [1]

2. the speed.

speed = m s^{-1} [3]

- (b) On Fig. 5.1, draw the wave pattern on the rope at a time 0.050 s later than that shown. [2]

- (c) State and explain whether the waves on the rope are

- (i) progressive or stationary,

.....
 [1]

- (ii) longitudinal or transverse.

.....
 [1]

- 10 (a) (i) Define, for a wave,
1. wavelength λ ,

..... [1]

2. frequency f .

..... [1]

- (ii) Use your definitions to deduce the relationship between λ , f and the speed v of the wave.

[1]

- (b) Plane waves on the surface of water are represented by Fig. 5.1 at one particular instant of time.

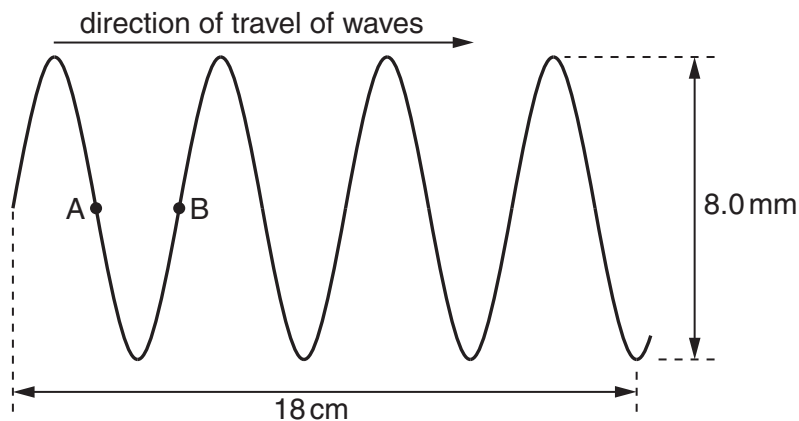


Fig. 5.1 (not to scale)

The waves have frequency 2.5 Hz.

Determine, for the waves,

- (i) the amplitude,

amplitude = mm [1]

- (ii) the speed,

speed = ms^{-1} [2]

- (iii) the phase difference between points A and B.

phase difference = unit [1]

- (c) The wave in (b) was produced in a ripple tank. Describe briefly, with the aid of a sketch diagram, how the wave may be observed.

.....

 [2]

11 (a) Explain what is meant by the following quantities for a wave on the surface of water:

(i) displacement and amplitude,

displacement

amplitude [2]

(ii) frequency and time period.

frequency

time period [2]

(b) Fig. 5.1 represents waves on the surface of water in a ripple tank at one particular instant of time.

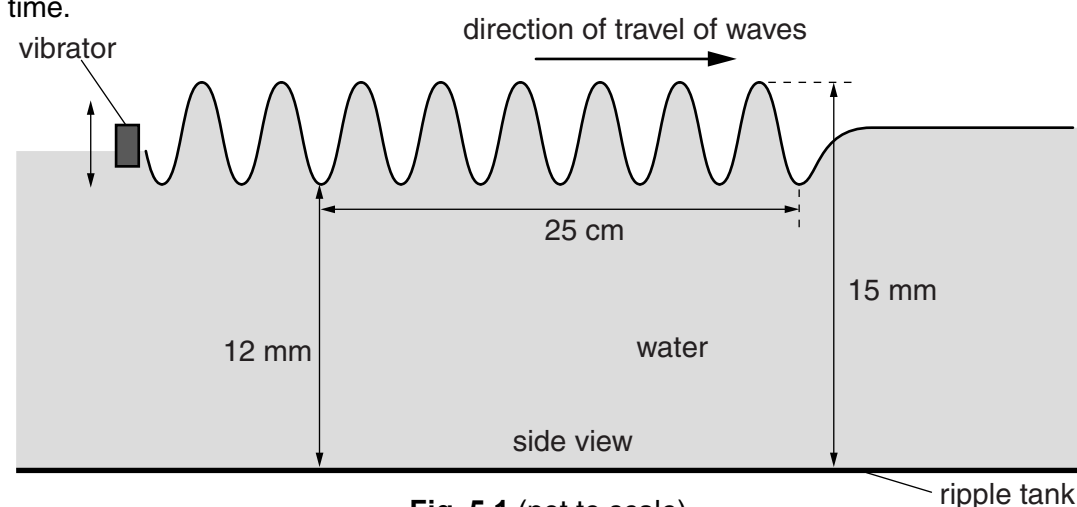


Fig. 5.1 (not to scale)

A vibrator moves the surface of the water to produce the waves of frequency f . The speed of the waves is 7.5 cm s^{-1} . Where the waves travel on the water surface, the maximum depth of the water is 15 mm and the minimum depth is 12 mm.

(i) Calculate, for the waves,

1. the amplitude,

amplitude = mm [1]

2. the wavelength.

wavelength = m [2]

(ii) Calculate the time period of the oscillations of the vibrator.

time period = s [2]

(c) State and explain whether the waves on the surface of the water shown in Fig. 5.1 are

(i) progressive or stationary,

..... [1]

(ii) transverse or longitudinal.

..... [1]

- 12 A hollow tube is used to investigate stationary waves. The tube is closed at one end and open at the other end. A loudspeaker connected to a signal generator is placed near the open end of the tube, as shown in Fig. 6.1.

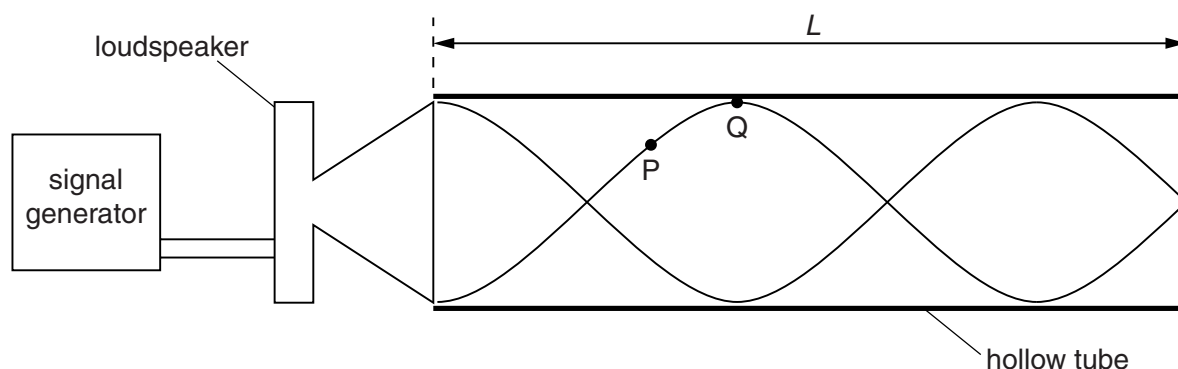


Fig. 6.1

The tube has length L . The frequency of the signal generator is adjusted so that the loudspeaker produces a progressive wave of frequency 440 Hz. A stationary wave is formed in the tube. A representation of this stationary wave is shown in Fig. 6.1.

Two points P and Q on the stationary wave are labelled.

- (a) (i) Describe, in terms of energy transfer, the difference between a progressive wave and a stationary wave.

.....[1]

- (ii) Explain how the stationary wave is formed in the tube.

.....

.....

.....[3]

- (iii) State the direction of the oscillations of an air particle at point P.

.....[1]

- (b) On Fig. 6.1 label, with the letter N, the nodes of the stationary wave. [1]

- (c) State the phase difference between points P and Q on the stationary wave.

phase difference = [1]

- (d) The speed of sound in the tube is 330 m s^{-1} .

Calculate

- (i) the wavelength of the sound wave,

wavelength = m [2]

- (ii) the length L of the tube.

length = m [2]

13 (a) Explain how stationary waves are formed.

9702/21/O/N/14/Q8

.....
.....
..... [2]

(b) The arrangement of apparatus used to determine the wavelength of a sound wave is shown in Fig. 8.1.

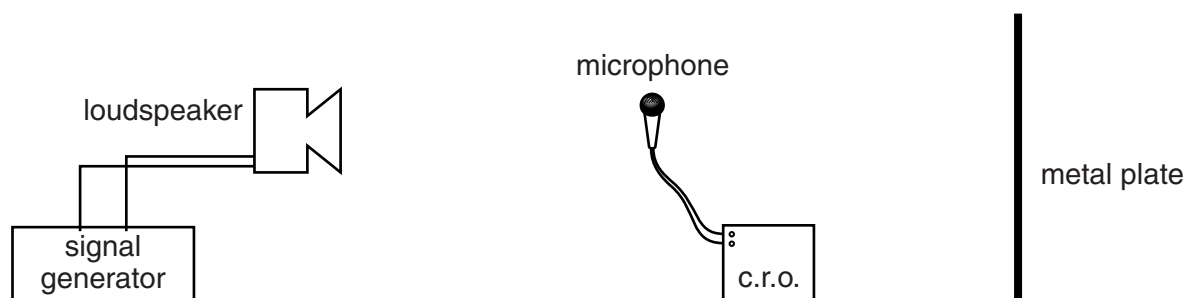


Fig. 8.1

The loudspeaker emits sound of one frequency. The microphone is connected to a cathode-ray oscilloscope (c.r.o.).

The waveform obtained on the c.r.o. for one position of the microphone is shown in Fig. 8.2.

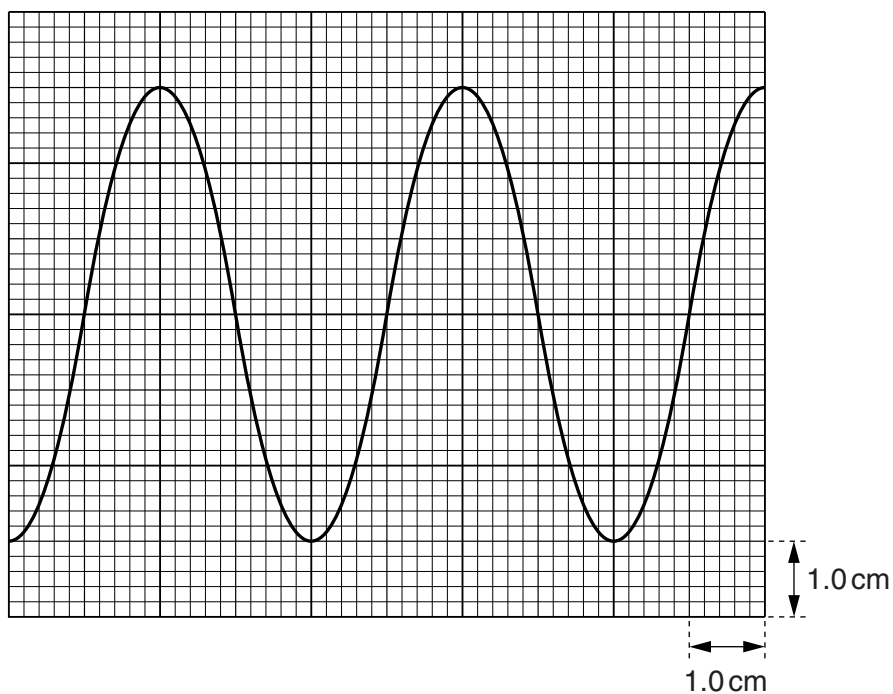


Fig. 8.2

The time-base setting of the c.r.o. is 0.20 ms cm^{-1} .

(i) Use Fig. 8.2 to show that the frequency of the sound is approximately 1300 Hz.

(ii) Explain how the apparatus is used to determine the wavelength of the sound. [2]

.....
.....
.....
..... [2]

(iii) The wavelength of the sound wave is 0.26 m. Calculate the speed of sound in this experiment.

speed = ms^{-1} [2]

9702/22/O/N/14/Q1

- 14 (a) The Young modulus of the metal of a wire is $1.8 \times 10^{11} \text{ Pa}$. The wire is extended and the strain produced is 8.2×10^{-4} . Calculate the stress in GPa.

stress = GPa [2]

(b) An electromagnetic wave has frequency 12 THz.

(i) Calculate the wavelength in μm .

wavelength = μm [2]

(ii) State the name of the region of the electromagnetic spectrum for this frequency.

..... [1]

15 (a) (i) Explain what is meant by a *progressive transverse wave*.

9702/23/O/N/14/Q7

progressive:

.....

transverse:

.....

[2]

(ii) Define frequency.

.....

.....[1]

(b) The variation with distance x of displacement y for a transverse wave is shown in Fig. 7.1.

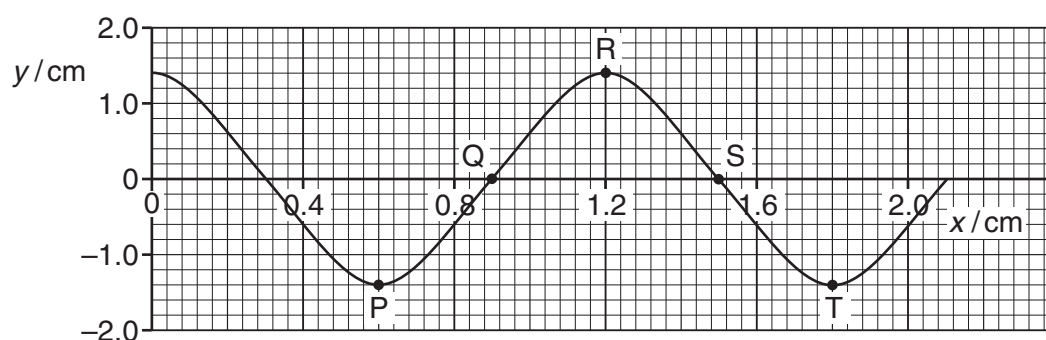


Fig. 7.1

On Fig. 7.1, five points are labelled.

Use Fig. 7.1 to state any two points having a phase difference of

(i) zero,

.....[1]

(ii) 270° .

.....[1]

(c) The frequency of the wave in (b) is 15 Hz.

Calculate the speed of the wave in (b).

speed = ms^{-1} [3]

(d) Two waves of the same frequency have amplitudes 1.4 cm and 2.1 cm.

Calculate the ratio

$$\frac{\text{intensity of wave of amplitude 1.4 cm}}{\text{intensity of wave of amplitude 2.1 cm}}$$

ratio = [2]

16 (a) State two differences between progressive waves and stationary waves. **9702/22/M/J/15/Q6**

1.

2.

..... [2]

(b) A source S of microwaves is placed in front of a metal reflector R, as shown in Fig. 6.1.

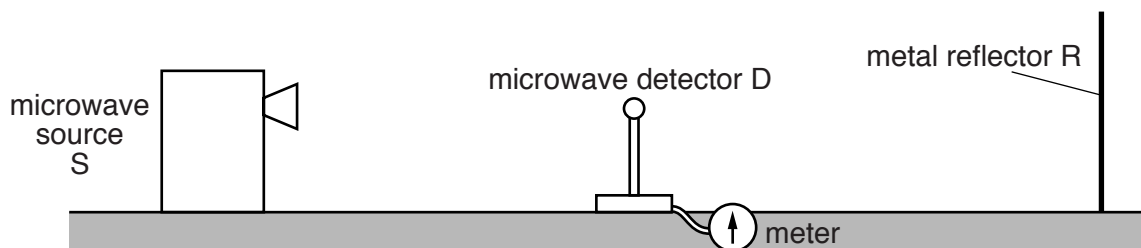


Fig. 6.1

A microwave detector D is placed between R and S.

Describe

(i) how stationary waves are formed between R and S,

.....

.....

..... [3]

(ii) how D is used to show that stationary waves are formed between R and S,

.....

..... [2]

(iii) how the wavelength of the microwaves may be determined using the apparatus in Fig. 6.1.

.....

..... [2]

(c) The wavelength of the microwaves in **(b)** is 2.8cm. Calculate the frequency, in GHz, of the microwaves.

frequency = GHz [3]

- 17 A wave is produced on the surface of a different liquid. At one particular time, the variation of the vertical displacement y with distance x along the surface of the liquid is shown in Fig. 4.2.

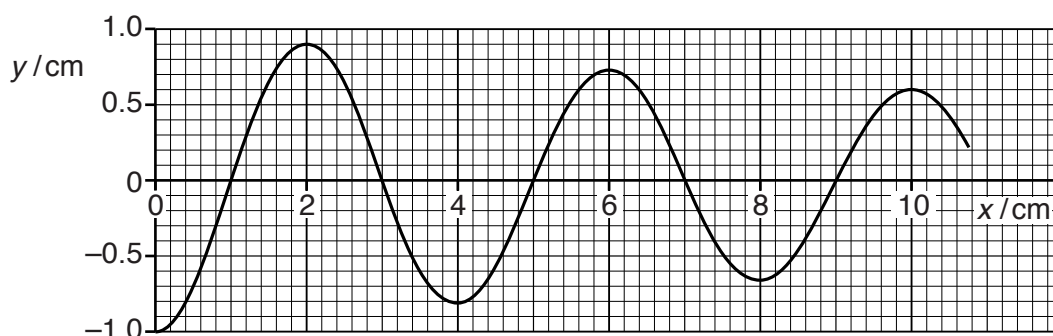


Fig. 4.2

- (i) The wave has intensity I_1 at distance $x = 2.0$ cm and intensity I_2 at $x = 10.0$ cm.

Determine the ratio

$$\frac{\text{intensity } I_2}{\text{intensity } I_1}.$$

ratio = [2]

- (ii) State the phase difference, with its unit, between the oscillations of the liquid particles at distances $x = 3.0$ cm and $x = 4.0$ cm.

phase difference = [1]

- 18 The variation with time t of the displacement y of a wave X, as it passes a point P, is shown in Fig. 5.1.

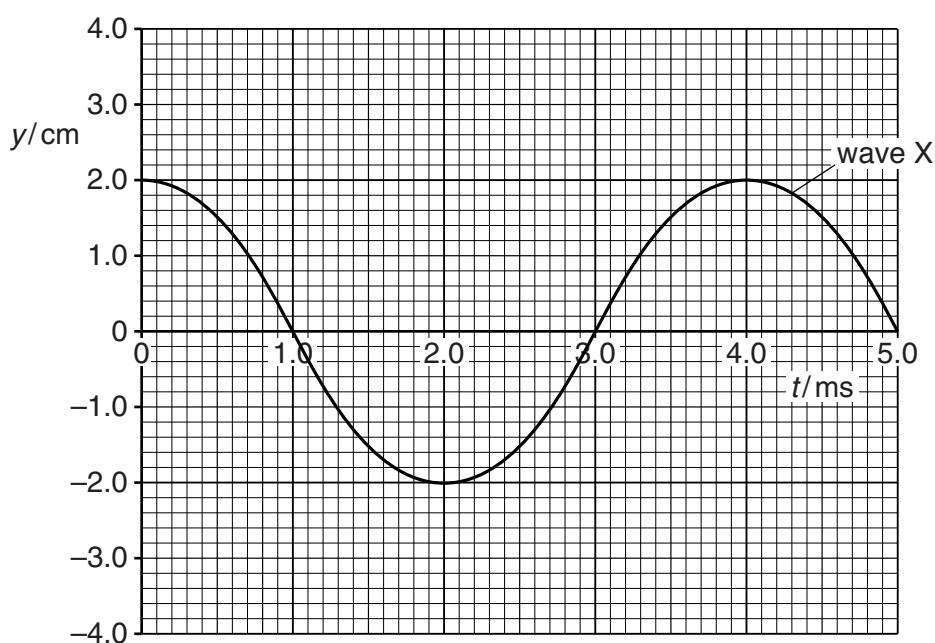


Fig. 5.1

The intensity of wave X is I .

- (a) Use Fig. 5.1 to determine the frequency of wave X.

frequency = Hz [2]

- (b) A second wave Z with the same frequency as wave X also passes point P. Wave Z has intensity $2I$. The phase difference between the two waves is 90° .

On Fig. 5.1, sketch the variation with time t of the displacement y of wave Z.

Show your working.

- 19 (a) By reference to the direction of the propagation of energy, state what is meant by a *longitudinal* wave and by a *transverse* wave.

longitudinal:

.....

.....

transverse:

.....

.....

[2]

- (b) The intensity of a sound wave passing through air is given by

$$I = K v \rho f^2 A^2$$

where I is the intensity (power per unit area),

K is a constant without units,

v is the speed of sound,

ρ is the density of air,

f is the frequency of the wave

and A is the amplitude of the wave.

Show that both sides of the equation have the same SI base units.

[3]

(c) (i) Describe the *Doppler effect*.

.....
.....[1]

(ii) A distant star is moving away from a stationary observer.

State the effect of the motion on the light observed from the star.

.....
.....
.....[1]

(d) A car travels at a constant speed towards a stationary observer. The horn of the car sounds at a frequency of 510 Hz and the observer hears a frequency of 550 Hz. The speed of sound in air is 340 m s^{-1} .

Calculate the speed of the car.

speed = m s^{-1} [3]

- 20 (a) Apparatus used to produce stationary waves on a stretched string is shown in Fig. 7.1.

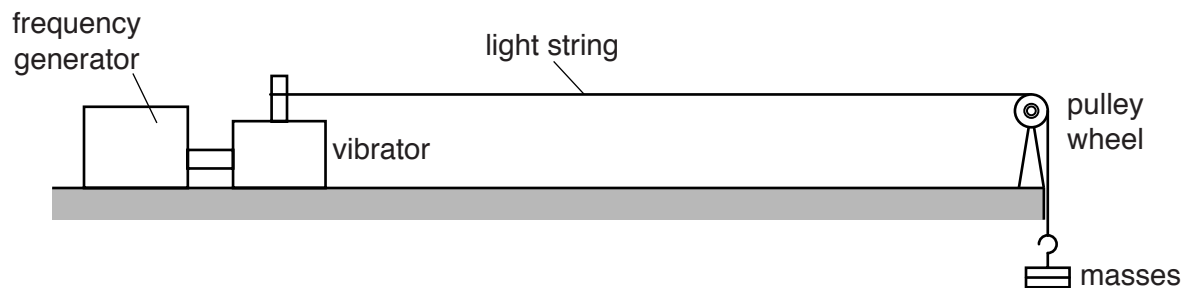


Fig. 7.1

The frequency generator is switched on.

- (i) Describe two adjustments that can be made to the apparatus to produce stationary waves on the string.

1.

.....

2.

.....

[2]

- (ii) Describe the features that are seen on the stretched string that indicate stationary waves have been produced.

..... [1]

- (b) The variation with time t of the displacement x of a particle caused by a progressive wave R is shown in Fig. 7.2. For the same particle, the variation with time t of the displacement x caused by a second wave S is also shown in Fig. 7.2.

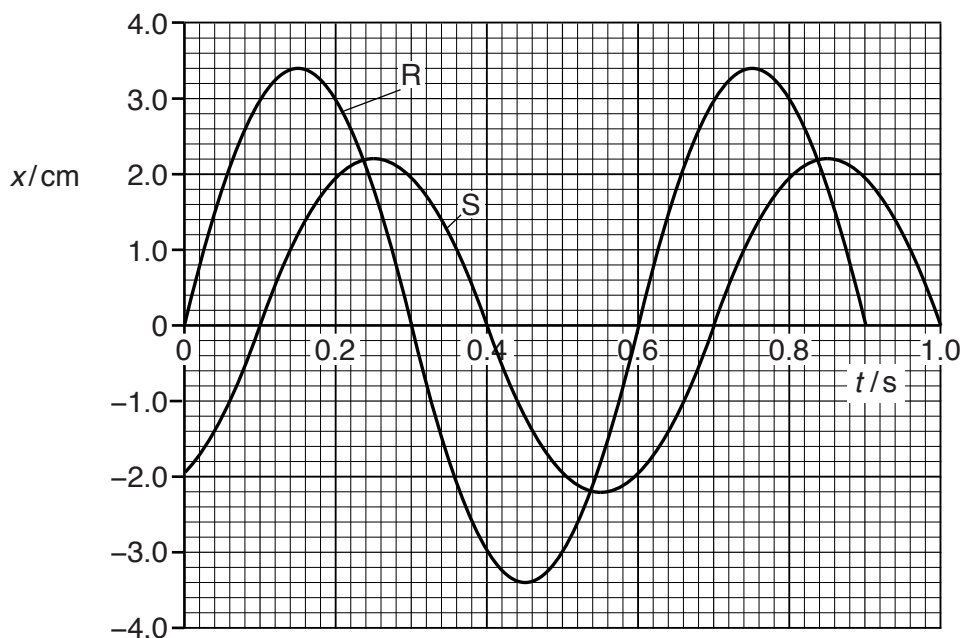


Fig. 7.2

- (i) Determine the phase difference between wave R and wave S. Include an appropriate unit.

phase difference = [1]

- (ii) Calculate the ratio

$$\frac{\text{intensity of wave R}}{\text{intensity of wave S}}.$$

ratio = [2]

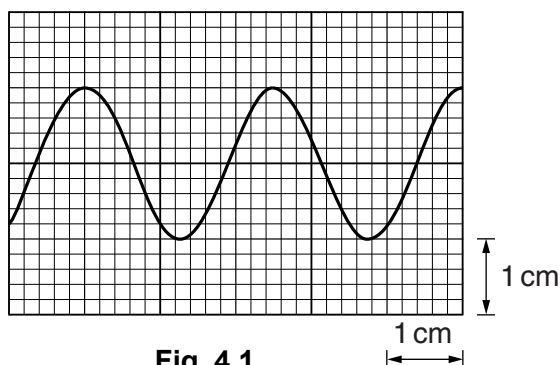
21 (a) State what is meant by the *frequency* of a progressive wave.

9702/21/O/N/16/Q4

.....

[2]

- (b) A cathode-ray oscilloscope (c.r.o.) is used to determine the frequency of the sound emitted by a loudspeaker. The trace produced on the screen of the c.r.o. is shown in Fig. 4.1.



The time-base setting of the c.r.o. is $250 \mu\text{s cm}^{-1}$.

Show that the frequency of the sound wave is 1600 Hz.

[2]

- (c) The loudspeaker in (b) emits the sound in all directions. A person attaches the loudspeaker to a string and then swings the loudspeaker at a constant speed in a horizontal circle above his head.

An observer, standing a large distance away from the loudspeaker, hears sound of maximum frequency 1640 Hz. The speed of sound in air is 330 m s^{-1} .

- (i) Determine the speed of the loudspeaker.

speed = m s^{-1} [2]

- (ii) Describe and explain, qualitatively, the variation in the frequency of the sound heard by the observer.

.....

[2]

22 (a) State the conditions required for the formation of stationary waves.
9702/21/M/J/17/Q4

.....
.....
.....[2]

(b) One end of a string is attached to a vibrator. The string is stretched by passing the other end over a pulley and attaching a load, as illustrated in Fig. 4.1.

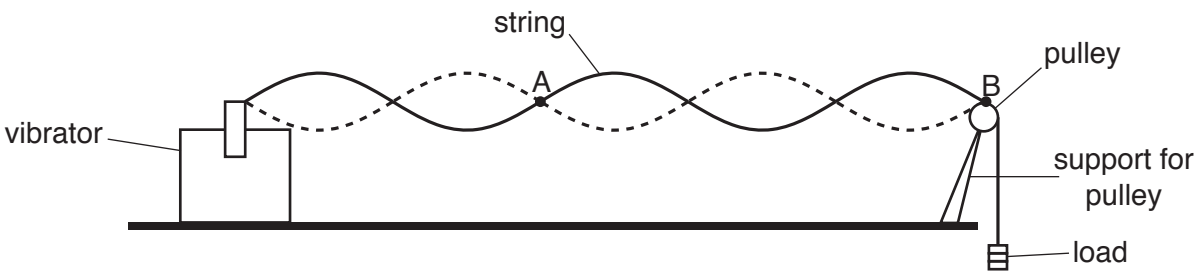


Fig. 4.1

The frequency of vibration of the vibrator is adjusted to 250 Hz and a transverse wave travels along the string with a speed of 12 m s⁻¹. The wave is reflected at the pulley and a stationary wave forms on the string.

Fig. 4.2 shows the string between points A and B at time $t = t_1$.

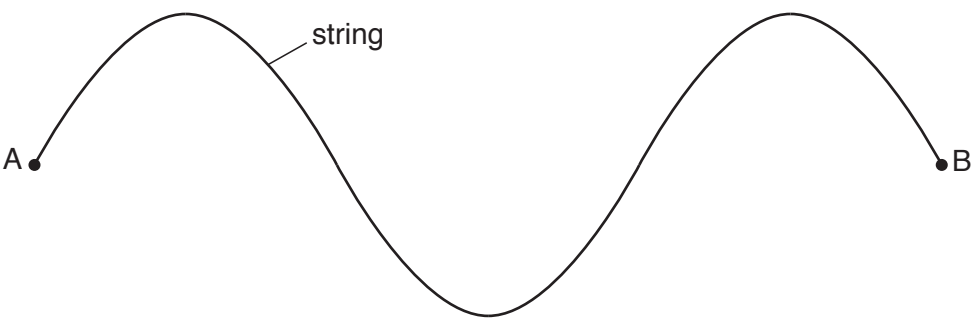


Fig. 4.2

At time $t = t_1$ the string has maximum displacement.

(i) Calculate the distance AB.

distance =m [2]

(ii) On Fig. 4.2, sketch the position of the string between A and B at times

1. $t = t_1 + 2.0 \text{ ms}$ (label this line P),
2. $t = t_1 + 5.0 \text{ ms}$ (label this line Q).

[3]

9702/21/M/J/017/Q5

23 (a) Describe the Doppler effect.

.....

[1]

- (b) A car travels with a constant velocity along a straight road. The car horn with a frequency of 400 Hz is sounded continuously. A stationary observer on the roadside hears the sound from the horn at a frequency of 360 Hz.
 The speed of sound is 340 m s^{-1} .

Determine the magnitude v , and the direction, of the velocity of the car relative to the observer.

$v = \dots\dots\dots \text{m s}^{-1}$

direction

[3]

24 (a) Define the *frequency* of a sound wave.

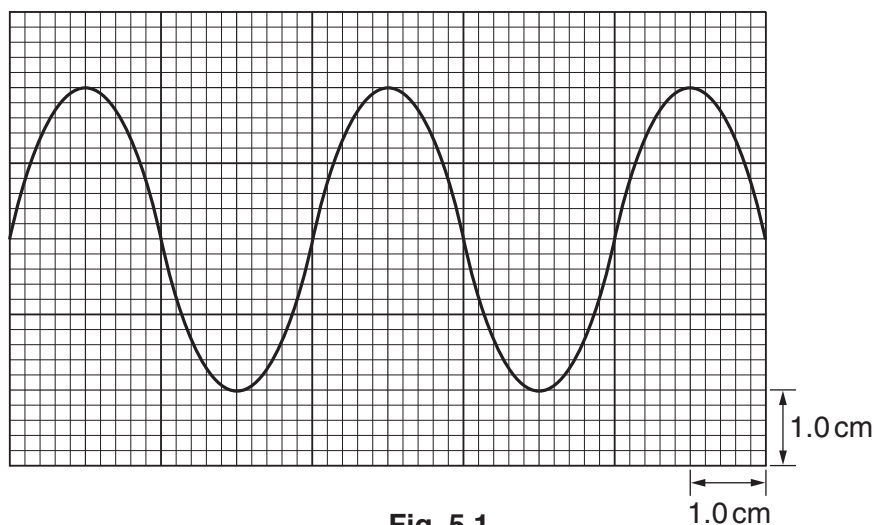
9702/22/M/J/017/Q5

.....[1]

(b) A sound wave travels through air. Describe the motion of the air particles relative to the direction of travel of the sound wave.

.....[1]

(c) The sound wave emitted from the horn of a stationary car is detected with a microphone and displayed on a cathode-ray oscilloscope (c.r.o.), as shown in Fig. 5.1.



The y -axis setting is 5.0 mV cm^{-1} .

The time-base setting is 0.50 ms cm^{-1} .

(i) Use Fig. 5.1 to determine the frequency of the sound wave.

frequency = Hz [2]

(ii) The horn of the car sounds continuously. Describe the changes to the trace seen on the c.r.o. as the car travels at constant speed

1. directly towards the stationary microphone,

.....

2. directly away from the stationary microphone.

.....[3]

25 (a) State the difference between a stationary wave and a progressive wave in terms of

(i) the energy transfer along the wave,

.....
[1]

(ii) the phase of two adjacent vibrating particles.

.....
[1]

(b) A tube is open at both ends. A loudspeaker, emitting sound of a single frequency, is placed near one end of the tube, as shown in Fig. 3.1.

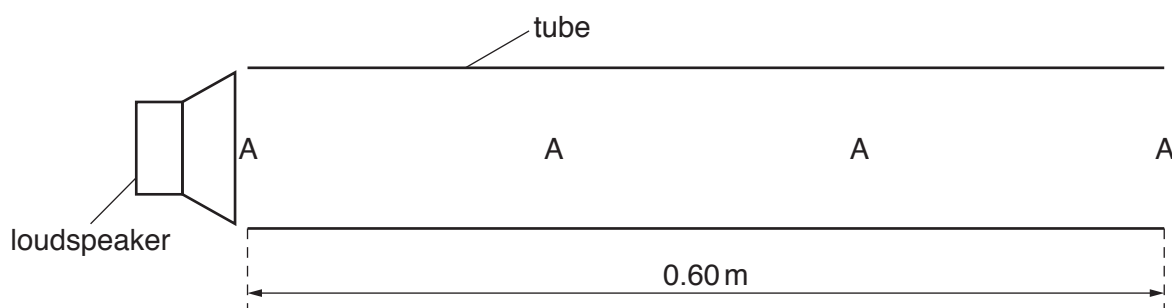


Fig. 3.1

The speed of the sound in the tube is 340 m s^{-1} . The length of the tube is 0.60 m . A stationary wave is formed with an antinode A at each end of the tube and two antinodes inside the tube.

(i) State what is meant by an *antinode* of the stationary wave.

.....
[1]

(ii) State the distance between a node and an adjacent antinode.

distance = m [1]

(iii) Determine, for the sound in the tube,

1. the wavelength,

wavelength = m [1]

2. the frequency.

frequency = Hz [2]

- (iv) Determine the minimum frequency of the sound from the loudspeaker that produces a stationary wave in the tube.

minimum frequency = Hz [2]

26 (a) State the conditions required for the formation of a stationary wave.

9702/22/O/N/17/Q4

.....
.....
.....[2]

- (b) A horizontal string is stretched between two fixed points X and Y. The string is made to vibrate vertically so that a stationary wave is formed. At one instant, each particle of the string is at its maximum displacement, as shown in Fig. 4.1.

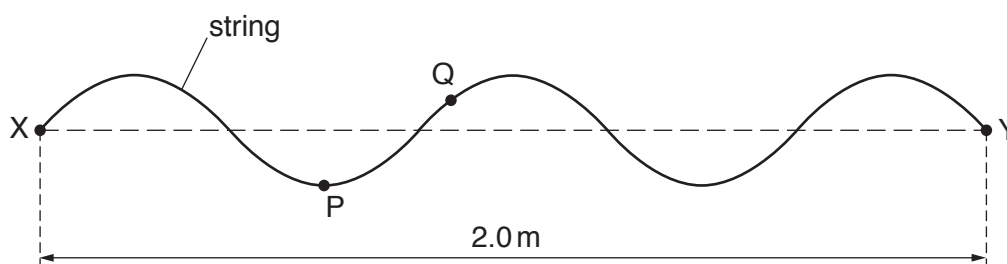


Fig. 4.1

P and Q are two particles of the string. The string vibrates with a frequency of 40 Hz. Distance XY is 2.0 m.

- (i) State the number of antinodes in the stationary wave.

number =[1]

- (ii) Determine the minimum time taken for the particle P to travel from its lowest point to its highest point.

time taken = s [2]

- (iii) State the phase difference, with its unit, between the vibrations of particle P and of particle Q.

phase difference =[1]

- (iv) Determine the speed of a progressive wave along the string.

speed = ms^{-1} [2]

- 27 (a) By reference to the direction of propagation of energy, explain what is meant by a *longitudinal* wave.

.....
[1]

- (b) A car horn emits a sound wave of frequency 800 Hz. A microphone and a cathode-ray oscilloscope (c.r.o.) are used to analyse the sound wave. The waveform displayed on the c.r.o. screen is shown in Fig. 4.1.

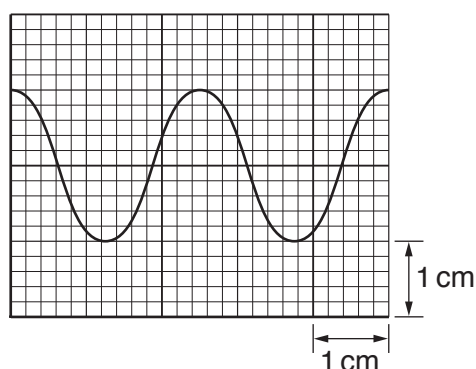


Fig. 4.1

Determine the time-base setting, in s cm^{-1} , of the c.r.o.

time-base setting = s cm^{-1} [3]

- (c) The intensity I of the sound at a distance r from the car horn in (b) is given by the expression

$$I = \frac{k}{r^2}$$

where k is a constant.

Fig. 4.2 shows the car in (b) on a road.

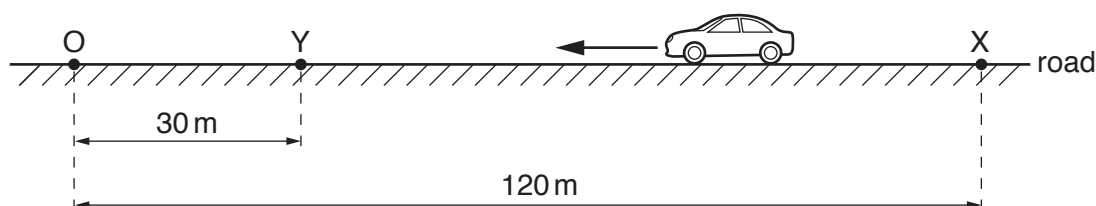


Fig. 4.2

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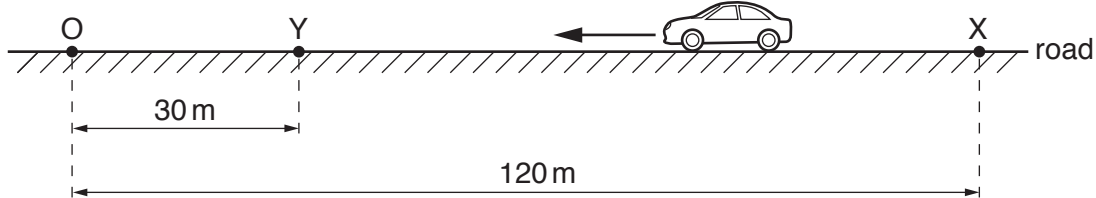


Fig. 4.2

An observer stands at point O. Initially the car is parked at point X which is 120 m away from point O. The car then moves directly towards the observer and stops at point Y, a distance of 30 m away from O.

The car horn continuously emits sound when the car is moving between points X and Y.

- (i) The sound wave at point O has amplitude A_X when the car is at X and has amplitude A_Y when the car is at Y.

Calculate the ratio $\frac{A_Y}{A_X}$.

ratio = [3]

- (ii) When the car is parked at X, the frequency of the sound from the horn that is detected by the observer is 800 Hz. As the car moves from X to Y, the maximum change in the detected frequency is 16 Hz. The speed of the sound in air is 330 m s^{-1} .

Determine, to two significant figures,

- the minimum wavelength of the sound detected by the observer,

wavelength = m [2]

- the maximum speed of the car.

speed = m s^{-1} [2]

28 (a) For a progressive wave, state what is meant by

9702/21/M/J/18/Q4

(i) the *period*,

.....
.....[1]

(ii) the *wavelength*.

.....
.....[1]

(b) Fig. 4.1 shows the variation with time t of the displacement x of two progressive waves P and Q passing the same point.

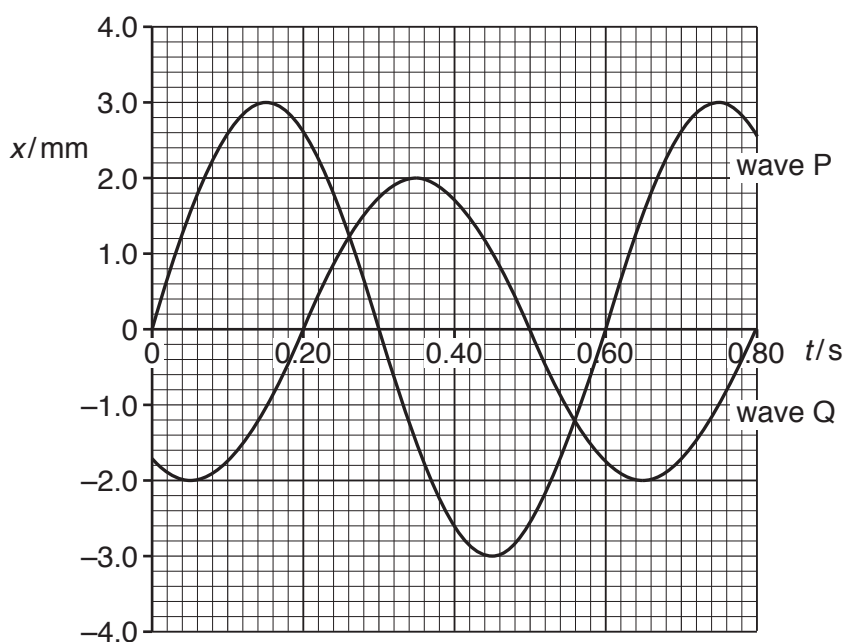


Fig. 4.1

The speed of the waves is 20 cm s^{-1} .

(i) Calculate the wavelength of the waves.

wavelength = cm [2]

(ii) Determine the phase difference between the two waves.

(iii) Calculate the ratio phase difference = ° [1]

$$\frac{\text{intensity of wave Q}}{\text{intensity of wave P}} .$$

ratio = [2]

(iv) The two waves superpose as they pass the same point. Use Fig. 4.1 to determine the resultant displacement at time $t = 0.45\text{ s}$.

displacement = mm [1]

- 29 (a) (i) Define the *wavelength* of a progressive wave.

.....
[1]

- (ii) State what is meant by an *antinode* of a stationary wave.

.....
[1]

- (b) A loudspeaker producing sound of constant frequency is placed near the open end of a pipe, as shown in Fig. 4.1.

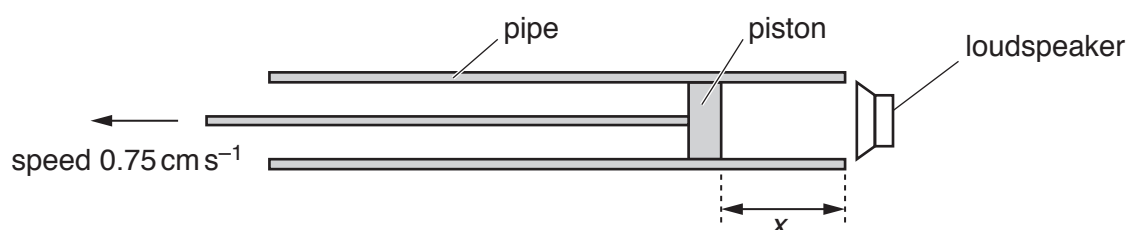


Fig. 4.1

A movable piston is at distance x from the open end of the pipe. Distance x is increased from $x = 0$ by moving the piston to the left with a constant speed of 0.75 cm s^{-1} .

The speed of the sound in the pipe is 340 m s^{-1} .

- (i) A much louder sound is first heard when $x = 4.5 \text{ cm}$. Assume that there is an antinode of a stationary wave at the open end of the pipe.

Determine the frequency of the sound in the pipe.

frequency = Hz [3]

- (ii) After a time interval, a second much louder sound is heard. Calculate the time interval between the first louder sound and the second louder sound being heard.

time interval = s [2]

- 30 (a) Sound waves are longitudinal waves. By reference to the direction of propagation of energy, state what is meant by a *longitudinal wave*.

.....
 [1]

- (b) A stationary sound wave in air has amplitude A . In an experiment, a detector is used to determine A^2 . The variation of A^2 with distance x along the wave is shown in Fig. 4.1.

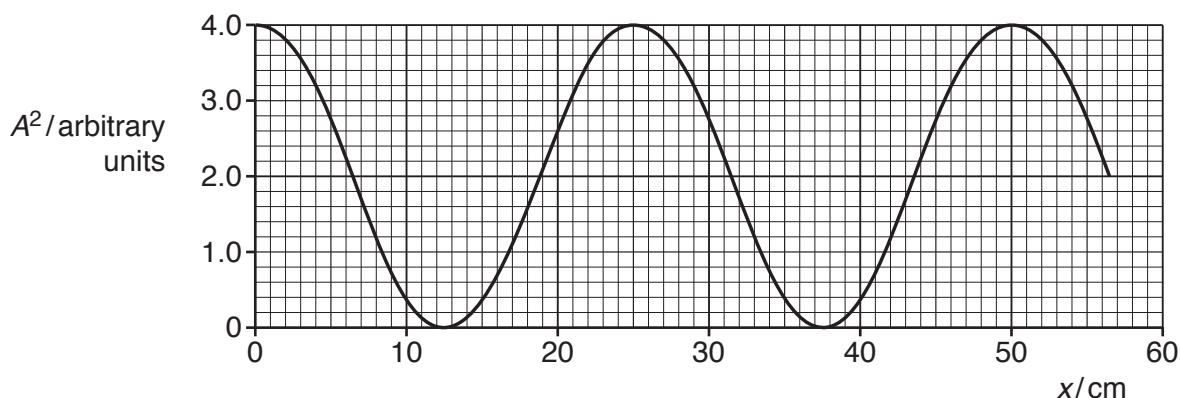


Fig. 4.1

- (i) State the phase difference between the vibrations of an air particle at $x = 25$ cm and the vibrations of an air particle at $x = 50$ cm.

phase difference = ° [1]

- (ii) The speed of the sound in the air is 330 ms^{-1} . Determine the frequency of the sound wave.

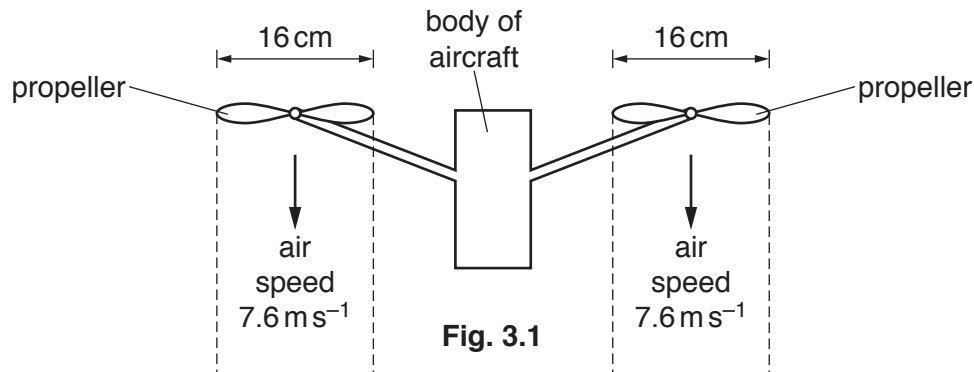
frequency = Hz [3]

- (iii) Determine the ratio

$$\frac{\text{amplitude } A \text{ of wave at } x = 20 \text{ cm}}{\text{amplitude } A \text{ of wave at } x = 25 \text{ cm}}$$

ratio = [2]

- 31 A small remote-controlled model aircraft has two propellers, each of diameter 16 cm. Fig. 3.1 is a side view of the aircraft when hovering.



Air is propelled vertically downwards by each propeller so that the aircraft hovers at a fixed position. The density of the air is 1.2 kg m^{-3} . Assume that the air from each propeller moves with a constant speed of 7.6 m s^{-1} in a uniform cylinder of diameter 16 cm. Also assume that the air above each propeller is stationary.

- (a) Show that, in a time interval of 3.0 s, the mass of air propelled downwards by **one** propeller is 0.55 kg.

[3]

- (b) Calculate:

- (i) the increase in momentum of the mass of air in (a)

increase in momentum = N s [1]

- (ii) the downward force exerted on this mass of air by the propeller.

force = N [1]

(c) State:

- (i) the upward force acting on **one** propeller

force = N [1]

- (ii) the name of the law that explains the relationship between the force in **(b)(ii)** and the force in **(c)(i)**.

..... [1]

(d) Determine the mass of the aircraft.

mass = kg [1]

- (e) In order for the aircraft to hover at a very high altitude (height), the propellers must propel the air downwards with a greater speed than when the aircraft hovers at a low altitude. Suggest the reason for this.

.....
..... [1]

- (f) When the aircraft is hovering at a high altitude, an electric fault causes the propellers to stop rotating. The aircraft falls vertically downwards. When the aircraft reaches a constant speed of 22 m s^{-1} , it emits sound of frequency 3.0 kHz from an alarm. The speed of the sound in the air is 340 m s^{-1} .

Determine the frequency of the sound heard by a person standing vertically below the falling aircraft.

frequency = Hz [2]

- 32 (a) State what is meant by the *wavelength* of a progressive wave.

.....
 [1]

- (b) A cathode-ray oscilloscope (CRO) is used to analyse a sound wave. The screen of the CRO is shown in Fig. 5.1.

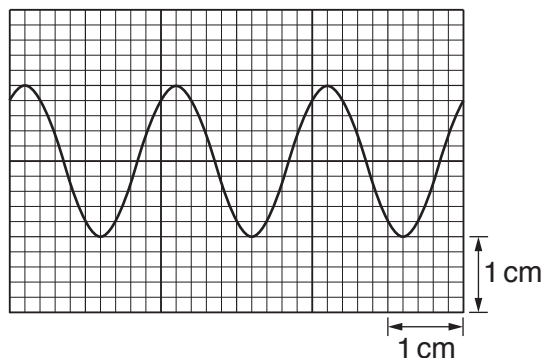


Fig. 5.1

The time-base setting of the CRO is 2.5 ms cm^{-1} .
 Determine the frequency of the sound wave.

frequency = Hz [2]

- (c) The source emitting the sound in (b) is at point A. Waves travel from the source to point C along two different paths, AC and ABC, as shown in Fig. 5.2.

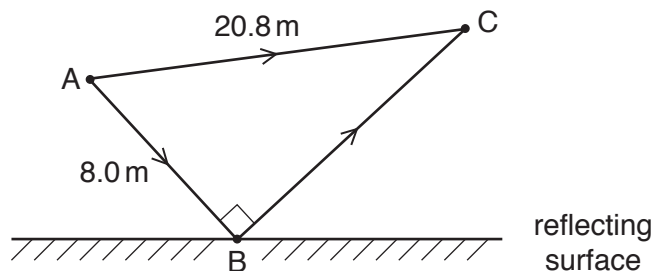


Fig. 5.2 (not to scale)

Distance AB is 8.0 m and distance AC is 20.8 m. Angle ABC is 90° . Assume that there is no phase change of the sound wave due to the reflection at point B. The wavelength of the waves is 1.6 m.

- (i) Show that the waves meeting at C have a path difference of 6.4 m.

[1]

- (ii) Explain why an intensity maximum is detected at point C.

.....

 [2]

- (iii) Determine the difference between the times taken for the sound to travel from the source to point C along the two different paths.

time difference = s [2]

- (iv) The wavelength of the sound is gradually increased. Calculate the wavelength of the sound when an intensity maximum is next detected at point C.

wavelength = m [1]

- 33 Two progressive sound waves Y and Z meet at a fixed point P. The variation with time t of the displacement x of each wave at point P is shown in Fig. 4.1.

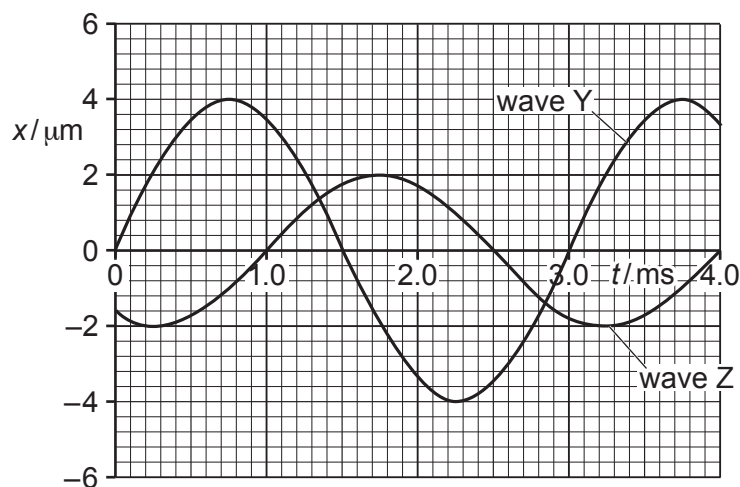


Fig. 4.1

- (a) Use Fig. 4.1 to state **one** quantity of waves Y and Z that is:

(i) the same

..... [1]

(ii) different.

..... [1]

- (b) State and explain whether waves Y and Z are coherent.

.....

..... [1]

- (c) Determine the phase difference between the waves.

phase difference = ° [1]

- (d) The two waves superpose at P. Use Fig. 4.1 to determine the resultant displacement at time $t = 0.75$ ms.

resultant displacement = μm [1]

- (e) The intensity of wave Y at point P is I .

Determine, in terms of I , the intensity of wave Z.

intensity = [2]

- (f) The speed of wave Z is 330 ms^{-1} .

Determine the wavelength of wave Z.

wavelength = m [3]

34. 9702/21/O/N/20/Q5

A progressive wave Y passes a point P. The variation with time t of the displacement x for the wave at P is shown in Fig. 5.1.

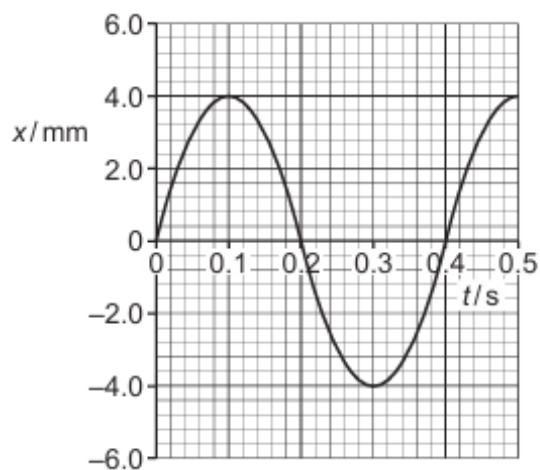


Fig. 5.1

The wave has a wavelength of 8.0 cm.

- (a) Determine the speed of the wave.

speed = ms^{-1} [2]

- (b) A second wave Z has wavelength 8.0 cm and amplitude 2.0 mm at point P. Waves Y and Z have the same speed.

For the waves at point P, calculate the ratio

$$\frac{\text{intensity of wave Z}}{\text{intensity of wave Y}}.$$

ratio = [3]

[Total: 5]

35. 9702/22/F/M/21/Q5

A source of sound is attached to a rope and then swung at a constant speed in a horizontal circle, as illustrated in Fig. 5.1.

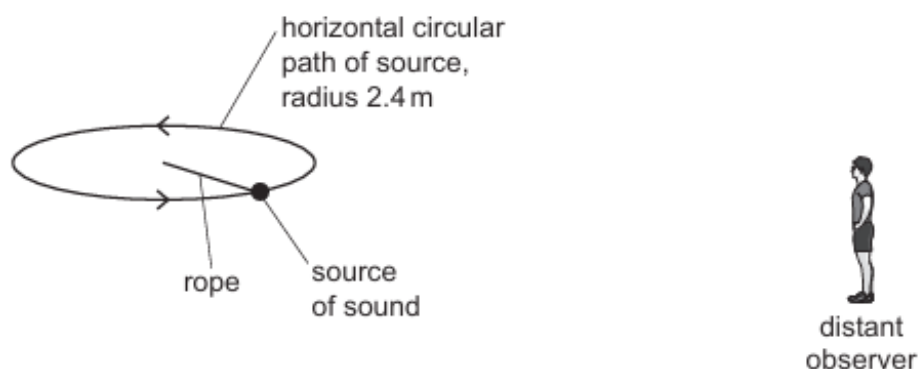


Fig. 5.1 (not to scale)

The source moves with a speed of 12.0 m s^{-1} and emits sound of frequency 951 Hz . The speed of the sound in the air is 330 m s^{-1} . An observer, standing a very long distance away from the source, hears the sound.

- (a) Calculate the minimum frequency, to three significant figures, of the sound heard by the observer.

minimum frequency = Hz [2]

- (b) The circular path of the source has a radius of 2.4 m .

Determine the shortest time interval between the observer hearing sound of minimum frequency and the observer hearing sound of maximum frequency.

time interval = s [2]

[Total: 4]

36. 9702/21/O/N/21/Q4

- (a) By reference to the direction of transfer of energy, state what is meant by a *longitudinal wave*.

.....
 [1]

- (b) A vehicle travels at constant speed around a wide circular track. It continuously sounds its horn, which emits a single note of frequency 1.2 kHz. An observer is a large distance away from the track, as shown in the view from above in Fig. 4.1.

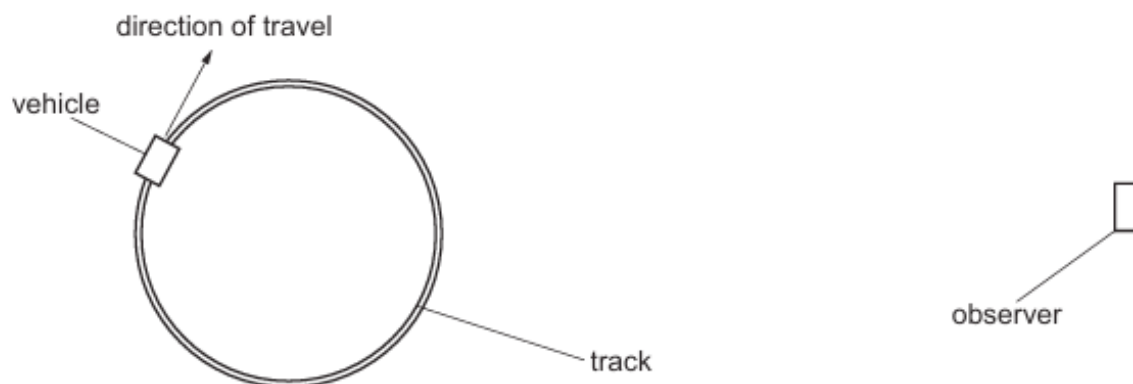


Fig. 4.1 (not to scale)

Fig. 4.2 shows the variation with time of the frequency f of the sound of the horn that is detected by the observer. The time taken for the vehicle to travel once around the track is T .

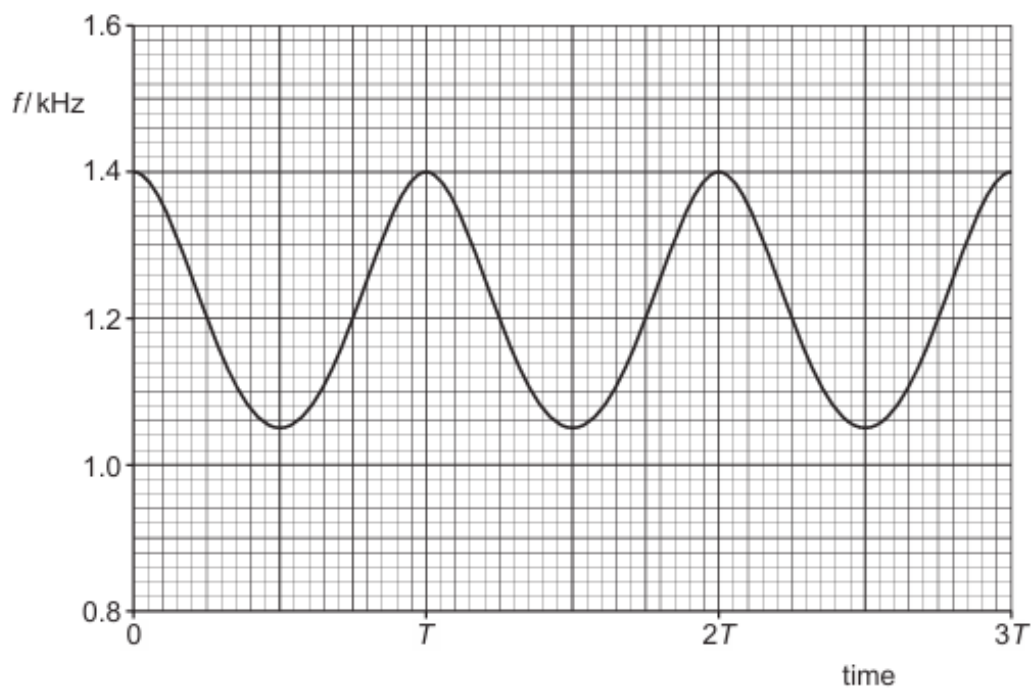


Fig. 4.2

- (i) Explain why the frequency of the sound detected by the observer is sometimes above and sometimes below 1.2 kHz.

.....
.....
.....
..... [2]

- (ii) State the name of the phenomenon in (b)(i).

..... [1]

- (iii) On Fig. 4.1, mark with a letter X the position of the vehicle when it emitted the sound that is detected at time T . [1]

- (iv) On Fig. 4.1, mark with a letter Y the position of the vehicle when it emitted the sound that is detected at time $\frac{9T}{4}$. [1]

- (c) The speed of the sound in the air is 320 m s^{-1} .

Use Fig. 4.2 to determine the speed of the vehicle in (b).

speed = m s^{-1} [3]

[Total: 9]

37. 9702/22/O/N/21/Q4

A child sits on the ground next to a remote-controlled toy car. At time $t = 0$, the car begins to move in a straight line directly away from the child. The variation with time t of the velocity of the car along this line is shown in Fig. 4.1.

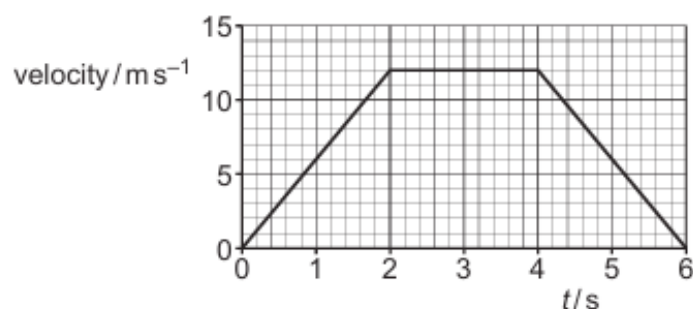


Fig. 4.1

The car's horn continually emits sound of frequency 925 Hz between time $t = 0$ and time $t = 6.0$ s. The speed of the sound in the air is 338 m s^{-1} .

- (a) Describe qualitatively the variation, if any, in the frequency of the sound heard, by the child, that was emitted from the car horn:

- (i) from time $t = 0$ to time $t = 2.0$ s

..... [1]

- (ii) from time $t = 4.0$ s to time $t = 6.0$ s.

..... [1]

- (b) Determine the frequency, to three significant figures, of the sound heard, by the child, that was emitted from the car horn at time $t = 3.0$ s.

frequency = Hz [2]

- (c) Determine the time taken for the sound emitted at time $t = 4.0$ s to travel to the child.

time taken = s [2]

[Total: 6]

38. 9702/22/F/M/22/Q5

- (a) (i) State the conditions required for the formation of a stationary wave.

.....
.....
.....
..... [2]

- (ii) State the phase difference between any two vibrating particles in a stationary wave between two adjacent nodes.

phase difference =° [1]

- (b) A motorcycle is travelling at 13.0 ms^{-1} along a straight road. The rider of the motorcycle sees a pedestrian standing in the road directly ahead and operates a horn to emit a warning sound. The pedestrian hears the warning sound from the horn at a frequency of 543 Hz . The speed of the sound in the air is 334 ms^{-1} .

- (i) Calculate the frequency, to three significant figures, of the sound emitted by the horn.

frequency = Hz [2]

- (ii) The motorcycle rider passes the stationary pedestrian and then moves directly away from her. As the rider moves away, he operates the horn for a second time. The pedestrian now hears sound that is increasing in frequency.

State the variation, if any, in the speed of the motorcycle when the rider operates the horn for the second time.

..... [1]

- (c) A beam of vertically polarised monochromatic light is incident normally on a polarising filter, as shown in Fig. 5.1.

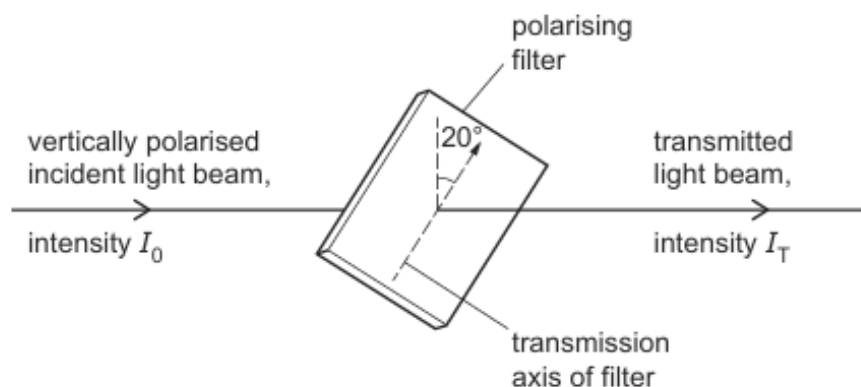


Fig. 5.1

The filter is positioned with its transmission axis at an angle of 20° to the vertical. The incident light has intensity I_0 and the transmitted light has intensity I_T .

- (i) By considering the ratio $\frac{I_T}{I_0}$, calculate the ratio

$$\frac{\text{amplitude of transmitted light}}{\text{amplitude of incident light}}.$$

Show your working.

ratio = [3]

- (ii) The filter is now rotated, about the direction of the light beam, from its starting position shown in Fig. 5.1. The direction of rotation is such that the angle of the transmission axis to the vertical initially increases.

Calculate the minimum angle through which the filter must be rotated so that the intensity of the transmitted light returns to the value that it had when the filter was at its starting position.

angle = $^\circ$ [1]

39. 9702/21/O/N/22/Q4

(a) Polarisation is a phenomenon associated with light waves but not with sound waves.

(i) State the meaning of polarisation.

.....
.....
..... [1]

(ii) State why light waves can be plane polarised but sound waves cannot.

.....
.....
..... [1]

(b) Two polarising filters A and B are positioned so that their planes are parallel to each other and perpendicular to a central axis line XY, as shown in Fig. 4.1.

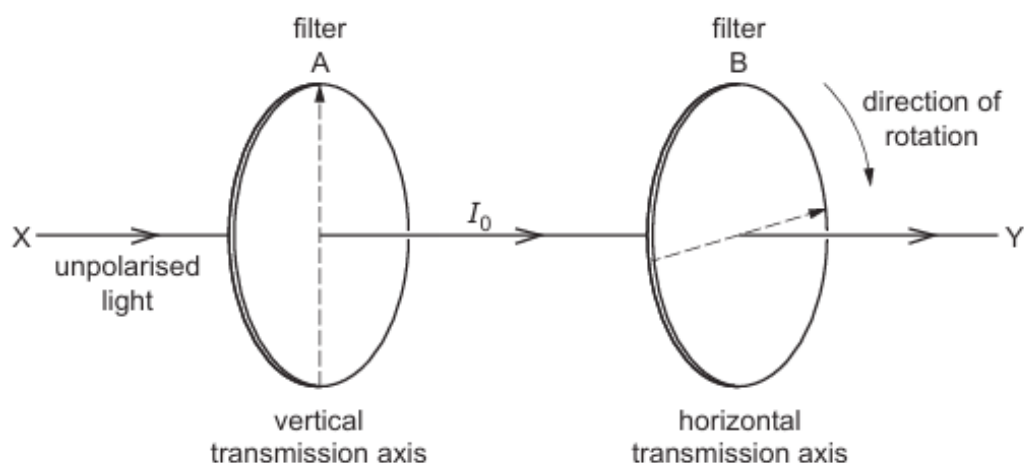


Fig. 4.1

The transmission axis of filter A is vertical and the transmission axis of filter B is horizontal.

Unpolarised light of a single frequency is directed along the line XY from a source positioned at X. The light emerging from filter A is vertically plane polarised and has intensity I_0 .

Filter B is rotated from its starting position about the line XY, as shown in Fig. 4.1.

After rotation, the intensity of the light emerging from filter B is $\frac{1}{4} I_0$.

Calculate the angle of rotation of filter B from its starting position.

angle of rotation = ° [3]

- (c) A microwave of intensity I_0 and amplitude A_0 meets another microwave of the same frequency and of intensity $\frac{1}{4} I_0$ travelling in the opposite direction. Both microwaves are vertically plane polarised and superpose where they meet.

- (i) Explain, without calculation, why these two waves cannot form a stationary wave with zero amplitude at its nodes.

.....
.....
..... [2]

- (ii) Determine, in terms of A_0 , the maximum amplitude of the wave formed.

maximum amplitude = A_0 [3]

[Total: 10]

40. 9702/22/O/N/22/Q5

- (a) Two progressive sound waves meet to form a stationary wave. The two waves have the same amplitude, wavelength, frequency and speed.

State the other condition that must be fulfilled by the two waves in order for them to produce the stationary wave.

..... [1]

- (b) A stationary wave is formed on a string that is stretched between two fixed points A and B. Fig. 5.1 shows the string at time $t = 0$ when each point is at its maximum displacement.



Fig. 5.1

Distance AB is 0.80 m. The period of the stationary wave is 0.016 s.

- (i) On Fig. 5.1, sketch a solid line to show the position of the string:

- at time $t = 0.004$ s (label this line P)
- at time $t = 0.024$ s (label this line Q).

[2]

- (ii) Determine the speed of a progressive wave along the string.

speed = ms^{-1} [3]

- (c) A beam of vertically polarised light of intensity I_0 is incident normally on a polarising filter that has its transmission axis at 30° to the vertical, as shown in Fig. 5.2.

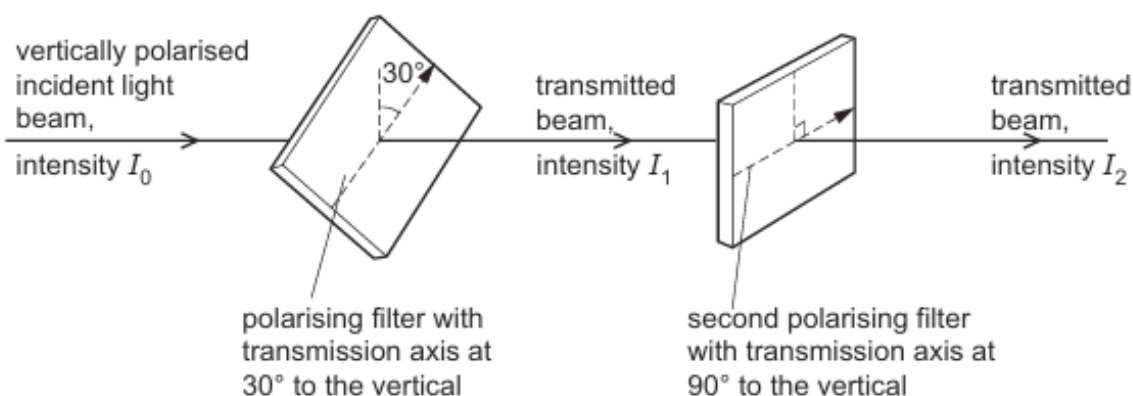


Fig. 5.2

The transmitted light from the first polarising filter has intensity I_1 . This light is then incident normally on a second polarising filter that has its transmission axis at 90° to the vertical. The transmitted light from the second filter has intensity I_2 .

Calculate:

- (i) the ratio $\frac{I_1}{I_0}$

$$\frac{I_1}{I_0} = \dots\dots\dots [2]$$

- (ii) the ratio $\frac{I_2}{I_0}$.

$$\frac{I_2}{I_0} = \dots\dots\dots [2]$$

[Total: 10]

41. 9702/23/O/N/22/Q4

- (a) A progressive longitudinal wave travels through a medium from left to right. Fig. 4.1 shows the positions of some of the particles of the medium at time t_0 and a graph showing the particle displacements at the same time t_0 .

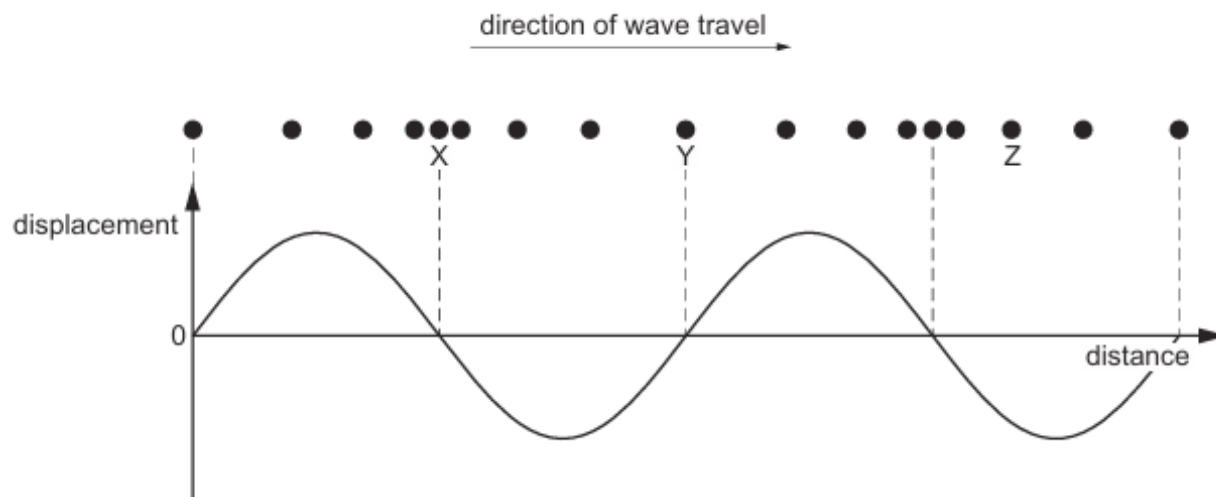


Fig. 4.1

Particle displacements to the right of their equilibrium positions are shown as positive on the graph and particle displacements to the left are shown as negative on the graph.

The period of the wave is T .

- (i) On Fig. 4.1, draw circles around two particles which are exactly one wavelength apart. [1]
- (ii) On Fig. 4.1, sketch a line on the graph to represent the displacements of the particles for the longitudinal wave at time $t_0 + \frac{T}{4}$. [3]
- (iii) State the direction of motion of particle Z at time $t_0 + \frac{T}{4}$. [1]

..... [1]

- (b) The frequency of the wave in (a) is 16 kHz. The distance between particles X and Y is 0.19 m.

Calculate the speed of the wave as it travels through the medium.

speed = ms^{-1} [3]

- (c) A longitudinal sound wave is travelling through a solid. The initial intensity of the wave is I_0 . The frequency of the wave remains constant and the amplitude falls to half of its original value.

Determine, in terms of I_0 , the final intensity of the wave.

intensity = I_0 [2]

- (d) The sound wave in (c) now meets another sound wave travelling in the opposite direction.

- (i) State a condition necessary for these two waves to form a stationary wave.

..... [1]

- (ii) State **two** ways in which a stationary wave differs from a progressive wave.

1

.....

2

.....

[2]

[Total: 13]

42. 9702/22/M/J/23/Q5

- (a) A progressive wave travels through a medium. The wave causes a particle of the medium to vibrate along a line P. The energy of the wave propagates along a line Q.

Compare the directions of lines P and Q if the wave is:

- (i) a transverse wave

..... [1]

- (ii) a longitudinal wave.

..... [1]

- (b) A tube is closed at one end. A loudspeaker is placed near the other end of the tube, as shown in Fig. 5.1.

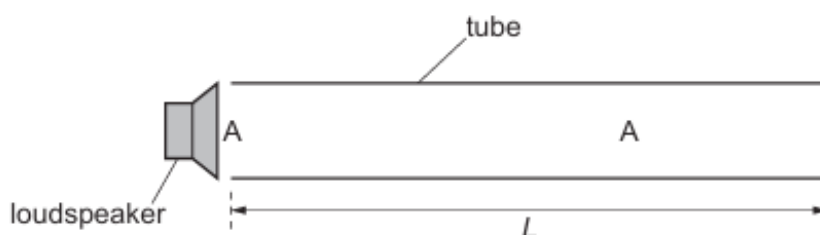


Fig. 5.1 (not to scale)

The loudspeaker emits sound of frequency 1.7 kHz . The speed of sound in the air in the tube is 340 m s^{-1} . A stationary wave is formed with an antinode A at the open end of the tube. There is only one other antinode A inside the tube, as shown in Fig. 5.1.

Determine:

- (i) the wavelength of the sound

wavelength = m [2]

- (ii) the length L of the tube

$L =$ m [1]

- (iii) the maximum wavelength of the sound from the loudspeaker that can produce a stationary wave in the tube.

maximum wavelength =m [1]

- (c) Two polarising filters are arranged so that their planes are vertical and parallel. The first filter has its transmission axis at an angle of 35° to the vertical and the second filter has its transmission axis at angle α to the vertical, as shown in Fig. 5.2.

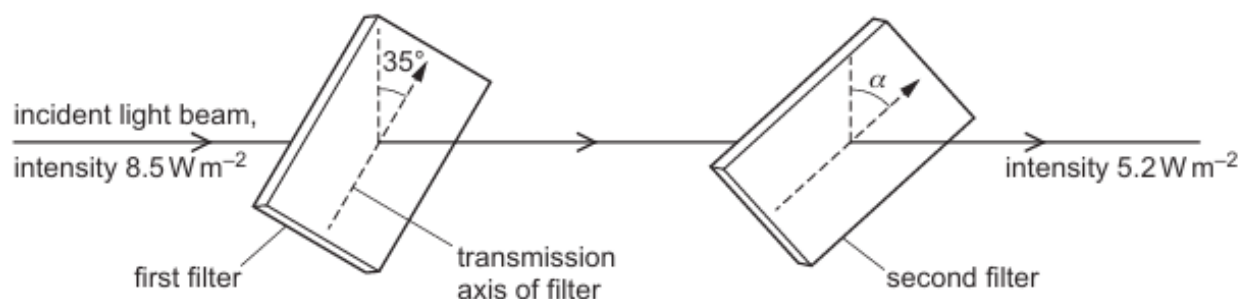


Fig. 5.2

Angle α is greater than 35° and less than 90° . A beam of vertically polarised light of intensity 8.5 W m^{-2} is incident normally on the first filter.

- (i) Show that the intensity of the light transmitted by the first filter is 5.7 W m^{-2} .

[1]

- (ii) The intensity of the light transmitted by the second filter is 5.2 W m^{-2} .

Calculate angle α .

$\alpha = \dots\dots\dots^\circ$ [2]

[Total: 9]

43. 9702/21/O/N/23/Q5

A train travels at a constant high speed along a straight horizontal track towards an observer standing adjacent to the track, as shown in Fig. 5.1.

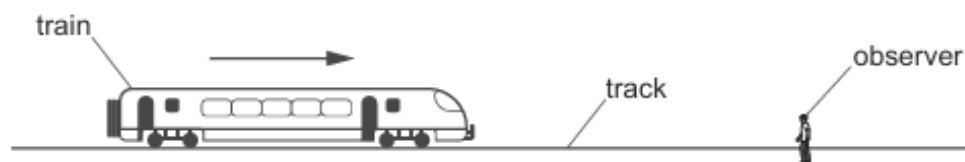


Fig. 5.1

The train sounds its horn continuously as it approaches the observer, from time $t = 0$ until it is well past the observer at time $t = t_2$. The train passes the observer at time $t = t_1$. The horn emits a sound wave of constant frequency f_s .

- (a) On Fig. 5.2, sketch the variation of the frequency of sound heard by the observer with time t , from time $t = 0$ to $t = t_2$.

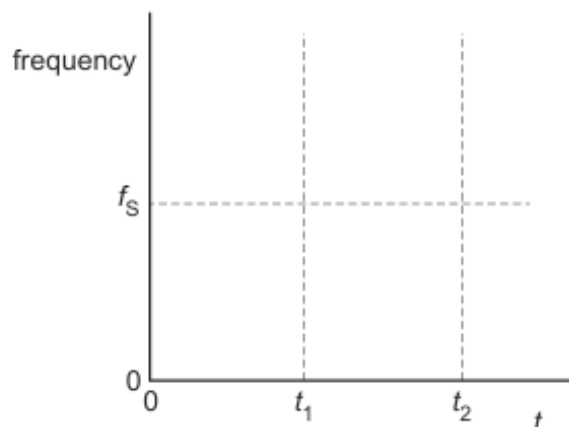


Fig. 5.2

[1]

- (b) At a particular time, the sound waves at the observer have an intensity of $4.7 \times 10^{-3} \text{ W m}^{-2}$. The waves at the observer are incident at right angles on a circular detector of radius 2.8 cm.

Calculate the power P of the waves incident on the detector.

$P = \dots\dots\dots \text{ W}$ [3]

[Total: 4]

44. 9702/23/O/N/23/Q6

A train travels at constant speed along a straight horizontal track towards an observer standing adjacent to the track, as shown in Fig. 6.1.

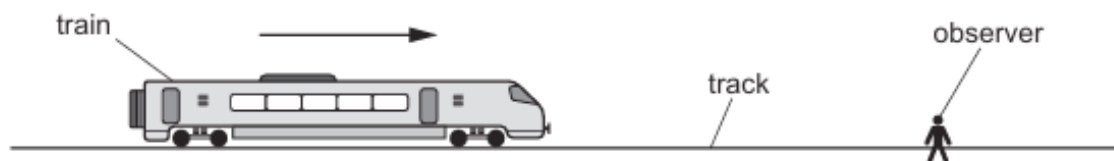


Fig. 6.1

The train sounds its horn continuously as it approaches the observer. The horn emits a sound of constant frequency 251 Hz. The frequency of sound heard by the observer is 291 Hz. The speed of sound in air is 340 m s^{-1} .

(a) Calculate the speed of the train.

speed = m s^{-1} [2]

- (b) The train approaches and then passes the observer. The intensity I of the sound heard by the observer varies with the distance d of the horn from the observer.

When the horn is at a distance x_0 from the observer, the intensity I of the sound heard is I_0 and the amplitude A of the sound wave at the observer is A_0 .

Fig. 6.2 shows the variation with d/x_0 of I/I_0 as the train moves away from the observer.

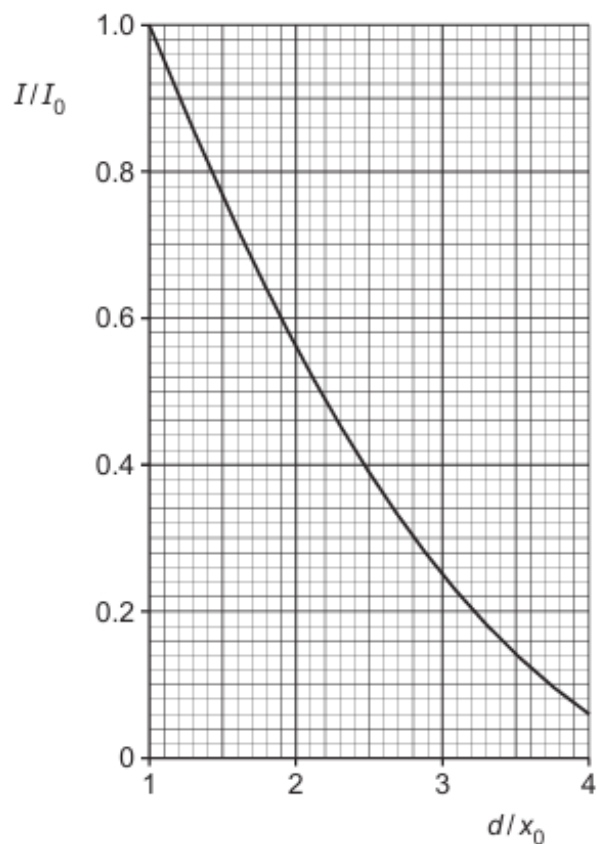


Fig. 6.2

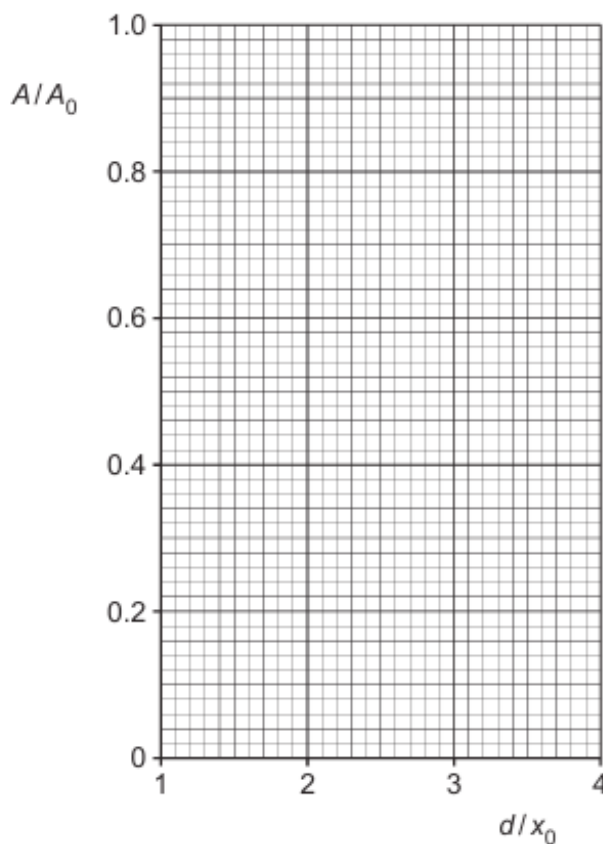


Fig. 6.3

- (i) State the relationship between amplitude A and intensity I for a progressive wave.

..... [1]

- (ii) On Fig. 6.3, sketch the variation with d/x_0 of A/A_0 . [2]

[Total: 5]

ANSWERS

1	(a)	transfer / propagation of energy	M1	
		as a result of oscillations / vibrations	A1	[2]
	(b)	(i) displacement / velocity / acceleration (of particles in the wave)	B1	[1]
		(ii) displacement etc. is normal to direction of energy transfer / travel of wave / propagation of wave(not 'wave motion')	B1	[1]
		(iii) displacement etc. along / same direction of energy transfer / travel of wave / propagation of wave(not 'wave motion')	B1	[1]
	(c)	diffraction: suitable object, means of observation	M1	
		either laser or lamp and aperture		
		or distant source	M1	
		light region where darkness expected	A1	
		interference: suitable object, means of observation and illumination	B1	
2		light and dark fringes observed	B1	
		appropriate reference to a dimension for diffraction or for interference	B1	[6]
	(a)	(i) frequency f	B1	[1]
		(ii) amplitude A	B1	[1]
	(b)	π rad or 180°(unit necessary)	B1	[1]
	(c)	(i) speed = $f \times L$	B1	[1]
		(ii) wave is reflected at end / at P	B1	
		either incident and reflected waves interfere		
		or two waves travelling in opposite directions interfere	M1	
		speed is the speed of incident or reflected wave / one of these waves	A1	[3]
3	(a)	e.g. no energy transfer		
		amplitude varies along its length/nodes <u>and</u> antinodes		
		neighbouring points (in inter-nodal loop) vibrate in phase, etc.		
		(any two, 1 mark each to max 2	B2	[2]
	(b)	(i) $\lambda = (330 \times 10^2)/550$	M1	
		$\lambda = 60$ cm	A0	[1]
		(ii) node labelled at piston	B1	
		antinode labelled at open end of tube	B1	
		additional node and antinode in correct positions along tube	B1	[3]
	(c)	at lowest frequency, length = $\lambda/4$	C1	
		$\lambda = 1.8$ m		
		frequency = $330/1.8$	C1	
		= 180 Hz	A1	[3]

4 (a)	(i) distance (of point on wave) from rest / equilibrium position	B1 [1]
	(ii) distance moved by wave energy / wavefront during one cycle of the source or minimum distance between two points with the same phase or between adjacent crests or troughs	B1 [1]
(b)	(i) $T = 0.60 \text{ s}$	B1 [1]
	(ii) $\lambda = 4.0 \text{ cm}$	B1 [1]
	(iii) either $v = \lambda/T$ or $v = f\lambda$ <u>and</u> $f = 1/T$ $v = 6.7 \text{ cm s}^{-1}$	C1 A1 [2]
(c)	(i) amplitude is decreasing so, it is losing power	M1 A1 [2]
	(ii) $\text{intensity} \sim (\text{amplitude})^2$ ratio = $2.0^2 / 1.1^2$ = 3.3	C1 C1 A1 [3]
5 (a)	two waves travelling (along the same line) in opposite directions overlap/meet same frequency / wavelength resultant displacement is the sum of displacements of each wave / produces nodes and antinodes	M1 A1 B1 [3]
(b)	apparatus: source of sound + detector + reflection system adjustment to apparatus to set up standing waves – how recognised measurements made to obtain wavelength	B1 B1 B1 [3]
(c)	(i) at least two nodes and two antinodes	A1 [1]
	(ii) node to node = $\lambda / 2 = 34 \text{ cm}$ (allow 33 to 35 cm) $c = f\lambda$ $f = 340 / 0.68 = 500$ (490 to 520) Hz	C1 C1 A1 [3]
6 (a)	waves (travels along tube) reflect at <u>closed end / end of tube</u> incident and reflected waves or these two waves are in <u>opposite directions</u> interfere or stationary wave formed if tube length equivalent to $\lambda / 4, 3\lambda / 4$, etc.	B1 M1 A1 [3]
(b)	(i) 1. no motion (as node) / zero amplitude 2. vibration backwards and forwards / maximum amplitude along length	B1 [1] B1 [1]
	(ii) $\lambda = 330 / 880 (= 0.375 \text{ m})$ $L = 3\lambda / 4$ $L = 3 / 4 \times (0.375) = 0.28$ (0.281) m	C1 C1 A1 [3]

7 (a)	travel through a vacuum / free space	B1	[1]
(b) (i)	B : name: microwaves wavelength: 10^{-4} to 10^{-1} m	B1	
	C : name: ultra-violet / UV wavelength: 10^{-7} to 10^{-9} m	B1	
	F : name: X-rays wavelength: 10^{-9} to 10^{-12} m	B1	[3]
(ii)	$f = \frac{3 \times 10^8}{500 \times 10^{-9}}$	C1	
	$f = 6(0) \times 10^{14}$ Hz	A1	[2]
(c)	vibrations are in one direction	M1	
	perpendicular to direction of propagation / energy transfer		
	or good sketch showing this	A1	[2]
8 (a) (i)	$v = f\lambda$	C1	
	$\lambda = 40 / 50 = 0.8(0)$ m	A1	[2]
(ii)	waves (travel along string and) reflect at Q / wall / fixed end	B1	
	incident and reflected waves interfere / superpose	B1	[2]
(b) (i)	nodes labelled at P, Q and the two points at zero displacement	B1	
	antinodes labelled at the three points of maximum displacement	B1	[2]
(ii)	$(1.5\lambda$ for PQ hence $PQ = 0.8 \times 1.5) = 1.2$ m	A1	[1]
(iii)	$T = 1 / f = 1/50 = 20$ ms	C1	
	5 ms is $\frac{1}{4}$ of cycle	A1	
	horizontal line through PQ drawn on Fig. 5.2	B1	[3]
9 (a) (i)	displacement is the distance the rope / particles are (above or below) from the equilibrium / mean / rest / undisturbed position (not 'distance moved')	B1	[1]
(ii) 1.	amplitude (= $80 / 4$) = 20 mm	B1	[1]
2.	$v = f\lambda$ or $v = \lambda / T$	C1	
	$f = 1 / T = 1 / 0.2$ (5 Hz)	C1	
	$v = 5 \times 1.5 = 7.5$ m s $^{-1}$	A1	[3]
(b)	point A of rope shown at equilibrium position	B1	
	same wavelength, shape, peaks / wave moved $\frac{1}{4}\lambda$ to right	B1	[2]
(c) (i)	progressive as energy OR peaks OR troughs is/are transferred/moved /propagated (by the waves)	B1	[1]
(ii)	transverse as particles/rope movement is perpendicular to direction of travel /propagation of the energy/wave velocity	B1	[1]

10 (a)	(i) 1.	wavelength: minimum distance between two points moving in phase OR distance between neighbouring or consecutive peaks or troughs OR wavelength is the distance moved by a wavefront in time T or one oscillation/cycle or period (of source)	B1	[1]
	2.	frequency: number of wavefronts / (unit) time OR number of oscillations per unit time or oscillations/time	B1	[1]
	(ii)	speed = $\frac{\text{distance}}{\text{time}} = \frac{\text{wavelength}}{\text{time period}}$ $= \lambda / T = \lambda f$	M1 A0	[1]
	(b) (i)	amplitude = 4.0 mm (allow 1 s.f.)	A1	[1]
	(ii)	wavelength = $18 / 3.75 (= 4.8)$ speed = $2.5 \times 4.8 \times 10^{-2} = 12 \times 10^{-2} \text{ ms}^{-1}$ unit consistent with numerical answer, e.g. in cm s^{-1} if cm used for λ and unit changed on answer line [if $18 \text{ cm} = 3.5\lambda$ used giving speed $13 (12.9) \text{ cm s}^{-1}$ allow max. 1].	C1 A1	[2]
	(iii)	180° or π rad	A1	[1]
(c)		light and screen and correct positions above and below ripple tank strobe or video camera	B1 B1	[2]

11 (a)	(i)	displacement is the distance from the equilibrium position / undisturbed position / midpoint / rest position amplitude is the maximum displacement	B1 B1	[2]
	(ii)	frequency is the number of wavefronts / crests passing a point per unit time / number of oscillations per unit time time period is the time between adjacent wavefronts or time for one oscillation	B1 B1	[2]
(b)	(i) 1.	amplitude = 1.5 mm	A1	[1]
	2.	wavelength = $25 / 6$ $= 4.2 \text{ cm}$ or $4.2 \times 10^{-2} \text{ m}$	C1 A1	[2]
	(ii)	$v = \lambda / T$ or $v = f \lambda$ <u>and</u> $T = 1 / f$ $T = 4.2 / 7.5 = 0.56 \text{ s}$	C1 A1	[2]
(c)	(i)	progressive wavefront / crests moving / energy is transferred by the waves	M0 A1	[1]
	(ii)	transverse the vibration is perpendicular to the direction of energy transfer / wave velocity or travel <u>of the wave / wavefronts</u>	M0 A1	[1]

12 (a) (i)	progressive wave transfers energy, stationary wave no transfer of energy/ keeps energy within wave	B1	[1]
(ii)	(progressive) wave / wave from loudspeaker reflects at end of tube reflected wave overlaps (another) progressive wave same frequency and speed hence stationary wave formed	B1 B1 B1	[3]
(iii)	(side to side) along length of tube / along axis of tube	B1	[1]
(b)	all three nodes clearly marked with N / clearly labelled at cross-over points	B1	[1]
(c)	phase difference = 0	A1	[1]
(d) (i)	$v = f\lambda$	C1	
	$\lambda = 330 / 440 = 0.75 \text{ m}$	A1	[2]
(ii)	$L = 5/4 \lambda$	C1	
	$= 5/4 \times 0.75 = 0.94 \text{ m}$	A1	[2]

13 (a)	two waves (of the same kind) travelling in opposite directions overlap waves have same frequency / wavelength and speed	B1 B1	[2]
(b) (i)	$T = 0.8 \text{ (ms)}$	C1	
	$f = 1 / (0.8 \times 10^{-3}) = 1250 \text{ (Hz)}$	A1	[2]
(ii)	microphone is moved from plate to loudspeaker or vice versa wavelength is the twice the distance between adjacent maxima or minima (seen on c.r.o.)	B1 B1	[2]
(iii)	$v = f\lambda$	C1	
	$= 1250 \times 0.26$		
	$= 330 \text{ (325) ms}^{-1}$	A1	[2]

14 (a)	stress = Young modulus $\times$ strain		
	$= 1.8 \times 10^{11} \times 8.2 \times 10^{-4} \text{ or } 1.476 \times 10^8$	C1	
	$= 0.15 \text{ (0.148) GPa}$	A1	[2]
(b) (i)	wavelength $= 3 \times 10^8 / 12 \times 10^{12}$	C1	
	$= 25 \mu\text{m}$	A1	[2]
(ii)	infra-red / IR	B1	[1]

15 (a)	(i) progressive: energy is moved / transferred / propagated from one place to another (without the bulk movement of the medium)	B1	
	transverse: (particles) oscillate / vibrate at right angles to the direction of travel of the energy / wavefront	B1	[2]
	(ii) number of oscillations per unit time / number of wavefronts passing a point per unit time	B1	[1]
	(b) (i) P and T	B1	[1]
	(ii) P and S <u>or</u> Q and T	B1	[1]
(c)	$\lambda = 1.2 \times 10^{-2} \text{ (m)}$	C1	
	$v = f\lambda = 15 \times 1.2 \times 10^{-2}$	C1	
	$= 0.18 \text{ ms}^{-1}$	A1	[3]
	(d) ratio $= (1.4)^2 / (2.1)^2$ $= 0.44$	C1 A1	[2]
16 (a)	progressive waves transfer/propagate energy and stationary waves do not <u>amplitude</u> constant for progressive wave and varies (from max/antinode to min/zero/node) for stationary wave	B1	
	adjacent particles in phase for stationary wave and out of phase for progressive wave	B1	
	(b) (i) wave / microwave <u>from source/S</u> reflects <u>at reflector/R</u>	(B1)	[2]
	reflected and (further) incident waves overlap/meet/superpose	B1	
	waves have same <u>frequency/wavelength/period</u> and <u>speed</u> (so stationary waves formed)	B1	[3]
	(ii) detector/D is moved between reflector/R and source/S (or v.v.)	B1	
	maximum, minimum/zero, (maximum... etc.) observed on meter/deflections/readings/measurements/recordings	B1	[2]
	(iii) <u>determine/measure</u> the distance between adjacent minima/nodes or maxima/antinodes or across <u>specific number</u> of nodes/antinodes	B1	
	wavelength is twice distance between <u>adjacent</u> nodes/minima or maxima/antinodes (or other correct method of calculation of wavelength from measurement)	B1	[2]
	(c) $v = f\lambda$	C1	
17	(i) intensity $\propto (\text{amplitude})^2$ ratio $= (0.60^2 / 0.90^2) = 0.44$	C1 A1	
	(ii) phase difference $= 90^\circ$	A1	
18 (a)	$T = 4 \text{ (ms)} \text{ or } 4 \times 10^{-3} \text{ (s)}$	C1	
	$f = 1/T = 1/0.004$		
	$= 250 \text{ Hz}$	A1	[2]
	(b) intensity $\propto (\text{amplitude})^2$ <u>and</u> amplitude $= 2.8 \text{ (2.83) (cm)}$	B1	
	curve with same period and with amplitude 2.8 cm curve shifted 1.0 ms to left or to right of wave X	B1 B1	[3]

19	(a)	longitudinal: vibrations/oscillations (of the particles/wave) are parallel to the direction or in the same direction (of the propagation of energy)	B1	
		transverse: vibrations/oscillations (of the particles/wave) are perpendicular to the direction (of the propagation of energy)	B1	[2]
	(b)	LHS: intensity = power / area units: $\text{kg m s}^{-2} \times \text{m} \times \text{s}^{-1} \times \text{m}^{-2}$ or $\text{kg m}^2 \text{s}^{-3} \times \text{m}^{-2}$	B1	
		RHS: units: $\text{m s}^{-1} \times \text{kg m}^{-3} \times \text{s}^{-2} \times \text{m}^2$	M1	
		LHS and RHS both kg s^{-3}	A1	[3]
	(c)	(i) change/difference in the <u>observed/apparent</u> frequency when the source is moving (relative to the observer)	B1	[1]
		(ii) wavelength increases/frequency decreases/red shift	B1	[1]
	(d)	observed frequency = $v f_s / (v - v_s)$	C1	
		$550 = (340 \times 510) / (340 - v_s)$	C1	
		$v_s = 25 \text{ (24.7) m s}^{-1}$	A1	[3]
20	(a)	(i) alter distance from vibrator to pulley alter frequency of generator (change tension in string by) changing value of the masses <i>any two</i>	B2	[2]
		(ii) points on string have <u>amplitudes</u> varying from maximum to zero/minimum	B1	[1]
	(b)	(i) 60° or $\pi/3$ rad	A1	[1]
		(ii) ratio = $[3.4 / 2.2]^2$	C1	
		= 2.4 (2.39)	A1	[2]
21	(a)	the number of oscillations per unit time	M1	
		of the source/of a point on the wave/of a particle (in the medium)	A1	[2]
		<i>or</i> the number of wavelengths/wavefronts per unit time passing a (fixed) point	(M1) (A1)	
	(b)	T or period = $2.5 \times 250 \text{ (}\mu\text{s)}$ (= 625 μs)	M1	
		frequency = $1 / (6.25 \times 10^{-4})$ or $1 / (2.5 \times 250 \times 10^{-6}) = 1600 \text{ Hz}$	A1	[2]
	(c)	(i) for maximum frequency: $f_o = f_s v / (v - v_s)$		
		$1640 = (1600 \times 330) / (330 - v_s)$	C1	
		$v_s = 8(.0) \text{ m s}^{-1}$ (8.049 m s^{-1})	A1	[2]
		(ii) loudspeaker moving towards observer causes rise in/higher frequency	B1	
		loudspeaker moving away from observer causes fall in/lower frequency	B1	[2]
		<i>or</i> <u>repeated</u> rise and fall/higher and then lower frequency caused by loudspeaker moving towards and away from observer	(M1) (A1)	

22 (a)	(two) waves travelling (at same speed) in opposite directions overlap waves (are same type and) have same frequency/wavelength	B1
		B1
(b) (i)	$\lambda = 12 / 250 (= 0.048 \text{ m})$	C1
	distance = 1.5×0.048	A1
	= 0.072 m	
(ii)	$T = 1 / 250$	C1
	= 0.004 (s) or 4 (ms)	
	1. curve drawn is mirror image of that in Fig. 4.2 and labelled P	A1
	2. horizontal line drawn between A and B and labelled Q	A1
23 (a)	observed frequency is different to source frequency when source moves relative to observer	B1
(b)	$360 = (400 \times 340) / (340 \pm v)$	C1
	$v = 38 (37.8) \text{ m s}^{-1}$	A1
	away (from the observer)	B1
24 (a)	frequency is the number of vibrations/oscillations per unit time or the number of wavefronts passing a point per unit time	
(b)	vibrations/oscillation of the air particles are parallel to the direction of it (the direction of travel of the sound wave)	B1
(c) (i)	$T = 2(.0) \text{ (ms)}$ $f = 500 \text{ Hz}$	A1
(ii)	1. amplitude increases (time) period decreases	B3
	2. amplitude decreases (time) period increases	<i>any 3 points</i>
25 (a) (i)	in a stationary wave energy is not transferred or in a progressive wave energy is transferred	B1
(ii)	in a stationary wave (adjacent) particles are in phase or in a progressive wave (adjacent) particles are out of phase/have a phase difference/not in phase	B1
(b) (i)	(position where) maximum amplitude	B1
(ii)	distance = 0.10 m	B1
(iii) 1.	$\lambda = 0.60 / 1.5$	A1
	= 0.40 m	
2.	$v = f\lambda$	C1
	$f = 340 / 0.40$	A1
	= 850 Hz	
(iv)	$\lambda = 2 \times 0.60$ or $\lambda = 3 \times 0.40$ or $f = 850 / 3$	C1
	$f = 280 (283) \text{ Hz}$	A1

26 (a)	(two) waves travelling (at same speed) in opposite directions overlap waves (are same type and) have same frequency/wavelength	B1 B1
(b) (i)	5	A1
(ii)	$T = 1/40 (= 2.5 \times 10^{-2})$ time taken $= 2.5 \times 10^{-2}/2$ $= 1.3 \times 10^{-2} \text{ s } (1.25 \times 10^{-2} \text{ s})$	C1 A1
(iii)	180°	A1
(iv)	$v = f\lambda$ $\lambda = 2.0/2.5 (= 0.80 \text{ m})$ $v = 0.80 \times 40$ $= 32 \text{ m s}^{-1}$	C1 A1
27 (a)	displacement of particles/vibration(s)/oscillation(s) is parallel to/along the direction of energy/propagation	B1
(b)	period $= 1/800 (= 1.25 \times 10^{-3} \text{ s})$ time-base setting $= 1.25 \times 10^{-3}/2.5$ $= 5.0 \times 10^{-4} \text{ s cm}^{-1}$	C1 C1 A1
(c) (i)	$I \propto A^2$ $(I_X/I_Y =) [r_Y/r_X]^2 = [A_X/A_Y]^2$ ratio $A_Y/A_X = 120/30$ $= 4.0$	C1 C1 A1
(ii) 1.	$v = f\lambda$ minimum $\lambda = 330/(800 + 16) = 0.40 \text{ m}$	C1 A1
2.	$f_o/f_s = v/(v - v_s)$ $816/800 = 330/(330 - v_s)$ $v_s = 6.5 \text{ m s}^{-1}$	C1 A1

- 28 (a) (i)** time for one oscillation/one vibration/one cycle **B1**
 or time between adjacent wavefronts/points in phase
 or shortest time between two wavefronts/points in phase
- (ii)** distance moved by wavefront/energy during one cycle/oscillation/period (of source) **B1**
 or minimum distance between two wavefronts
 or distance between two adjacent wavefronts
 or minimum distance between two points having the same displacement
 and moving in the same direction
- (b) (i)** $v = \lambda / T$ or $v = f\lambda$ and $f = 1/T$ **C1**
 $\lambda = 20 \times 0.60$ **A1**
 $= 12 \text{ cm}$
- (ii)** phase difference $= 360^\circ \times (0.20/0.60)$ or $360^\circ \times (0.40/0.60)$ **A1**
 $= 120^\circ$ or 240°
- (iii)** $I \propto A^2$ **C1**
 $I_Q/I_P = A_Q^2/A_P^2$ **A1**
 $= 2.0^2/3.0^2$
 $= 0.44$
- (iv)** displacement $= 1.00 - 3.00$ **A1**
 $= -2.00 \text{ mm}$

- 29 (a) (i)** distance moved by wavefront/energy during one cycle/oscillation/period (of source) **B1**
 or minimum distance between two wavefronts
 or distance between two adjacent wavefronts
- (ii)** (position where) maximum amplitude **B1**
- (b) (i)** $\lambda = 4 \times 0.045$ **C1**
 $(= 0.18 \text{ (m) or } 18 \text{ (cm)})$
 $v = f\lambda$ **C1**
 $f = 340/0.18$ **A1**
 $= 1900 \text{ Hz}$
- (ii)** distance $= \lambda/2$ **C1** or $t = 4.5/0.75$ and $t = 13.5/0.75$ **(C1)**
 $(= 0.09 \text{ (m) or } 9 \text{ (cm)})$ time $= 18 - 6$ **(A1)**
time $= 0.09 / 0.0075$ **A1** $= 12 \text{ s}$
 $= 12 \text{ s}$

30 (a)	vibration(s)/oscillation(s) (of particles) parallel to direction of propagation of energy	B1
(b)(i)	phase difference = 180°	A1
(ii)	$v = f\lambda$	C1
	$\lambda/2 = 25 \text{ (cm) or } 0.25 \text{ (m)}$	C1
	$f = 330/0.50$	A1
	$= 660 \text{ Hz}$	
(iii)	(readings from graph =) <u>2.6</u> and 4.0	C1
	ratio = $(2.6/4.0)^{1/2}$	A1
	$= 0.81$	

31 (a)	$\rho = m / V$	C1
	$V = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 \text{ (= } 0.458 \text{ m}^3\text{)}$	C1
	$m = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 \times 1.2 = 0.55 \text{ kg}$	A1
(b)(i)	$\Delta p = 0.55 \times 7.6$ $= 4.2 \text{ N s}$	A1
(ii)	$F = 4.2 / 3.0 \text{ or } 0.55 \times 7.6 / 3.0$ $= 1.4 \text{ N}$	A1
(c)(i)	$F = 1.4 \text{ N}$	A1
(ii)	Newton's third law (of motion)	B1
(d)	$2 \times 1.4 = m \times 9.81$	A1
	$m = 0.29 \text{ kg}$	
(e)	the density of air is less at high altitude	B1
(f)	$f_o = f_s v / (v - v_s)$	C1
	$= 3000 \times 340 / (340 - 22)$	
	$= 3200 \text{ Hz}$	A1

- 32 (a)** distance moved by wavefront/energy during one cycle/oscillation/period (of source) **B1**
or
minimum distance between two wavefronts
or
distance between two adjacent wavefronts
- (b)** $(T =) 2.0 \times 2.5 (= 5.0 \text{ ms})$ **or** $2.0 \times 2.5 \times 10^{-3} (= 5.0 \times 10^{-3} \text{ s})$ **C1**
 $f = 1 / (5.0 \times 10^{-3})$ **A1**
 $= 200 \text{ Hz}$
- (c)(i)** (path difference =) $8.0 + (20.8^2 - 8.0^2)^{0.5} - 20.8 = 6.4 \text{ (m)}$ **A1**
- (ii)**
- path difference = 4λ **B2**
 - waves (meet at C) in phase
 - constructive interference (of waves)
- any two points, one mark each*
- (iii)** $v = 200 \times 1.6$ **C1**
 $= 320 \text{ (m s}^{-1}\text{)}$
 $\Delta t = 6.4 / 320$ **or** $27.2 / 320 - 20.8 / 320$ **A1**
 $= 0.020 \text{ s}$
- (iv)** $3\lambda = 6.4$ **A1**
 $\lambda = 2.1 \text{ m}$

- 33 (a)(i)** frequency **or** period **B1**
- (ii)** amplitude **B1**
- (b)** constant phase difference so coherent **B1**
- (c)** 120° **B1**
- (d)** resultant displacement $= 4.0 \mu\text{m} - 1.0 \mu\text{m}$ **B1**
 $= 3.0 \mu\text{m}$
- (e)** $I \propto A^2$ **C1**
intensity of Z $= (2^2 / 4^2) I$ **A1**
 $= 0.25 I$
- (f)** $v = \lambda / T$ **C1**
or
 $v = f\lambda$ **and** $f = 1 / T$
 $330 = \lambda / 3.0 \times 10^{-3}$ **C1**
 $\lambda = 0.99 \text{ m}$ **A1**

34.

5(a)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$v = 8.0 \times 10^{-2} / 0.40$ $= 0.20 \text{ m s}^{-1}$	A1
5(b)	$I \propto A^2$	C1
	ratio $= 2^2 / 4^2$	C1
	$= 0.25$	A1

35.

5(a)	$f_o = f_s v / (v + v_s)$ $f_o = 951 \times 330 / (330 + 12)$	C1
	$= 918 \text{ Hz}$	A1
5(b)	$t = d / 12$ $= (\pi \times 2.4) / 12$	C1
	$= 0.63 \text{ s}$	A1

36.

4(a)	oscillations (of particles) are parallel to (the direction of) energy transfer	B1
4(b)(i)	(frequency varies as) vehicle moves relative to (stationary) observer	C1
	(vehicle) moving towards (observer) gives higher (observed) frequency (than 1.2 kHz) <u>and</u> (vehicle) moving away (from observer) gives lower (observed) frequency (than 1.2 kHz)	A1
4(b)(ii)	Doppler effect	B1
4(b)(iii)	position of vehicle labelled 'X' at top (12 o'clock) position on track	B1
4(b)(iv)	position of vehicle labelled 'Y' at right-hand edge (3 o'clock) position on track	B1
4(c)	maximum frequency = 1.40 (kHz) or 1.40×10^3 (Hz)	C1
	$1.40 = (1.2 \times 320) / (320 - v)$	C1
	$v = 46 \text{ m s}^{-1}$	A1
	or	
	minimum frequency = 1.05 (kHz) or 1.05×10^3 (Hz)	(C1)
	$1.05 = (1.2 \times 320) / (320 + v)$	(C1)
	$v = 46 \text{ m s}^{-1}$	(A1)

37.

4(a)(i)	decrease(s)	B1
4(a)(ii)	increase(s)	B1
4(b)	$f_o = f_s v / (v + v_s)$ $= 925 \times 338 / (338 + 12)$	C1
	$= 893 \text{ Hz}$	A1
4(c)	distance $= (\frac{1}{2} \times 2 \times 12) + (2 \times 12)$ $(= 36 \text{ m})$	C1
	time taken $= 36 / 338$ $= 0.11 \text{ s}$	A1

38.

5(a)(i)	(two) waves travelling (at same speed) in opposite directions overlap	B1
	waves (of the same type) have same frequency/wavelength	B1
5(a)(ii)	phase difference = 0	A1
5(b)(i)	$f_o = f_s v / v - v_s$ $543 = f \times 334 / (334 - 13)$	C1
	$f = 522 \text{ Hz}$	A1
5(b)(ii)	(the speed is) decreasing	B1
5(c)(i)	$I \propto A^2$	B1
	$I_T / I_0 = \cos^2 20^\circ$ or $A_T / A_0 = \cos 20^\circ$	C1
	ratio = 0.94	A1
5(c)(ii)	angle = 140°	A1

39.

4(a)(i)	oscillations are in a single direction, which is perpendicular to the direction of propagation (of the wave) or oscillations are in a single plane, which contains the direction of propagation (of the wave)	B1
4(a)(ii)	light waves are transverse and sound waves are longitudinal	B1
4(b)	$I = I_0 \cos^2 \theta$	C1
	$\cos^2 \theta = 1/4$ so $\cos \theta = 1/2$ $\theta = 60^\circ$ or 120° or 240° or 300°	C1
	angle of rotation = $(120^\circ - 90^\circ)$ or $(240^\circ - 90^\circ)$ or $(300^\circ - 90^\circ)$ $= 30^\circ$ or 150° or 210° or 330°	A1
4(c)(i)	the waves have different amplitudes	B1
	cannot have resultant displacement that is always zero or cannot have (complete) destructive interference (at nodes) or (at nodes resultant) amplitude is the difference of the amplitudes	B1
4(c)(ii)	$I \propto A^2$	C1
	$A^2 / A_0^2 = (I_0 / 4) / I_0$ $A = 0.5 A_0$	C1
	maximum amplitude = $A_0 + 0.5 A_0$ $= 1.5 A_0$	A1

40.

5(a)	(they travel in) opposite directions	B1
5(b)(i)	straight line from A to B, labelled P	B1
	line that is 'mirror image' of given line, labelled Q	B1
5(b)(ii)	$\lambda = 0.80 / 2$ (= 0.40 m)	C1
	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$v = 0.40 / 0.016$ or 62.5×0.40 = 25 m s ⁻¹	A1
5(c)(i)	$I_1 / I_0 = \cos^2 30^\circ$	C1
	= 0.75	A1
5(c)(ii)	$I_2 / I_1 = \cos^2 60^\circ$ $I_2 / I_0 = \cos^2 30^\circ \times \cos^2 60^\circ$ or $0.75 \times \cos^2 60^\circ$	C1
	= 0.19	A1

41.

4(a)(i)	circles drawn around any two particles with seven (uncircled) particles in between	A1
4(a)(ii)	curve has an initial negative displacement and initial amplitude same as original curve	B1
	curve has same amplitude as original curve throughout	B1
	curve has same wavelength as original curve throughout, with constant (non-zero) phase difference	B1
4(a)(iii)	(to the) right / rightwards	A1
4(b)	$\lambda = 2 \times 0.19$ = 0.38 m	C1
	$v = f\lambda$	C1
	$v = (16 \times 10^3) \times 0.38$ $v = 6100 \text{ m s}^{-1}$	A1
4(c)	$I \propto A^2$	C1
	$I = (1 / 2)^2 I_0$ intensity = 0.25 I_0	A1
4(d)(i)	same frequency / wavelength / period	B1
4(d)(ii)	- a stationary wave has nodes/antinodes (and a progressive wave does not) - a stationary wave does not transfer/propagate energy (and a progressive wave does transfer/propagate energy) - different points on a stationary wave have different amplitudes (and all points on a progressive wave have the same/constant amplitude) - stationary wave has adjacent particles that are in phase (and adjacent particles on progressive wave are out of phase) Any two points, 1 mark each. Allow the reverse statement for each marking point.	B2

42.

5(a)(i)	(they are) perpendicular	B1
5(a)(ii)	(they are) parallel	B1
5(b)(i)	$\lambda = v / f$	C1
	$= 340 / 1700$	A1
	$= 0.20 \text{ m}$	
5(b)(ii)	$L = \frac{3}{4} \times \lambda = \frac{3}{4} \times 0.20$ $= 0.15 \text{ m}$	A1
5(b)(iii)	$\lambda = 4 \times 0.15$ or 0.20×3 $= 0.60 \text{ m}$	A1
5(c)(i)	$(I \Rightarrow) 8.5 \times \cos^2 35^\circ = 5.7 \text{ (W m}^{-2}\text{)}$	A1
5(c)(ii)	$5.2 = 5.7 \cos^2 \theta$ $(\theta = 17^\circ)$	C1
	$\alpha = 35^\circ + 17^\circ$	A1
	$= 52^\circ$	

43.

5(a)	sketch: approximately horizontal line above horizontal dashed line from $t = 0$ to $t = t_1$ and approximately horizontal line below horizontal dashed line from $t = t_1$ to $t = t_2$	A1
5(b)	$I = P / A$	C1
	$A = \pi \times 0.028^2$ or $\pi \times 2.8^2$ $(= 2.46 \times 10^{-3}$ or $24.6)$	C1
	$P = 4.7 \times 10^{-3} \times 2.46 \times 10^{-3}$ $= 1.2 \times 10^{-5} \text{ W}$	A1

44.

6(a)	$291 = 251 \times 340 / (340 - v_{(s)})$	C1
	$v_s = 47 \text{ m s}^{-1}$	A1
6(b)(i)	$I \propto A^2$	B1
6(b)(ii)	sketch: line starts at (1.0, 1.0)	B1
	approximately straight line drawn with negative gradient and line ends at (4.0, 0.25)	B1